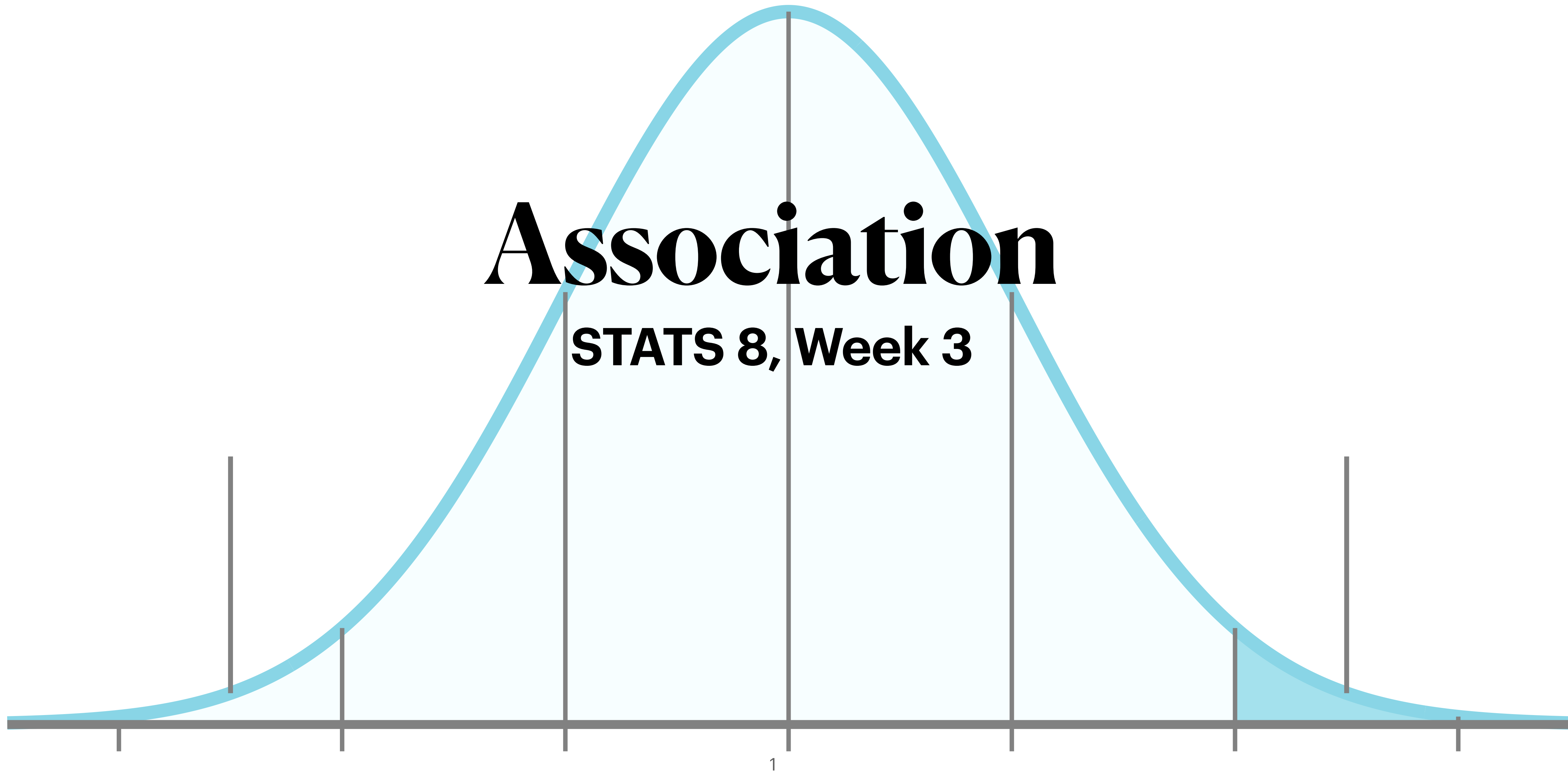
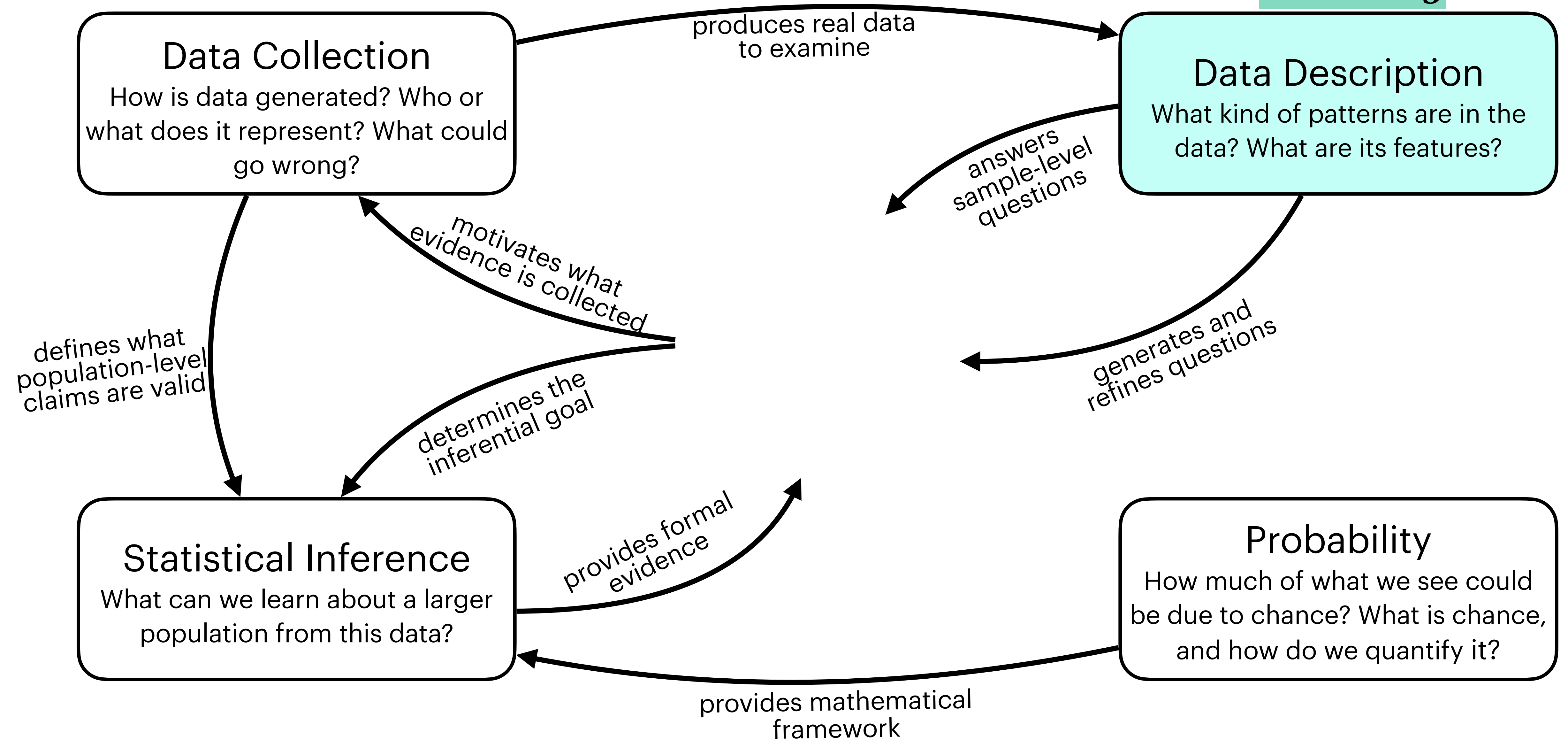


Association

STATS 8, Week 3

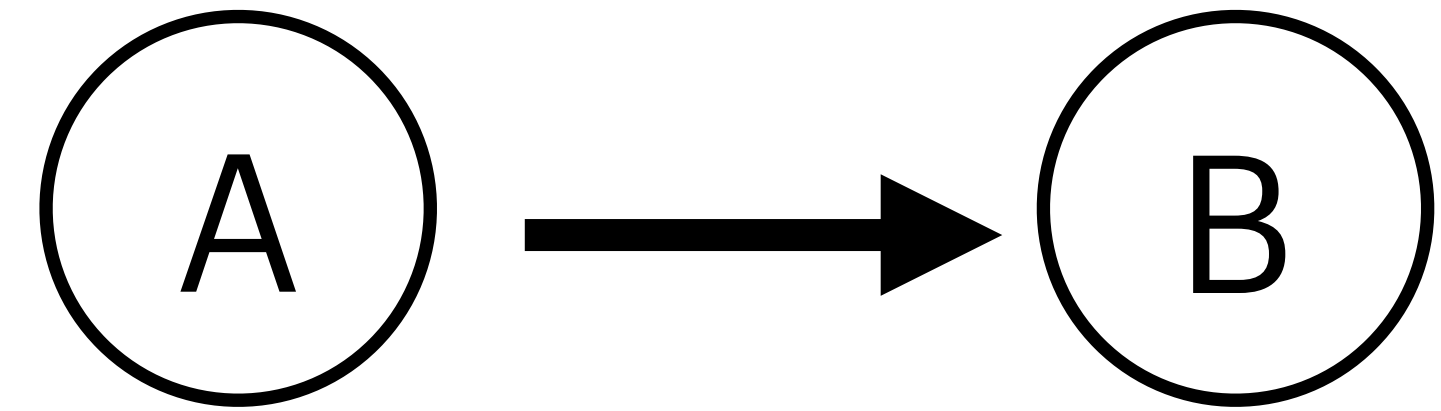


Week 2 - 3

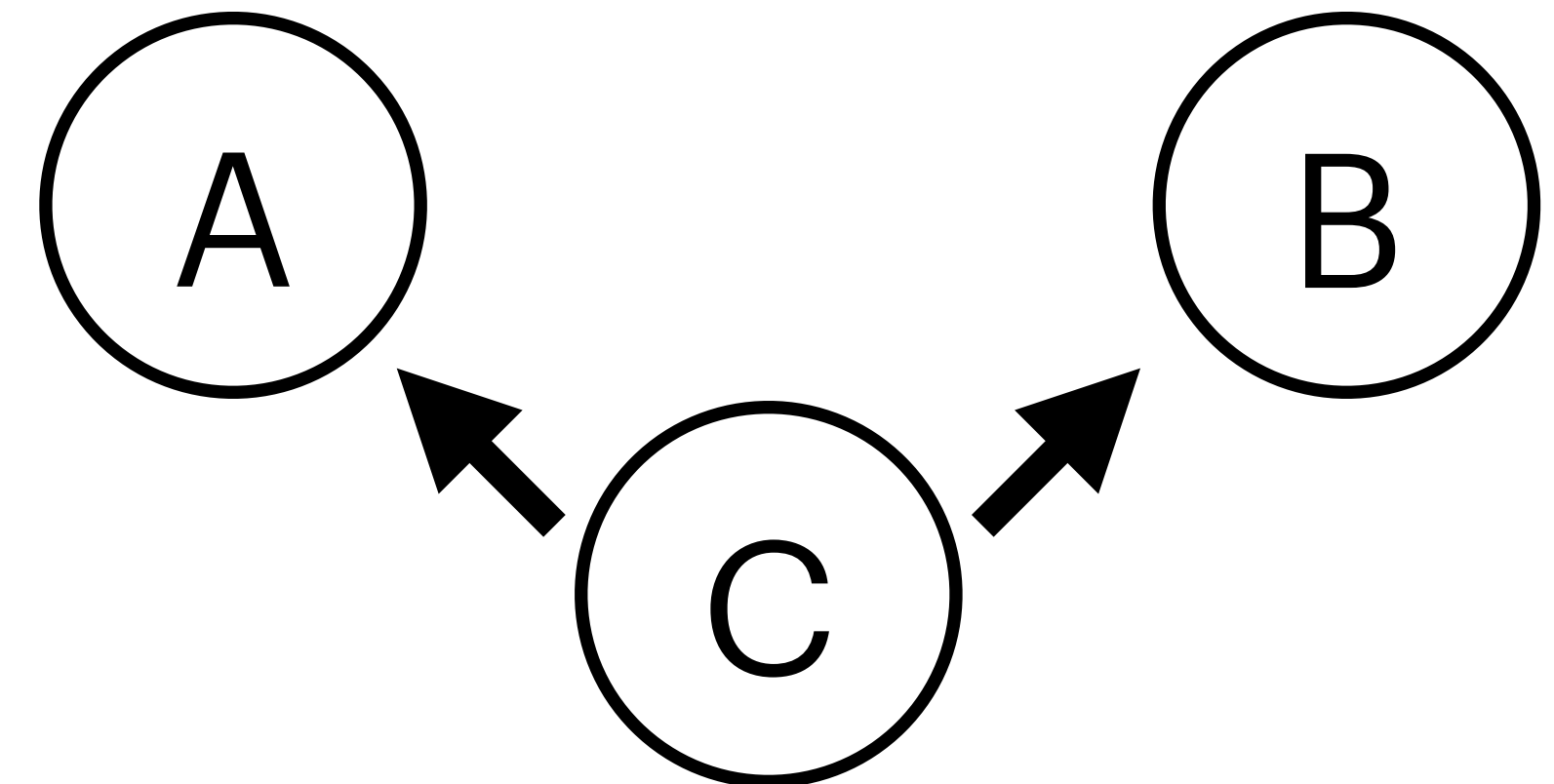
Association

What is it?

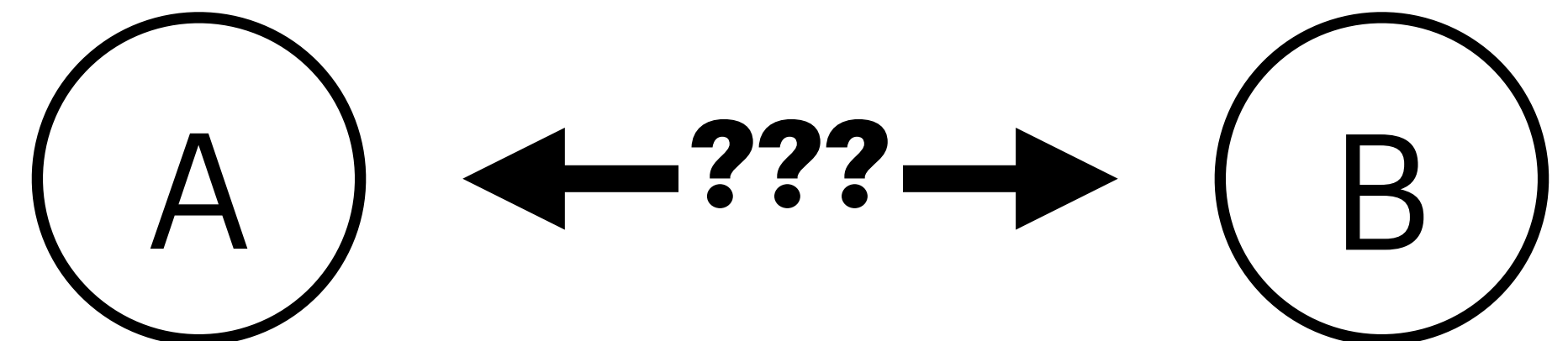
- In statistics, **association** (**relationship**) refers to the idea that variables can vary together with some level of synchronicity
- link or pattern between two variables, where information about the value or category of one variable can provide information about the other
- implies that variables are related, but it does not imply causation!



or



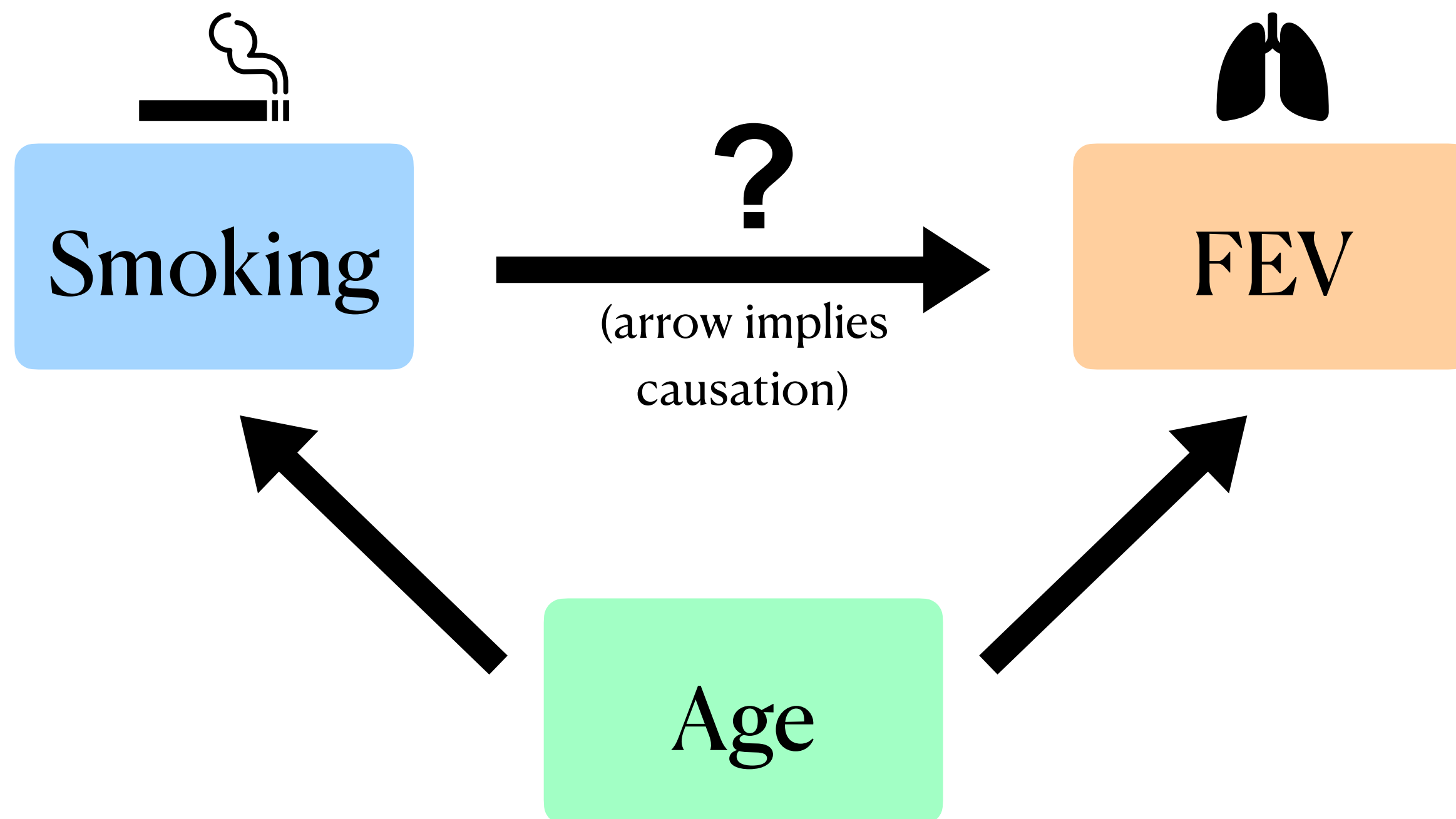
or



Smoking & Lung Function

(Recall from Topic 1)

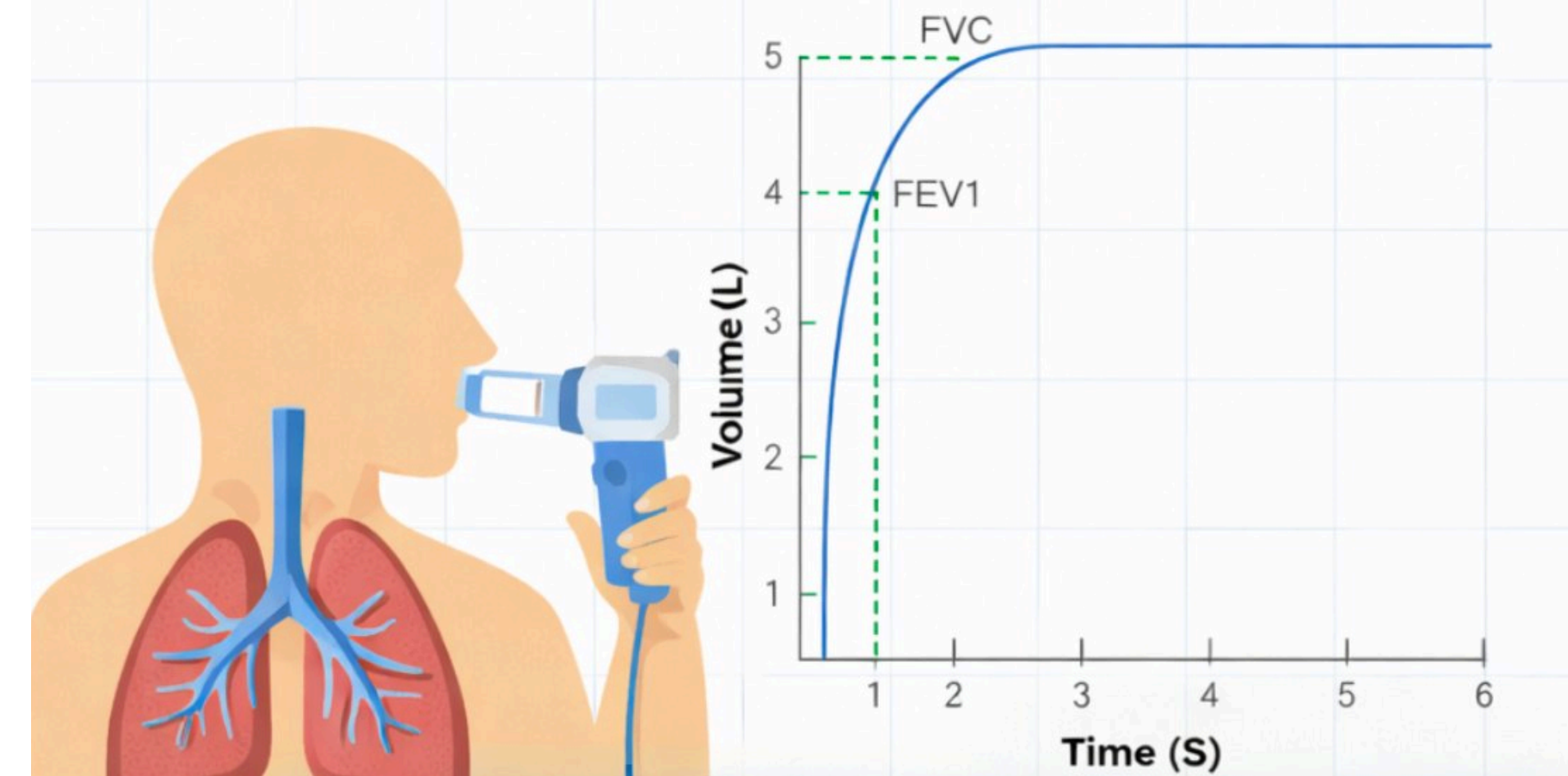
- Smoking in children/adolescents and FEV (Kahn, 2005)
- Is smoking associated with lung function in young people?
 - Result: kids who smoke have better lung function??



FEV1

(Forced Expiratory Volume in One Second)

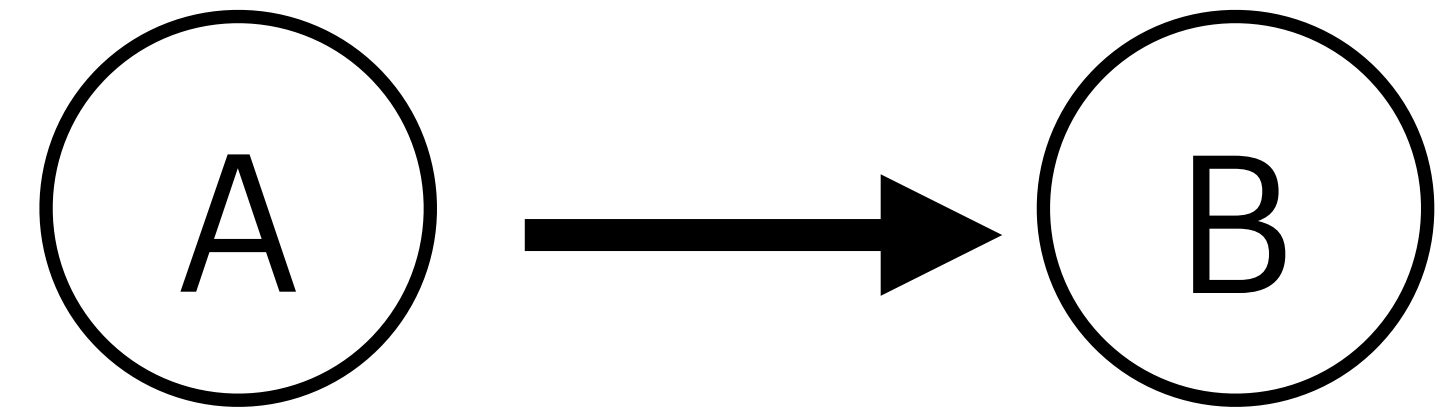
FEV1 is the volume of air a person can forcibly exhale during the **first second** of a forced vital capacity maneuver. It is measured during spirometry after the patient inhales fully to total lung capacity and then exhales as quickly and forcefully as possible.



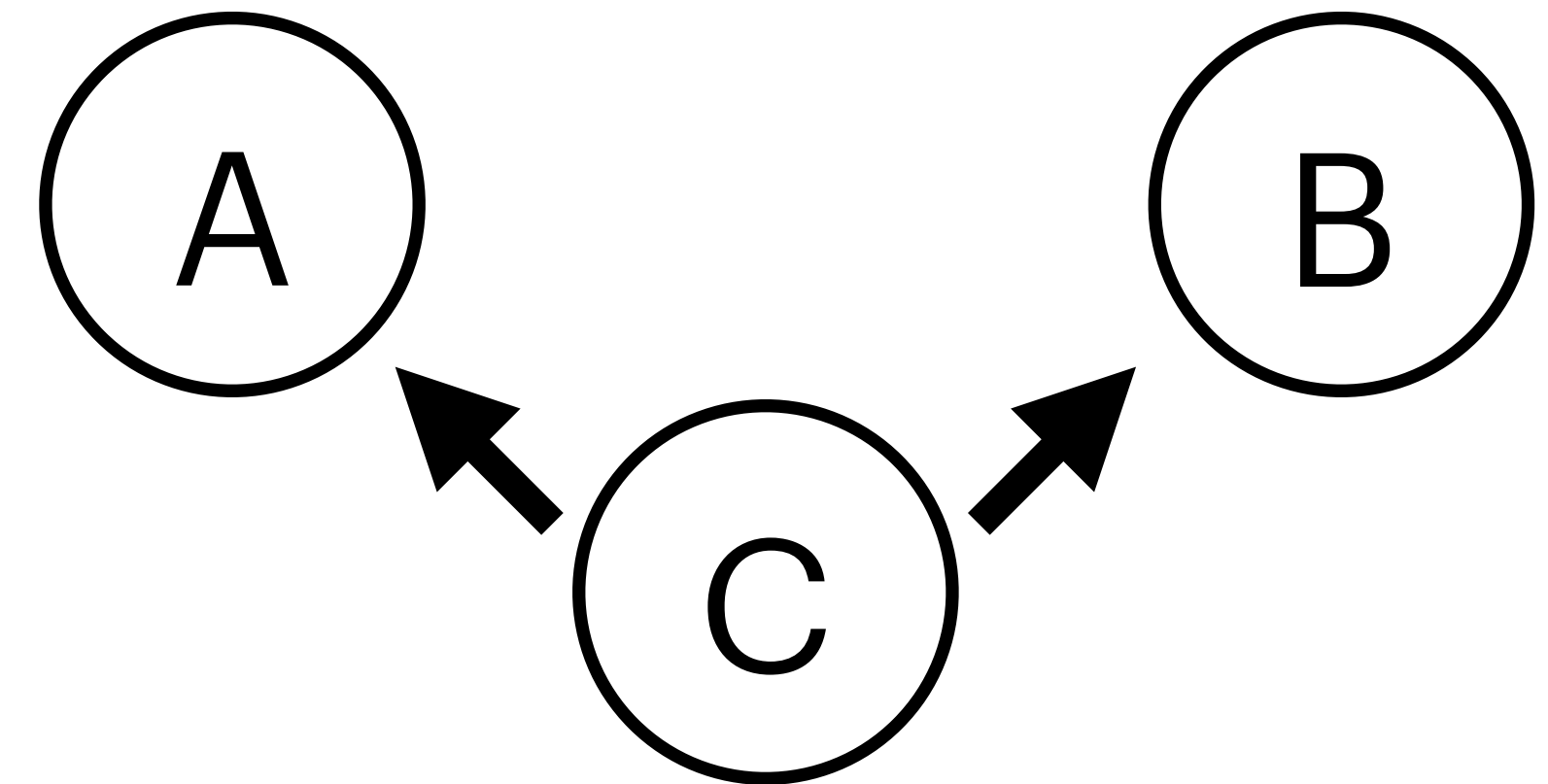
Association

What is it?

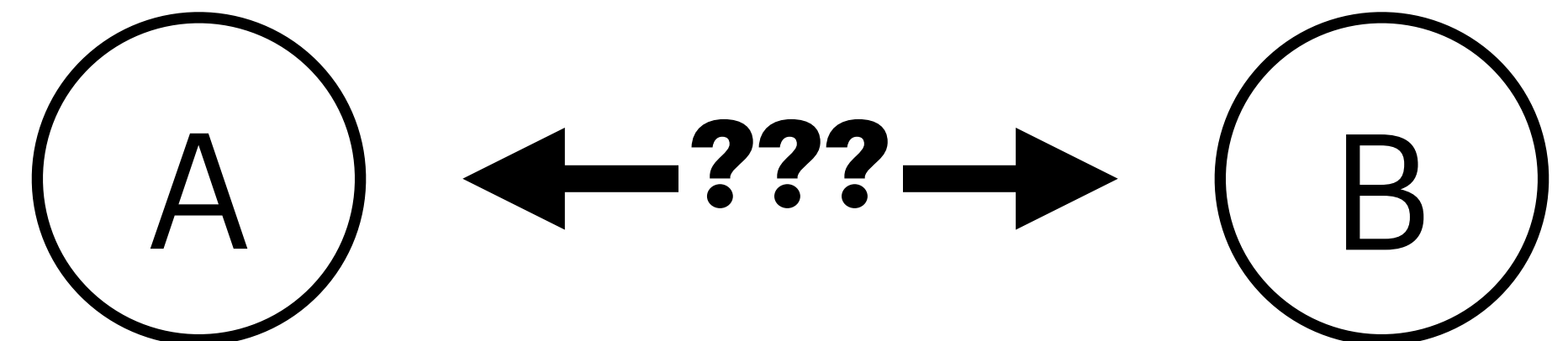
- In statistics, **association (relationship)** refers to the idea that variables can vary together with some level of synchronicity
- e.g., in children and adolescents,
 - Smoking is associated with FEV
 - Age is associated with FEV
 - Smoking is associated with age



or



or



Describing Associations

Variable types

- Between **one categorical and one quantitative variable**
 - Graphical: side-by-side boxplots or histograms
 - Numeric: comparing means and medians, spread
- Between **two categorical variables**
 - Graphical: bar graphs
 - Numeric: contingency/two-way tables, conditional proportions
- Between **two quantitative variables**
 - Graphical: scatterplots
 - Numeric: correlation, regression

Describing Associations

Variable types

- Between **one categorical and one quantitative variable**

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- Numeric: comparing means and medians, spread

- Between **two categorical variables**

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- Between **two quantitative variables**

- Graphical: scatterplots
- Numeric: correlation, regression



Started this during Topic 2, so let's start here

Two-Way Tables

Categorical + Categorical

- **Two-way tables (contingency tables)** are used to organize the counts of joint outcomes of two categorical variables (sometimes called **factors**)
- Seatbelt usage by 12th graders (CDC YRBSS)
 - Seatbelt usage (categorical, ordinal)
 - Sex (categorical, nominal)
- Raw data looks like:

Identifier	Seatbelt usage	Sex
Person 1	Always	Female
Person 2	Sometimes	Male
Person 3	Always	Male
Person 4	Most times	Male
Person 5	Most times	Female

⋮

	Male	Female	Total
Always	771	915	1686
Most times	302	276	578
Sometimes	247	167	414
Rarely	165	84	249
Never	90	25	115
Total	1575	1467	3042

Two-Way Tables

Categorical + Categorical

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★ This data set has 3042 individuals

★ 1686 of those individuals said that they always wear a seatbelt

★ 771 of those individuals said that they always wear a seatbelt and are male

Two-Way Tables

Categorical + Categorical

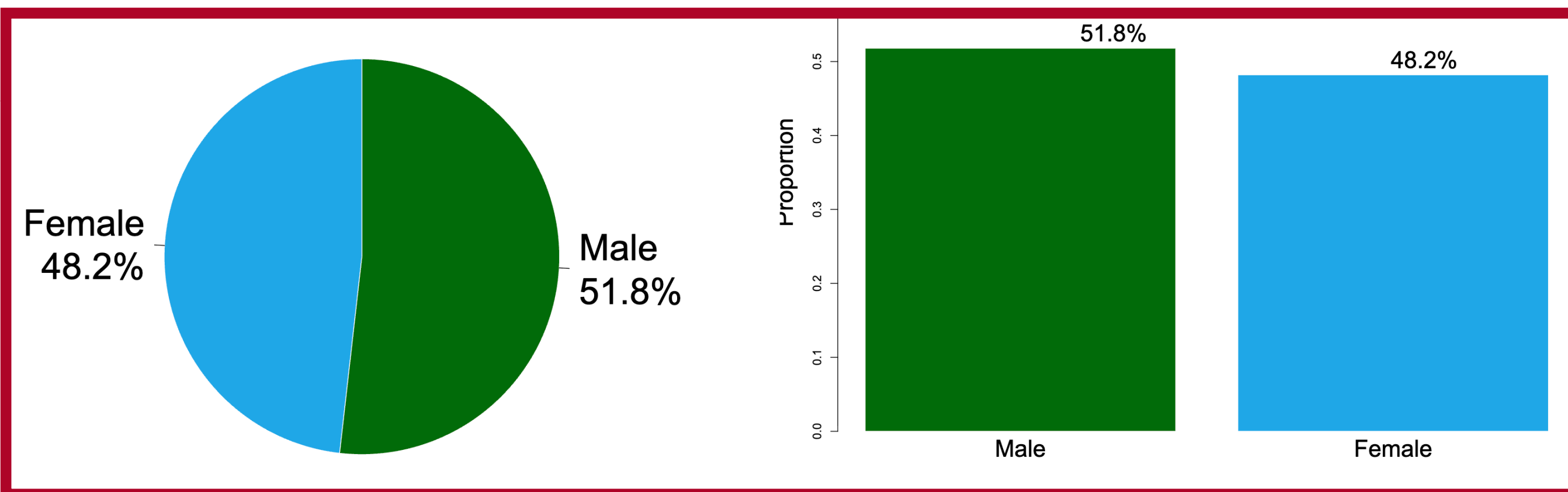
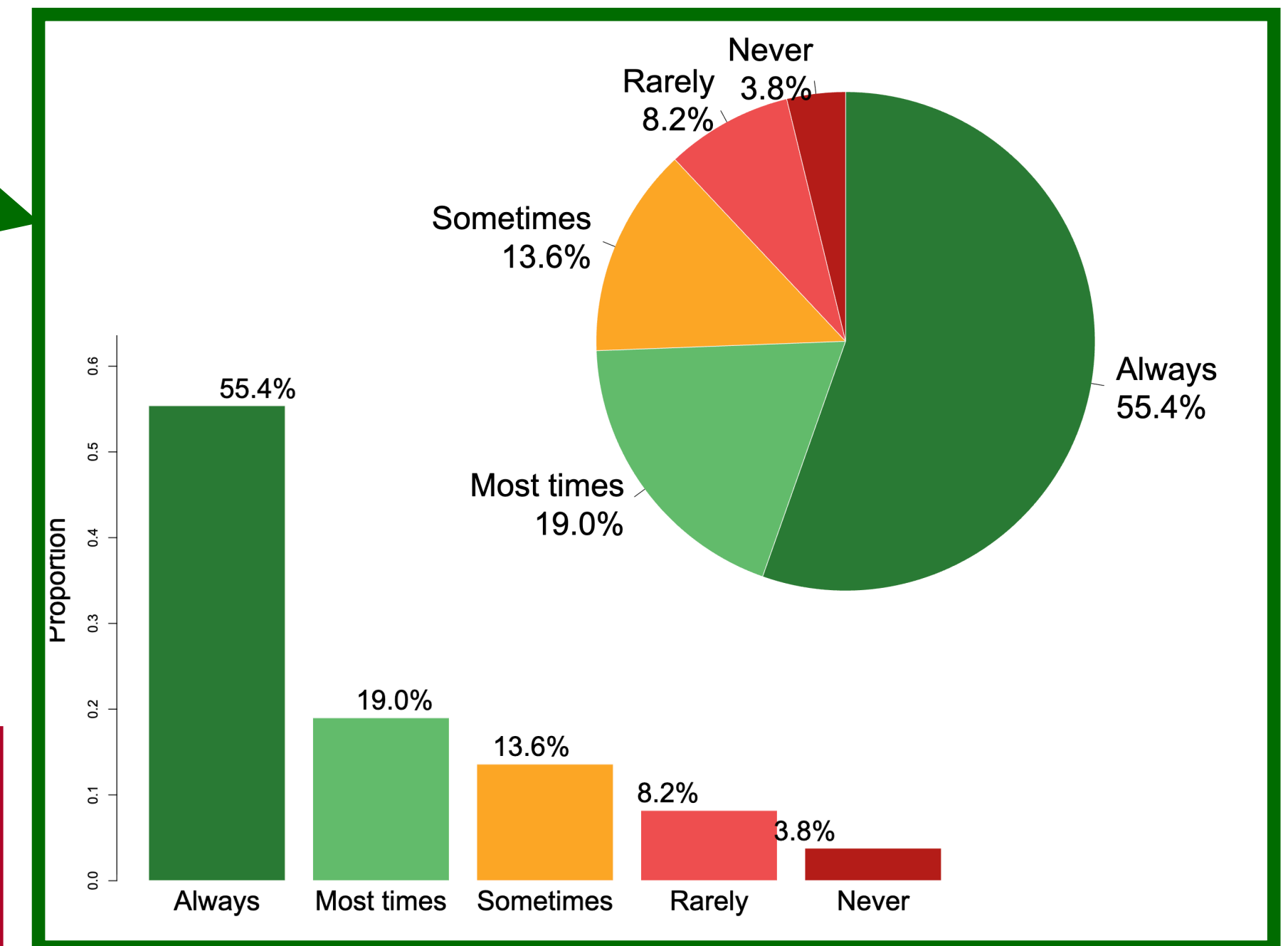
	Male	Female	Total
Always	771	915	1686 0.554
Most times	302	276	578 0.190
Sometimes	247	167	414 0.136
Rarely	165	84	249 0.082
Never	90	25	115 0.038
Total	1575 0.518	1467 0.482	3042

The **marginal distributions** (in the “margins” of the table) summarize each factor independently, with proportions or percents

Two-Way Tables

	Male	Female	Total
Always	771	915	1686 0.554
Most times	302	276	578 0.190
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The **marginal distributions** summarize each factor independently

Notice that these proportions sum to 1, and the counts sum to 3042

Notice that these proportions sum to 1, and the counts sum to 3042

Two-Way Tables

Categorical + Categorical

	Male	Female	Total
Always	771 0.253	915 0.301	1686
Most times	302 0.099	276 0.091	578
Sometimes	247 0.081	167 0.055	414
Rarely	165 0.054	84 0.028	249
Never	90 0.030	25 0.008	115
Total	1575	1467	3042

The **joint distribution** (in the center of the table) shows how the two factors vary together, with proportions or percents

Two-Way Tables

Categorical + Categorical

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Always	771 0.253	915 0.301	1686
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The **joint distribution** (in the center of the table) shows how the two factors vary together, with proportions or percents

The proportion of people surveyed who are female **and** always wear a seatbelt is 0.301



Two-Way Tables

Categorical + Categorical

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Notice that these proportions sum to 1, and the counts sum to 3042

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Most times	302 0.192	276 0.188	578 0.190
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Total	1575 1	1467 1	3042 1

The **conditional distributions**, given by row or column percentages/proportions, give the distribution of one variable conditional on one category of the other variable

Conditional distribution of seatbelt usage, given that the individual is **female**.

Conditional distribution of seatbelt usage, given that the individual is **male**.

Two-Way Tables

Categorical + Categorical

	Male	Female	Total
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The **conditional distributions**, given by row or column percentages/proportions, give the distribution of one variable conditional on one category of the other variable

You can think of conditioning as temporarily pretending the rest of the data doesn't exist

i.e., if we want the conditional distribution of seatbelt usage for females, we can

- pretend that instead of surveying 12th graders we surveyed only 12th grade females
- drop the non-female data and compute our proportions

Two-Way Tables

Categorical + Categorical

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The **conditional distributions**, given by row or column percentages/proportions, give the distribution of one variable conditional on one category of the other variable

62.4% of the females surveyed said they always wear a seatbelt.

49% of the males surveyed said they always wear a seatbelt.

Two-Way Tables

Categorical + Categorical

	Male	Female	Total
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PollEv.com/stats8: Which is the best interpretation of the value 0.114?

Two-Way Tables

Categorical + Categorical

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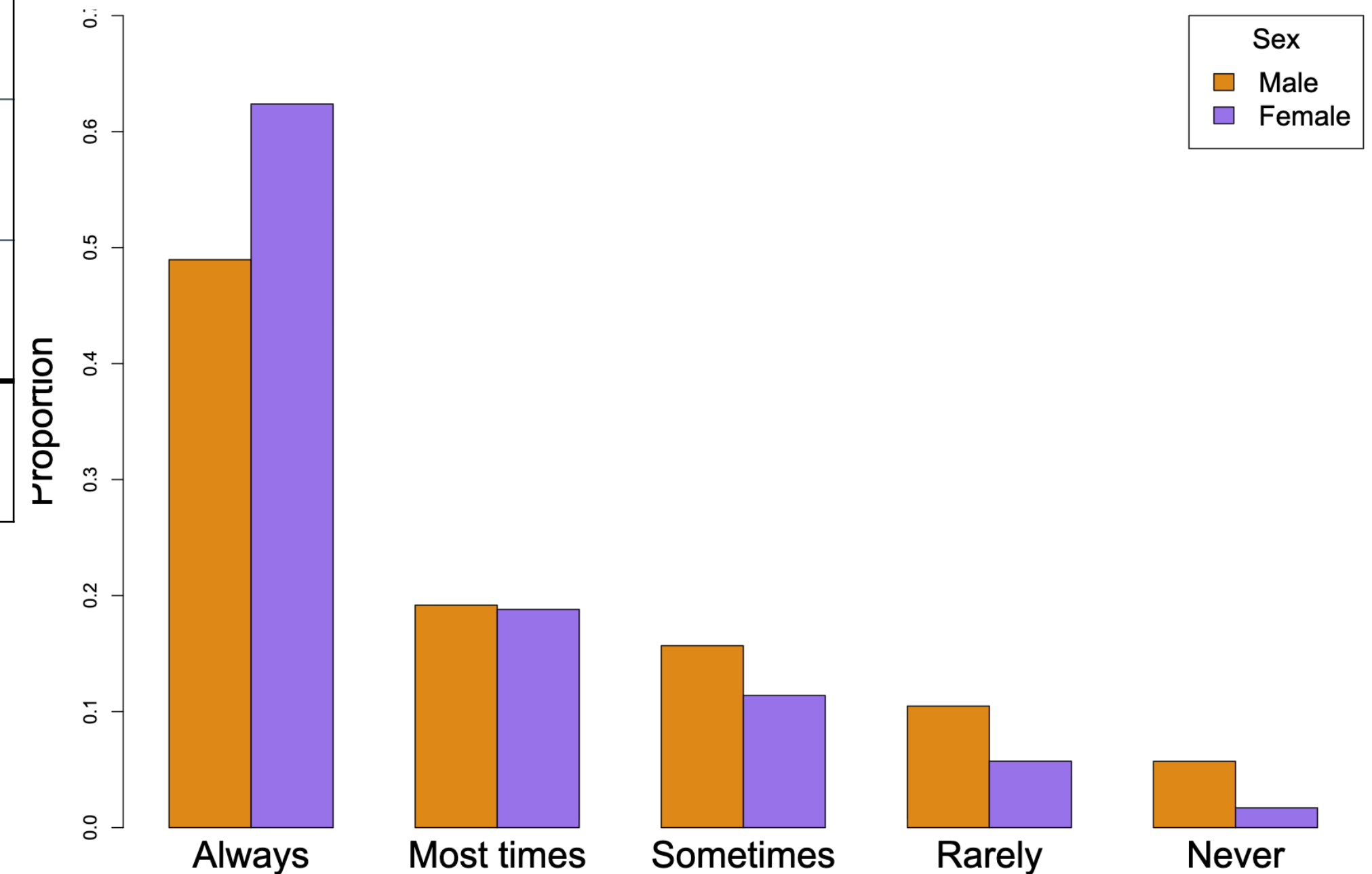
The proportion of female 12th graders surveyed who said they sometimes wear a seatbelt

Two-Way Tables

Categorical + Categorical

	Male	Female	Total
Always	771 0.490	915 0.624	1686 0.554
Most times	302 0.192	276 0.188	578 0.190
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The **conditional distributions**, given by row or column percentages/proportions, give the distribution of one variable conditional on one category of the other variable



Two-Way Tables

Categorical + Categorical

	Male	Female	Total
Always	771 0.457	915 0.543	1686 1
Most times	302 0.522	276 0.478	578 1
Sometimes	247 0.597	167 0.403	414 1
Rarely	165 0.663	84 0.337	249 1
Never	90 0.783	25 0.217	115 1
Total	1575 0.518	1467 0.482	3042 1

The **conditional distributions**, given by row or column percentages/proportions, give the distribution of one variable, conditional on one category of the other variable

The conditional distribution of sex given that the individual said they **always** wear a seatbelt

The conditional distribution of sex given that the individual said they wear a seatbelt **most times**

The conditional distribution of sex given that the individual said they **sometimes** wear a seatbelt



Two-Way Tables

Categorical + Categorical

	Male	Female	Total
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The **conditional distributions**, given by row or column percentages/proportions, give the distribution of one variable, conditional on one category of the other variable

52.2% of people surveyed who said they wear a seatbelt **most times** were male.

59.7% of people surveyed who said they **sometimes** wear a seatbelt were male.

Is sex associated with seatbelt usage?

- Let's start thinking about how we might use data to provide evidence for or against certain claims (note that this is essentially the whole point of this class)
- Claim:** 12th grade females are more likely to always wear a seatbelt than 12th grade males.
 - Which numbers might you pull from the table to provide support for or against this claim?

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Is sex associated with seatbelt usage?

- **Claim:** 12th grade females are more likely to always wear a seatbelt than 12th grade males.
 - Which numbers might you pull from the table to provide support for or against this claim?
 - Is it tempting to say the data support this claim because $0.543 > 0.457$?

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Is sex associated with seatbelt usage?

- **Claim:** 12th grade females are more likely to always wear a seatbelt than 12th grade males.
 - Which numbers might you pull from the table to provide support for or against this claim?
 - Think about which groups we are comparing... (remember confusion of the inverse?)

	Male	Female	Total
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Two-Way/Contingency Table

A survey of 479 children found that those who had slept with a nightlight or in a fully lit room before the age of 2 had a higher incidence of nearsightedness (myopia) later in childhood (Quinn et al. 1999)

	No Myopia	Myopia	Total
Darkness	155	17	172
Night light	153	79	232
Full light	34	41	75
Total	342	137	479

Practice

[PollEv.com/stats8](https://pollEv.com/stats8)

- How many variables are shown in this table?
 - A. 1
 - B. 2
 - C. 5
 - D. 7

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Practice

[PollEv.com/stats8](https://pollEv.com/stats8)

- How would you compute the marginal distribution of sleeptime lighting?
 - A. By taking the bottom row and dividing by 479
 - B. By taking the rightmost column and dividing by 479
 - C. By taking the entire table and dividing by 479

Two-Way/Contingency Table

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Practice

[PollEv.com/stats8](https://www.pollEv.com/stats8)

- What proportion of children in this survey slept with a night light before the age of 2?
 - A. $232 / 479$
 - B. $79 / 479$
 - C. $79 / 137$
 - D. $153 / 479$
 - E. $153 / 342$
 - F. None of the above

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- What proportion of children in this survey slept with a night light before the age of 2?

A. $232 / 479$

B. $79 / 479$

C. $79 / 137$

D. $153 / 479$

E. $153 / 342$

F. None of the above

Two-Way/Contingency Table

Practice

[PollEv.com/stats8](https://www.pollEv.com/stats8)

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- What proportion of children who slept in darkness before the age of 2 have myopia?
 - A. $137 / 479$
 - B. $17 / 479$
 - C. $17 / 137$
 - D. $17 / 172$
 - E. None of the above

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Practice

[PollEv.com/stats8](https://poll-ev.com/stats8)

- The proportion of children who slept in darkness, that now have myopia (the proportion from the previous slide) is a proportion from what kind of distribution?
 - Joint
 - Marginal
 - Conditional
 - Total
 - Other

Two-Way/Contingency Table

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 - E. Other

Two-Way/Contingency Table

Practice

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- Someone wants to use this study to claim that those who slept with a night light are more likely to have myopia in later childhood than those who slept in darkness.
- How would you provide evidence for or against this claim?

Two-Way/Contingency Table

Practice

A survey of 479 children found that those who had slept with a nightlight or in a fully lit room before the age of 2 had a higher incidence of nearsightedness (myopia) later in childhood (Quinn et al. 1999)

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- How would you provide evidence for or against this claim?

I would compare the proportion of myopia in the night light group to the proportion of myopia in the darkness group to see which is higher (compare conditional distributions)

Two-Way/Contingency Table

Practice

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- How would you provide evidence for or against this claim?

I would redo the study as an experiment rather than as an observational study :)

Two-Way/Contingency Table

Practice

[PollEv.com/stats8](https://poll-ev.com/stats8)

Main news source

age group

	TV	Internet	Newspapers	Other	Total
18-39 years old	71		5	51	434
40-64 years old		207	22	55	
65+ years old	123		33	14	202
Total	401	546		120	1127

- What value should go in the red cell?

2018 general social survey ([source](#))

Two-Way/Contingency Table

Practice

[PollEv.com/stats8](https://poll-ev.com/stats8)

Main news source

age group

	TV	Internet	Newspapers	Other	Total
18-39 years old	71		5	51	434
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- What proportion of people surveyed get their news from the internet?
- What kind of distribution is this proportion from?

2018 general social survey ([source](#))

Two-Way/Contingency Table

Practice

Main news source						
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	40-64 years old	207	207	22	55	
	65+ years old	123		33	14	202
	Total	401	546	60	120	1127

- What proportion of people surveyed get their news from the internet?
- $546 / 1127$
- What kind of distribution is this proportion from?
- **Marginal**

2018 general social survey ([source](#))

Two-Way/Contingency Table

Practice

Main news source						
age group		TV	Internet	Newspapers	Other	Total
	18-39 years old	71		5	51	434
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	Total	401	546	60	120	1127

- What proportion of 18-39 year olds get their news from TV?
- What kind of distribution is this proportion from?

2018 general social survey ([source](#))

Two-Way/Contingency Table

Practice

age group	Main news source					
	TV	Internet	Newspapers	Other	Total	
	18-39 years old	71		5	51	434
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	65+ years old	123		33	14	202
	Total	401	546	60	120	1127

- What proportion of 18-39 year olds get their news from TV?
- 71 / 434
- What kind of distribution is this proportion from?
- Conditional

2018 general social survey ([source](#))

Two-Way/Contingency Table

Practice

Main news source						
age group		TV	Internet	Newspapers	Other	Total
	18-39 years old	71		5	51	434
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- What proportion of 65+ year olds get their news from the internet?
- What kind of distribution is this proportion from?

2018 general social survey ([source](#))

Two-Way/Contingency Table

Practice

age group	Main news source					
	TV	Internet	Newspapers	Other	Total	
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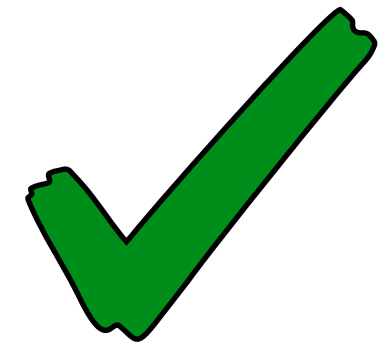
- What proportion of 65+ year olds get their news from the internet?
- 32 / 202
- What kind of distribution is this proportion from?
- Conditional

2018 general social survey ([source](#))

Describing Associations

Variable types

- Between **one categorical and one quantitative variable**
 - Graphical: side-by-side boxplots or histograms
 - Numeric: comparing means and medians, spread
- Between **two categorical variables**
 - Graphical: bar graphs
 - Numeric: contingency/two-way tables, conditional proportions
- Between **two quantitative variables**
 - Graphical: scatterplots
 - Numeric: correlation, regression



Describing Associations

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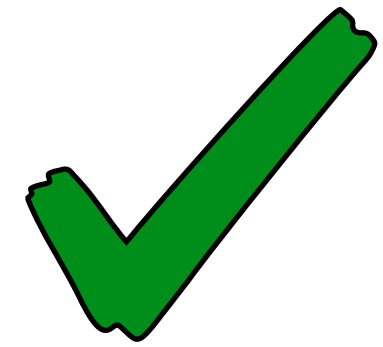
(Just briefly)

- Between **two categorical variables**

- Graphical: bar graphs
- Numeric: contingency/two-way tables, conditional proportions

- Between **two quantitative variables**

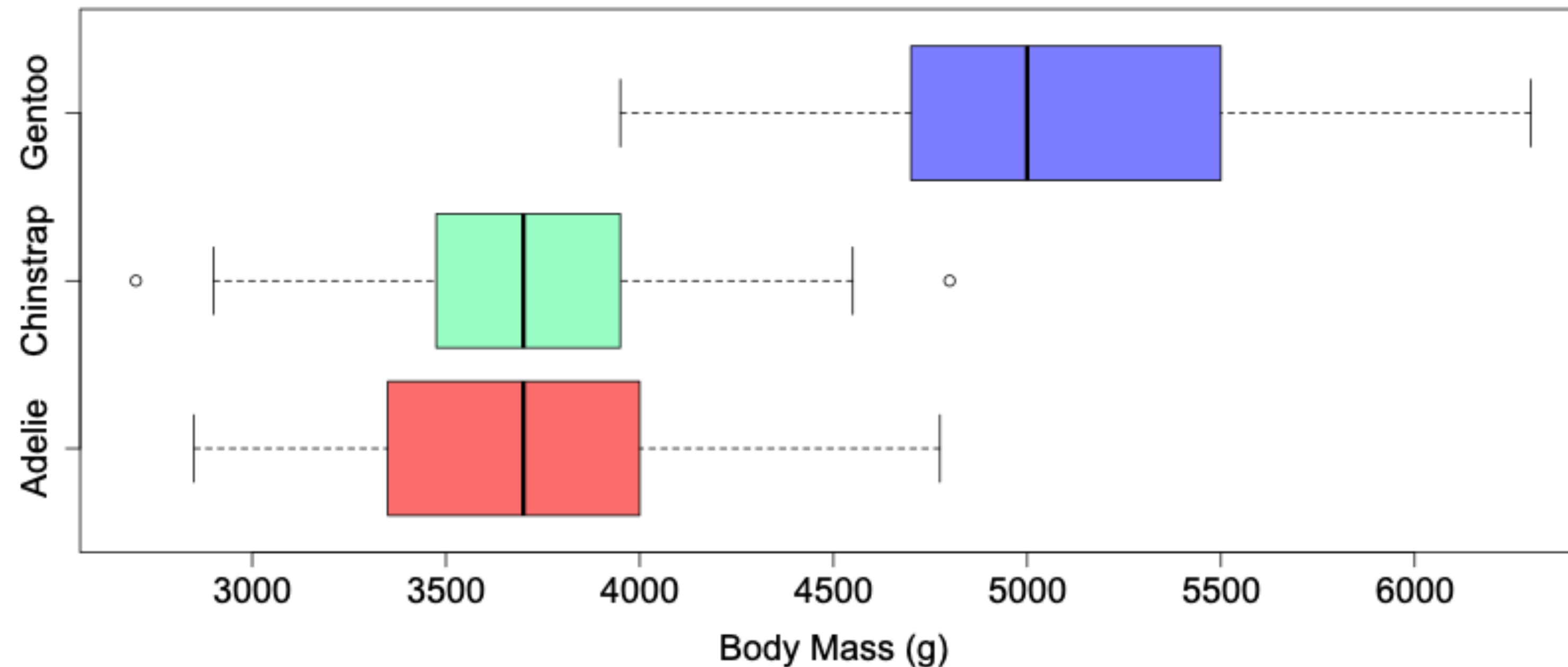
- Graphical: scatterplots
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Graphical Comparisons

Categorical + Quantitative

- How do distributions of quantitative variables differ between different categories of a categorical variable?
- Comparative boxplots of the quantitative variable split by categories of the categorical variable can help:



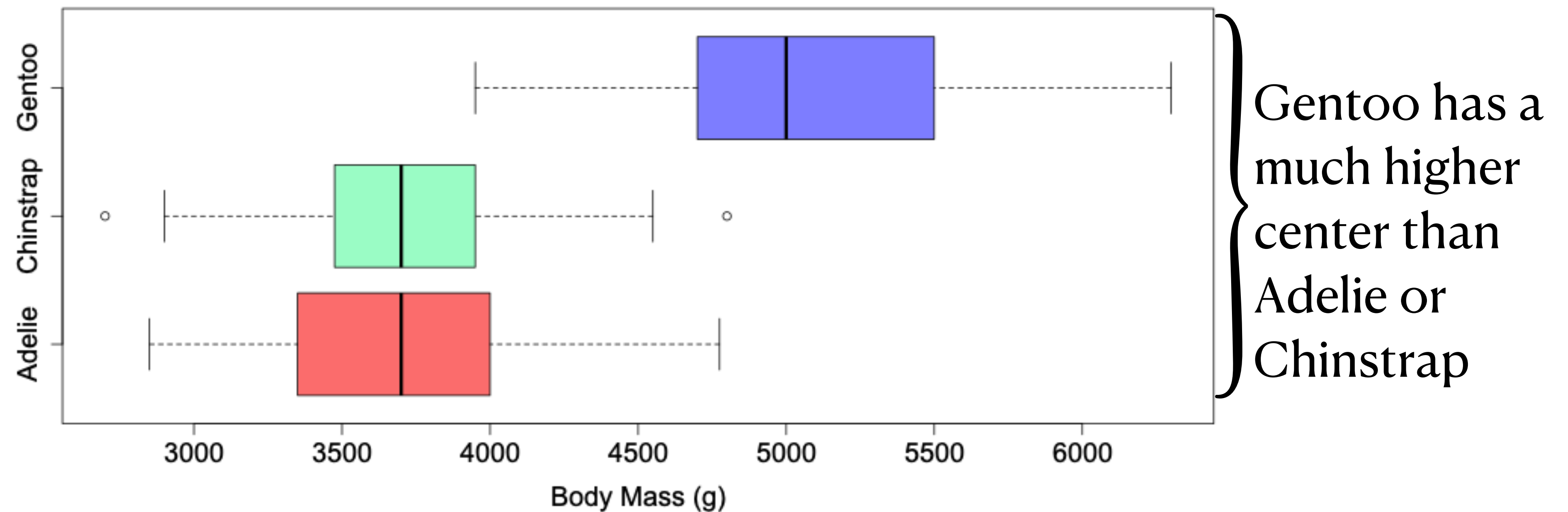
- Above: body mass of penguins split by species



Graphical Comparisons

Categorical + Quantitative

- Important note: best to compare both center *and* spread
- Comparative boxplots of the quantitative variable split by categories of the categorical variable can help:

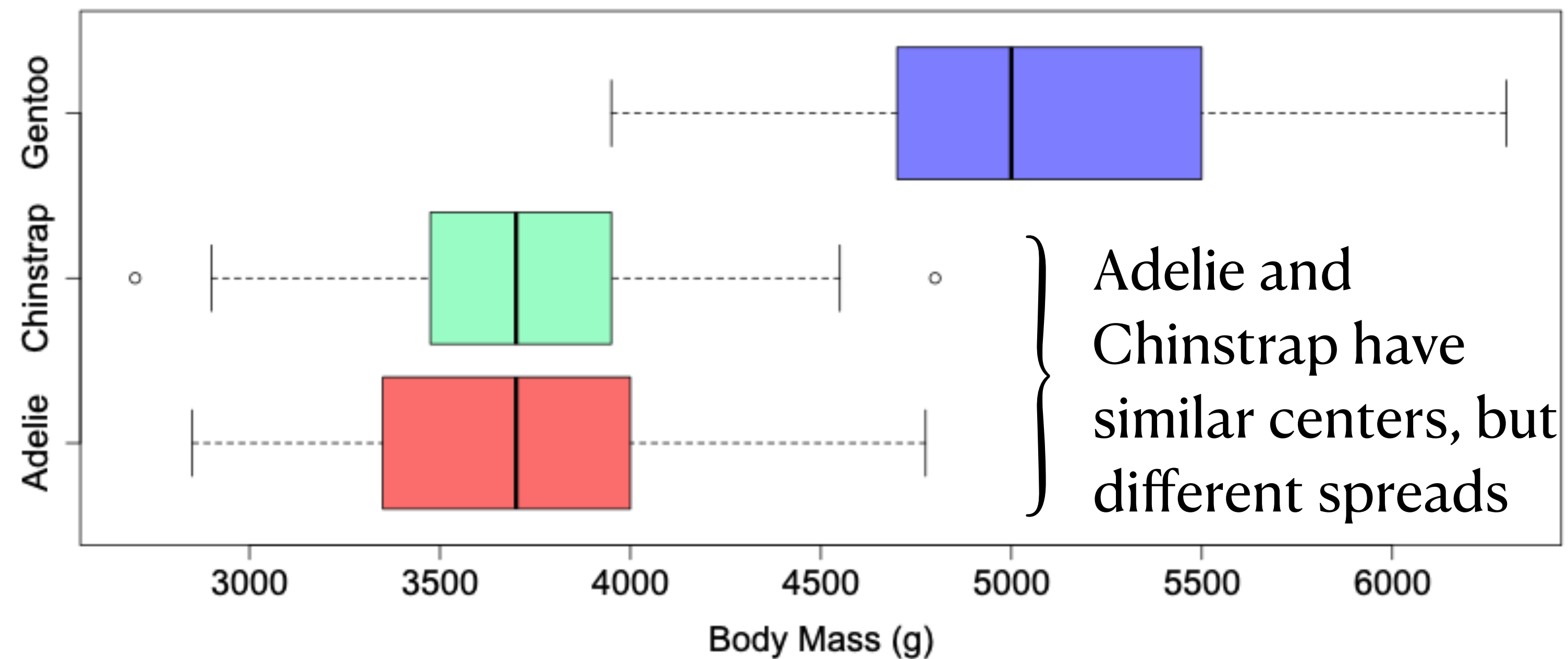


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Graphical Comparisons

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- Important note: best to compare both center *and* spread
- Comparative boxplots of the quantitative variable split by categories of the categorical variable can help:



- Above: body mass of penguins split by species

Graphical Comparisons

Categorical + Quantitative

- Important note: best to compare both center *and* spread
- Comparative histograms (plotted on the same axis) of the quantitative variable split by categories of the categorical variable can also help:



- Above: body mass of penguins split by species

Numeric Comparisons

Categorical + Quantitative

- Directly compare the values computed from the sample, often in a table:

Species	n (sample size)	Median	IQR	Range
Gentoo	123	5000.0	800.0	2350.0
Chinstrap	68	3700.0	462.5	2100.0
Adelie	151	3700.0	650.0	1925.0
Overall	342	4050.0	1200.0	3600.0

Numeric Comparisons

Categorical + Quantitative

- Directly compare the values computed from the sample, often in a table:

Species	n (sample size)	Mean	Variance (SD)
Gentoo	123	5076.0	254,133.2 (504.1)
Chinstrap	68	3733.1	147,713.5 (384.3)
Adelie	151	3700.6	210,282.9 (458.6)
Overall	342	4201.8	643,131.1 (802.0)

Mean is in grams, standard deviation is in grams, variance is in **grams²

Numeric Comparisons

Categorical + Quantitative

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Think intuitively about why the combined variance/SD is larger than the species-level ones...

Mean is in grams, standard deviation is in grams, variance is in **grams²

Group Comparisons

Categorical + Categorical and Categorical + Quantitative

- **Big idea:** when looking at associations, if one variable is categorical we make comparisons between categories of that variable
 - If the other variable is also **categorical**, this is usually with **proportions**
 - e.g. The proportion of males who always wear a seatbelt vs. proportion of females who always wear a seatbelt
 - If the other variable is **quantitative**, this can be many things but is most often comparing **means** or **averages**
 - e.g. The mean body mass of adelic vs. gentoo vs. chinstrap penguins
- **Are the proportions/means different between different groups? How different are they?**

Describing Associations

Variable types

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- Graphical: bar graphs
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- Between **two quantitative variables**

- Graphical: scatterplots
- Numeric: correlation, regression

Describing Associations

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- Graphical: bar graphs
- Numeric: contingency/two-way tables, conditional proportions

- Between **two quantitative variables**

- Graphical: scatterplots
- Numeric: correlation, regression



The biggest part of
Topic 3

Group Comparisons

Quantitative + Quantitative

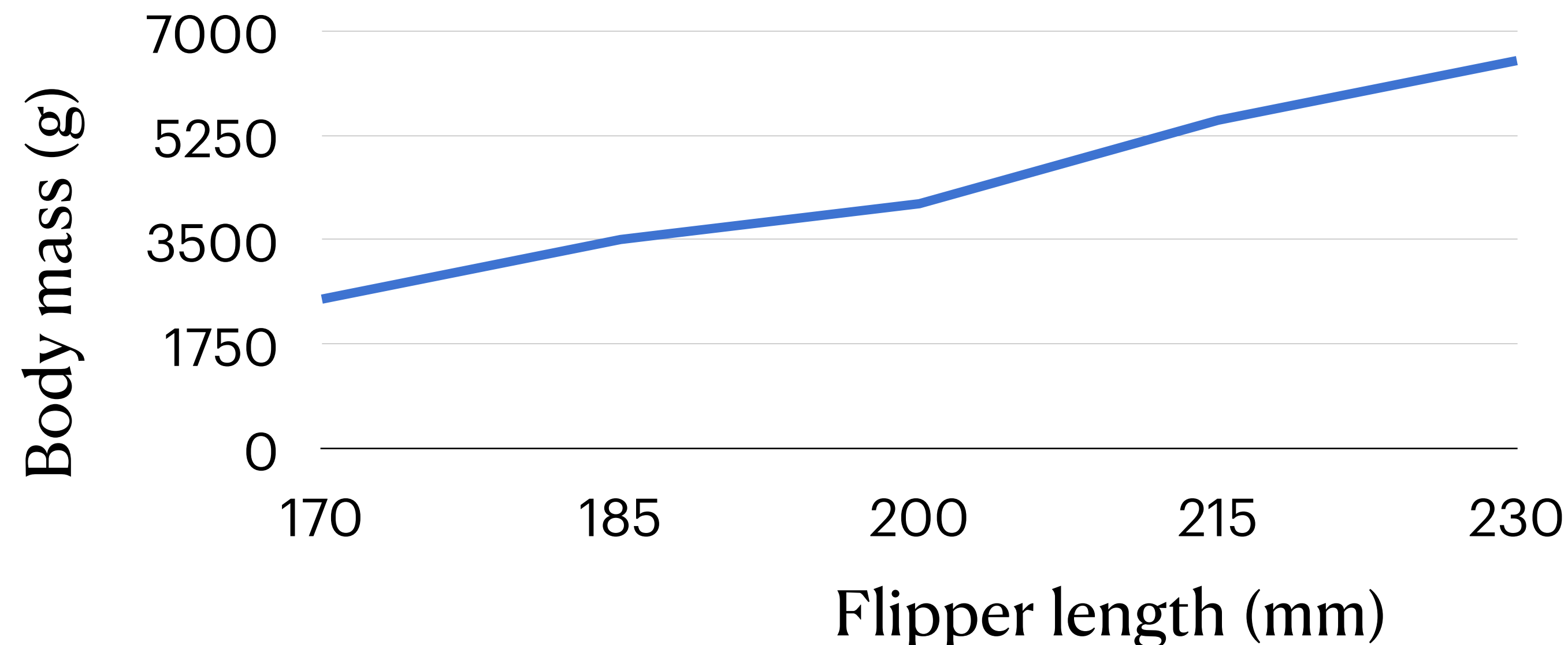
- A few slides ago, we said...
 - **Big idea:** when looking at associations, if one variable is categorical we make comparisons (of means or proportions) between categories of that variable
 - **Are the proportions/means different between different groups? How different are they?**
- Now, what if *both* variables are quantitative?
 - Don't have natural groups to make comparisons between...
 - Need some new tools

Explanatory vs. Response Variables

Quantitative + Quantitative

- We want to model one quantitative variable as a function of the other one
- e.g., suppose we want to model penguins' body masses (grams) as a function of their flipper length (millimeters)

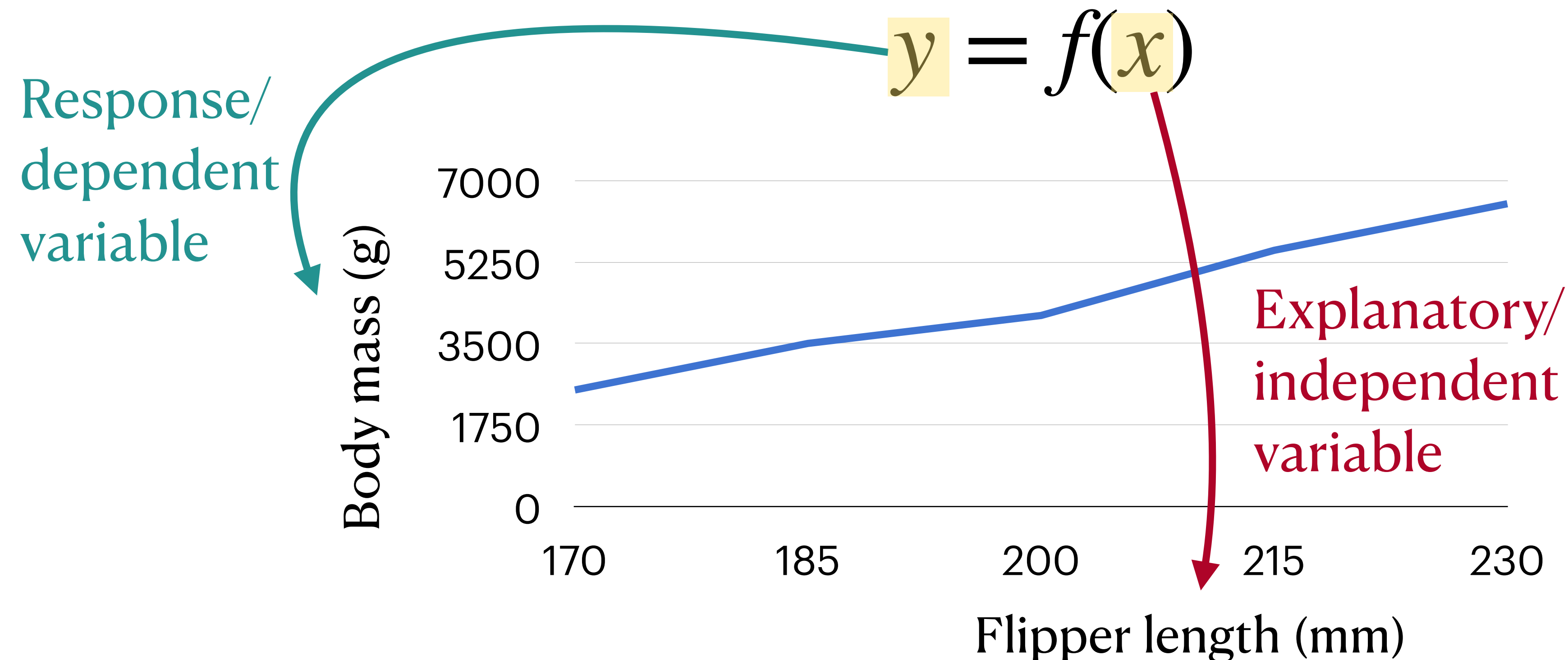
$$y = f(x)$$



Explanatory vs. Response Variables

Quantitative + Quantitative

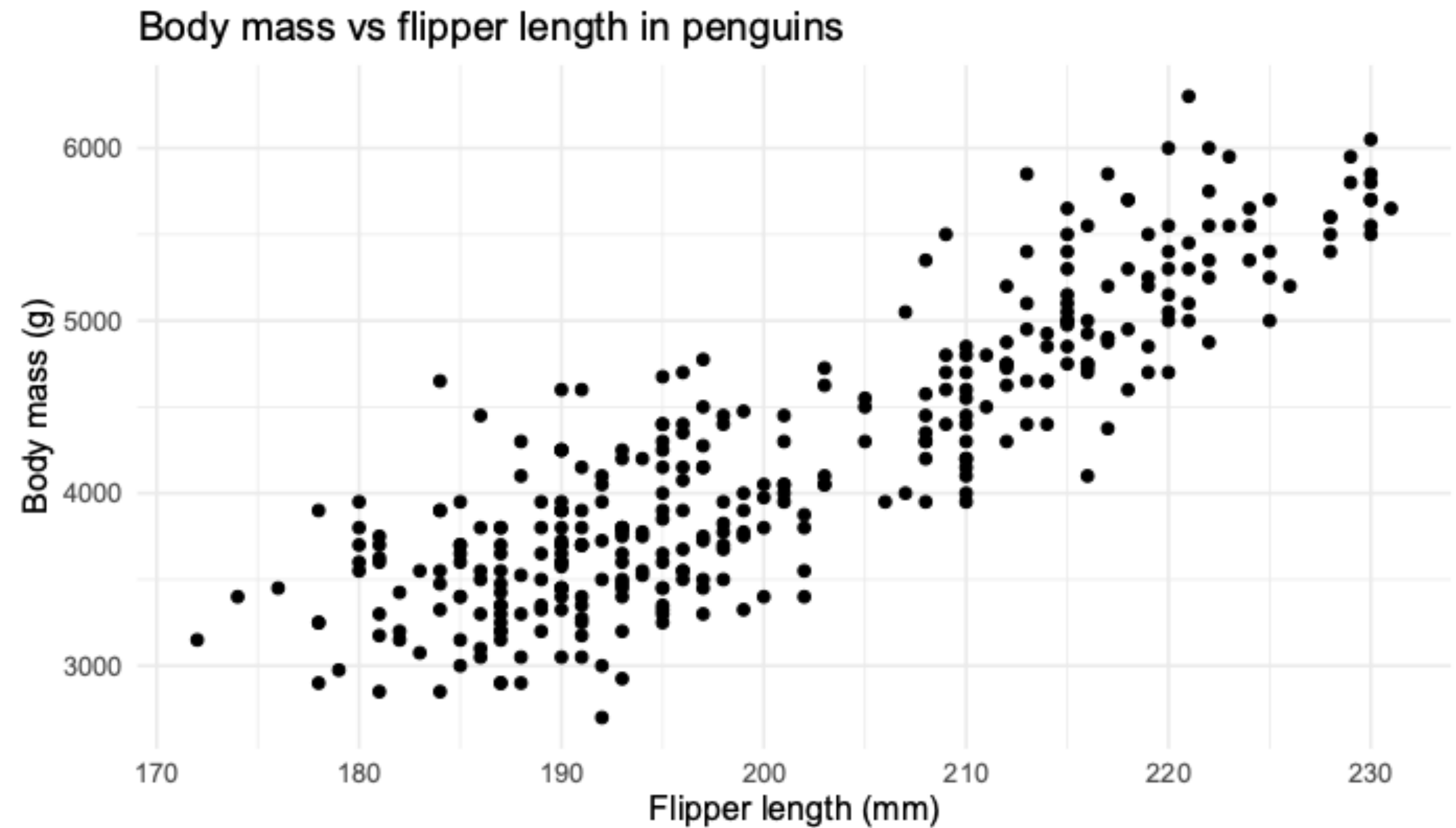
- We want to model one quantitative variable as a function of the other one
- e.g., suppose we want to model penguins' body masses (grams) as a function of their flipper length (millimeters)



Scatterplots

Quantitative + Quantitative

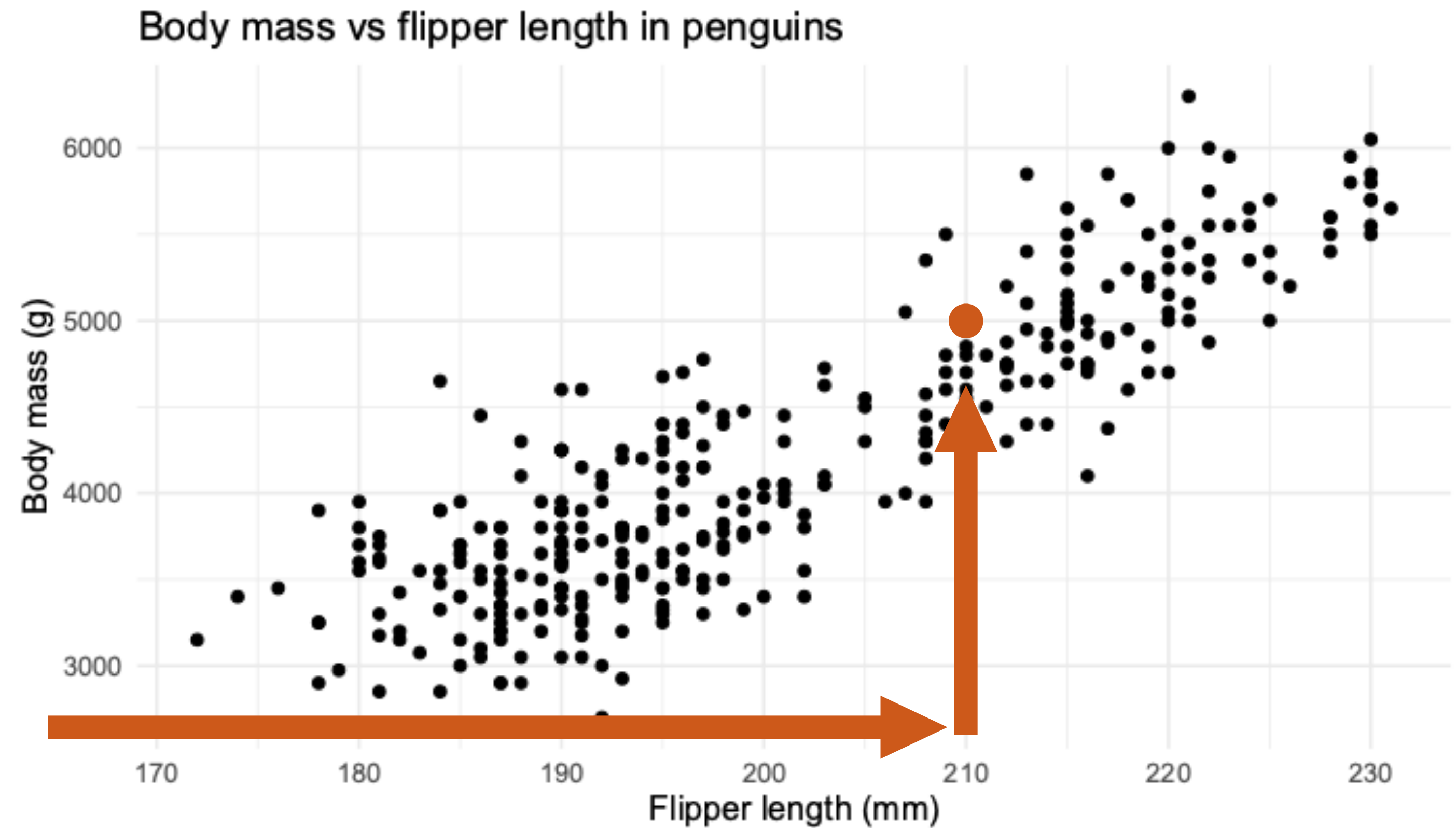
- Each **observational unit** represents a **dot**
- Each observational unit has (at least) two measured quantitative variables
 - One is on the horizontal axis (explanatory)
 - One is on the vertical axis (response)



Scatterplots

Quantitative + Quantitative

- Each **observational unit** represents a **dot**
- Each observational unit has (at least) two measured quantitative variables
 - One is on the horizontal axis (explanatory)
 - One is on the vertical axis (response)
- e.g., say we measure a penguin with flipper length 210 mm and body mass 5000 g



Describing Associations

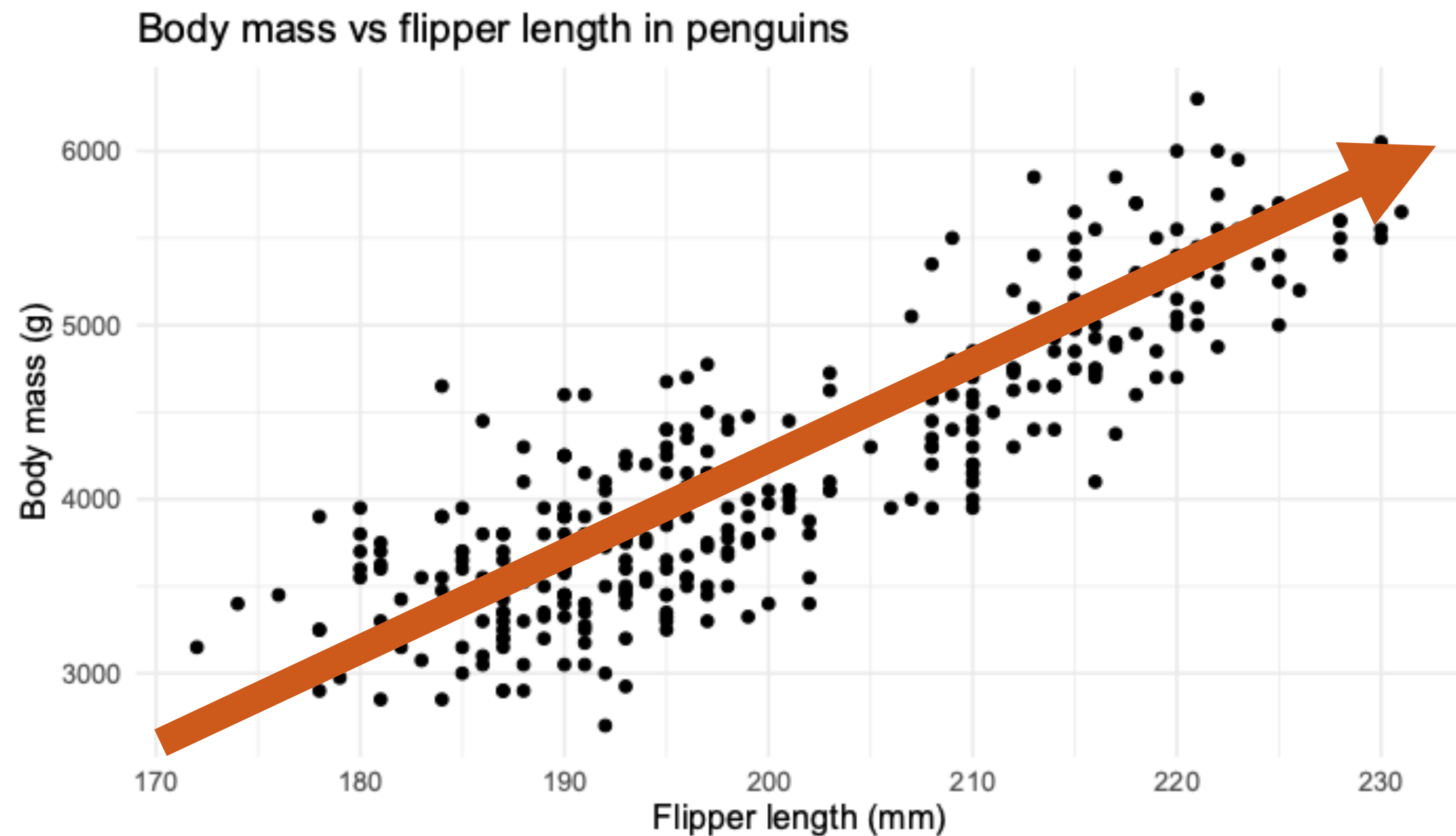
Quantitative + Quantitative

- Recall that in Topic 2 we talked about the **distribution** of a quantitative variable in terms of **center**, **spread**, and **shape** (and discussed **outliers**)
- We will do the same thing when describing associations between two quantitative variables:
 - Form: what is the “shape” of the relationship?
 - Strength: how closely do the data follow the shape?
 - Direction: what direction does the relationship go?
 - Outliers: are there any points that don't fit the main pattern?

Describing Associations

Form

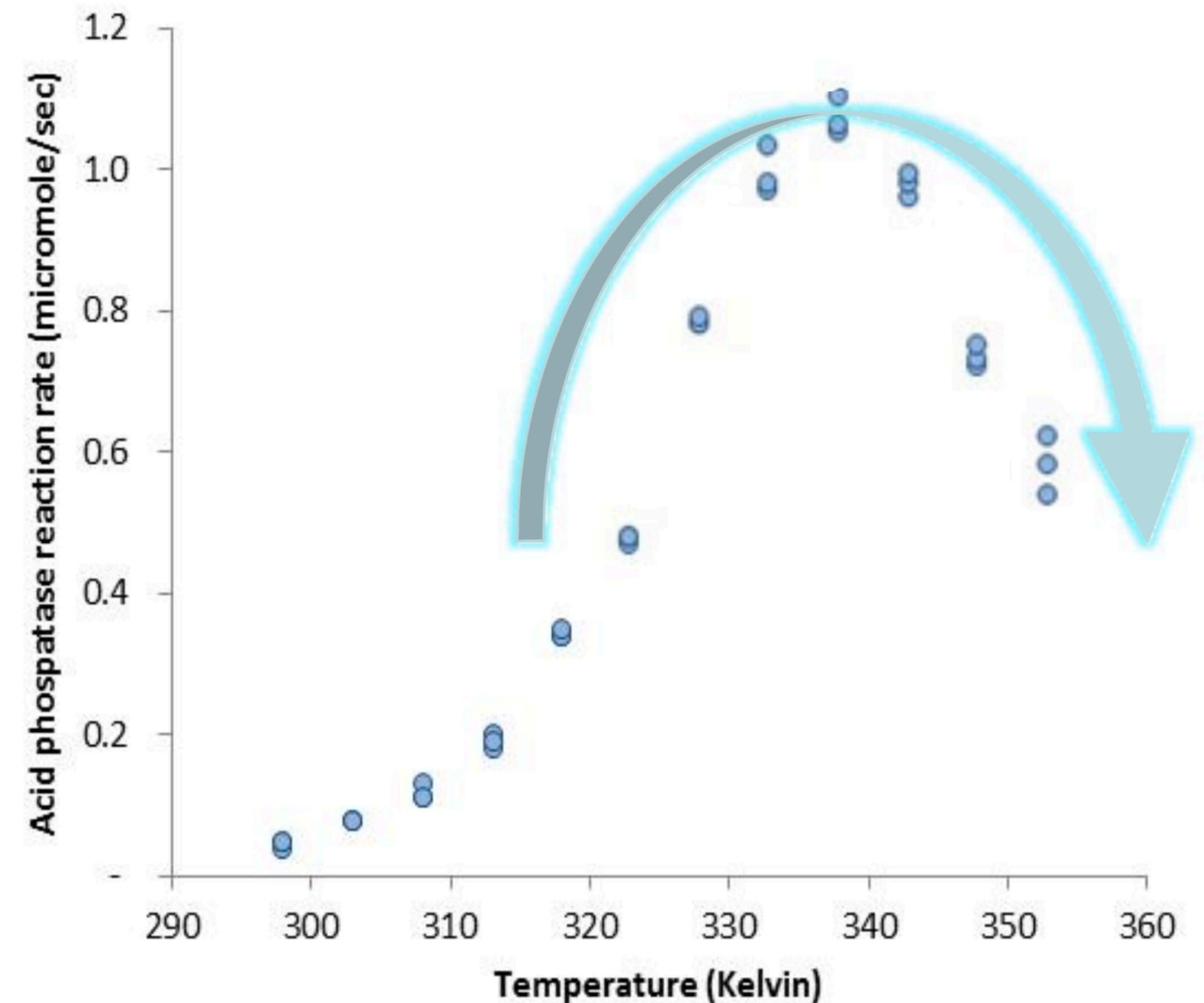
- Relationships can take many different shapes
- Penguins example looks approximately **linear**, meaning that on average the points follow a line
- Line + what some statisticians call “noise”: variability that comes from elsewhere



Describing Associations

Form

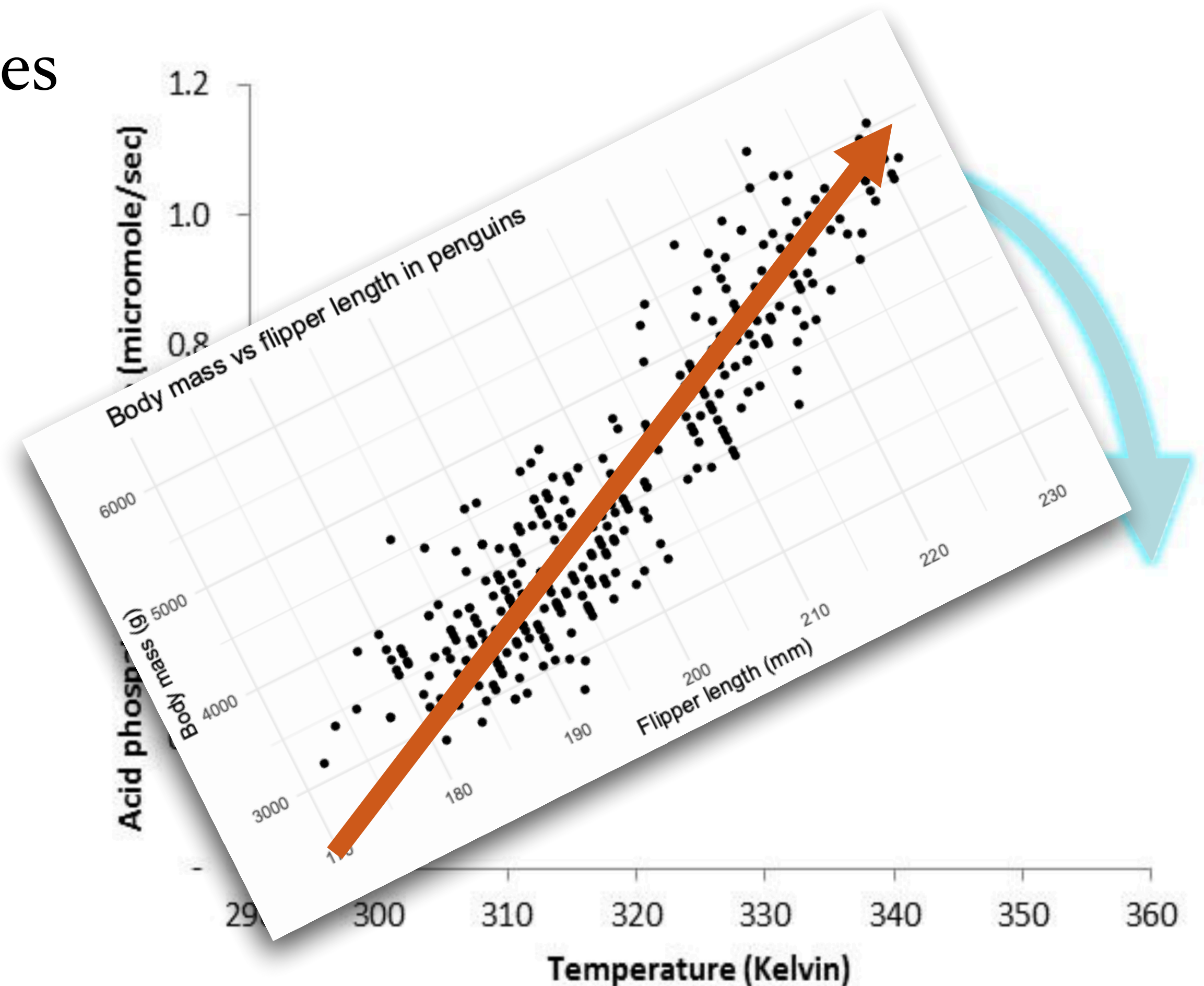
- Relationships can take many different shapes
- This temperature vs. acid phosphatase reaction rate is very nonlinear →



Describing Associations

Form

- Relationships can take many different shapes
- This temperature vs. acid phosphatase reaction rate is very nonlinear →
- In this class we will focus on linear relationships (this is common in science as well)



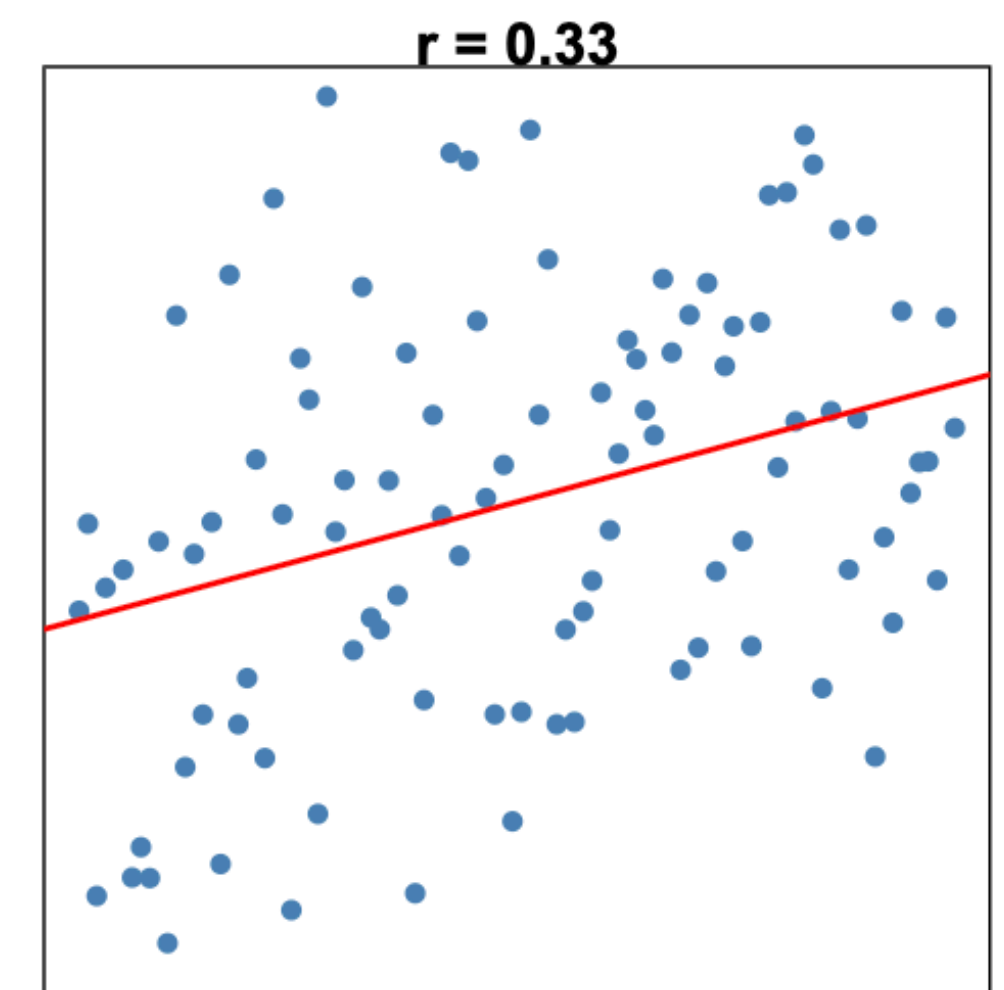
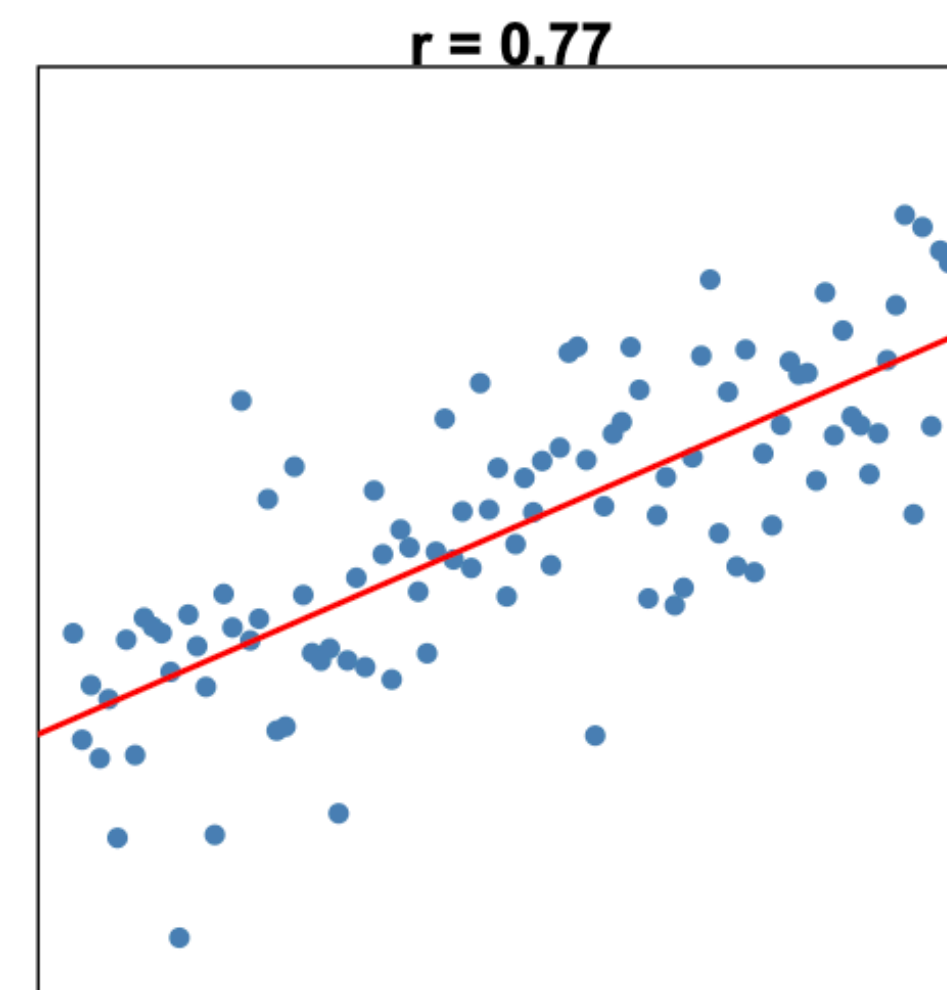
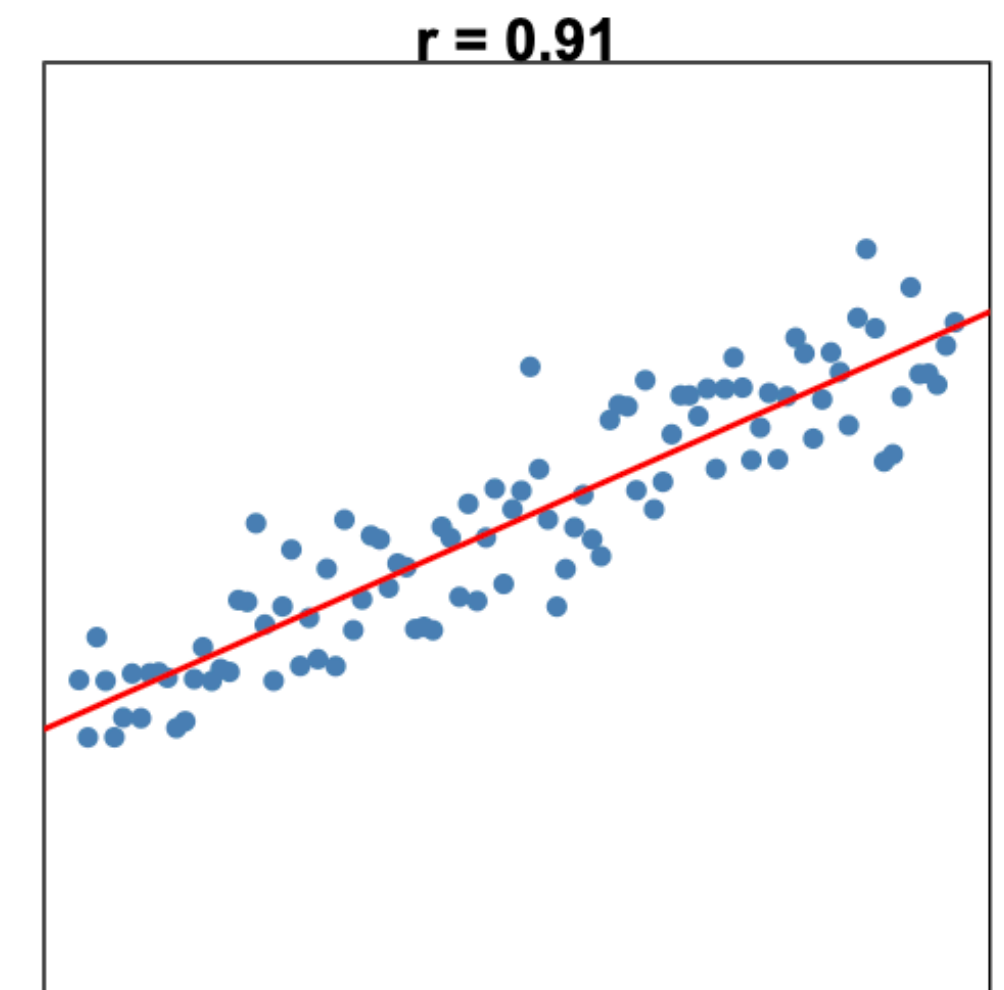
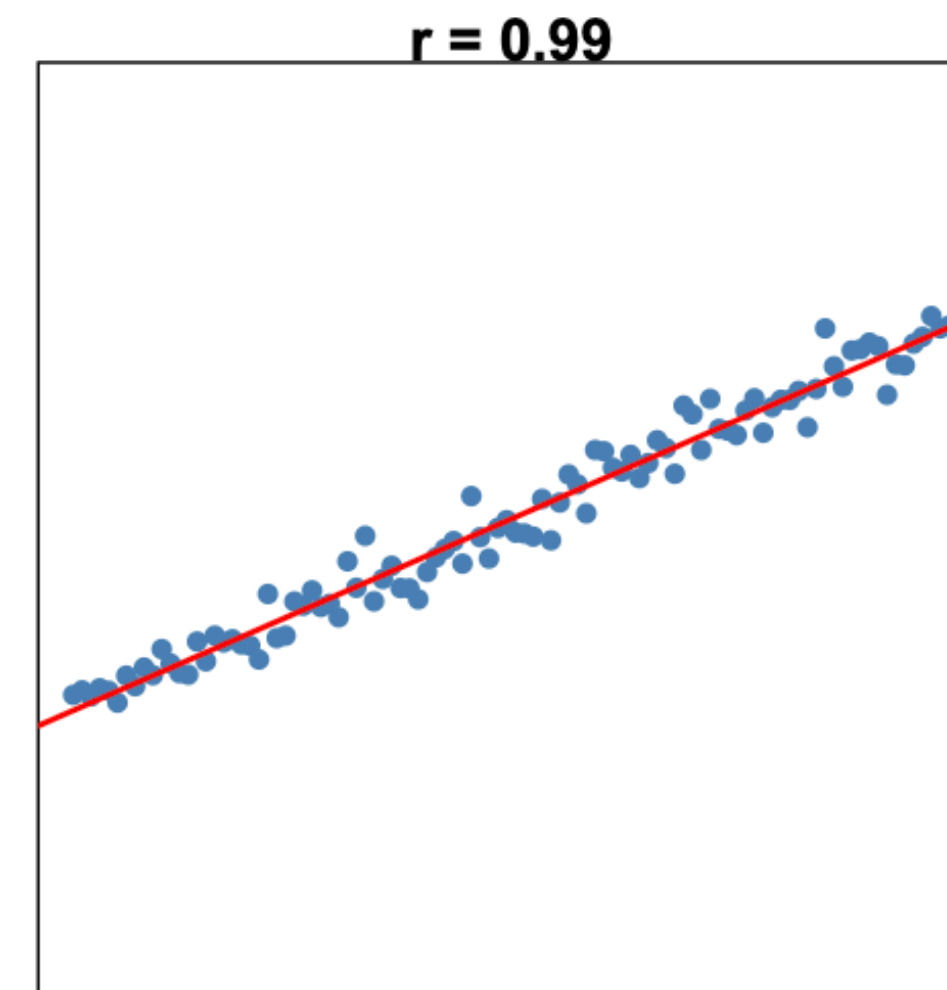
Describing Associations

Strength

- The strength of the relationship refers to how closely the data follows the relationship (less noise/scatter vs. more noise/scatter)
- In a linear relationship, we use the **correlation coefficient (r)** to express the strength of the relationship, where **r** is always between -1 and 1

$$r = \frac{1}{n - 1} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right)$$

(I could not care less if you know how to compute this but you should know what it means)



Describing Associations

Strength

- In a linear relationship, we use the **correlation coefficient (r)** to express the strength of the relationship

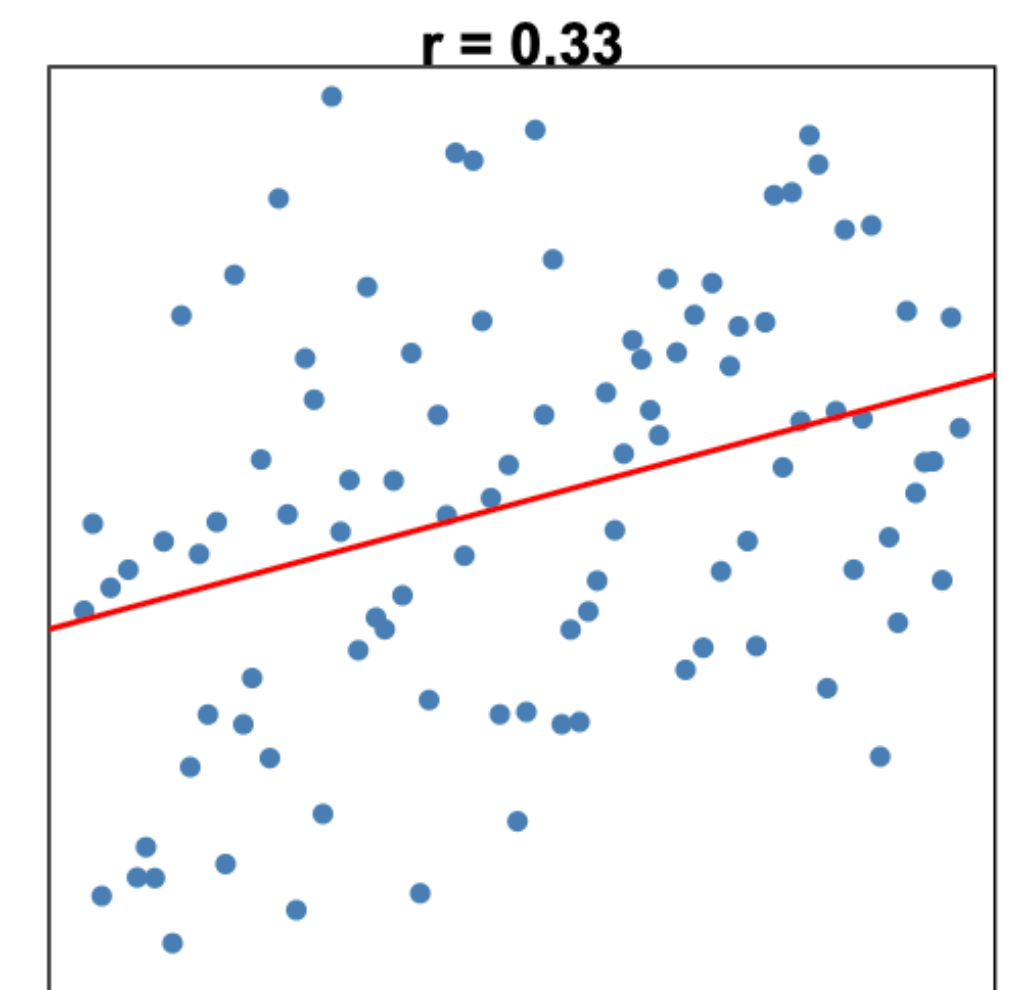
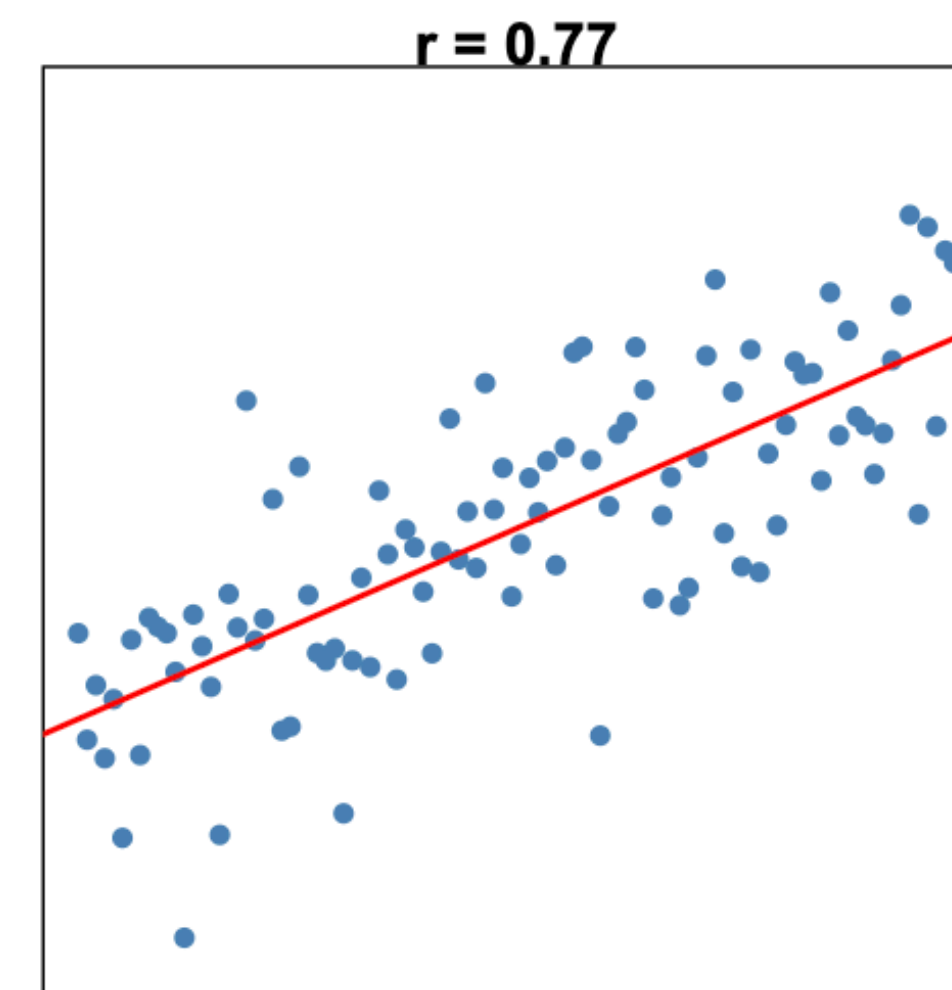
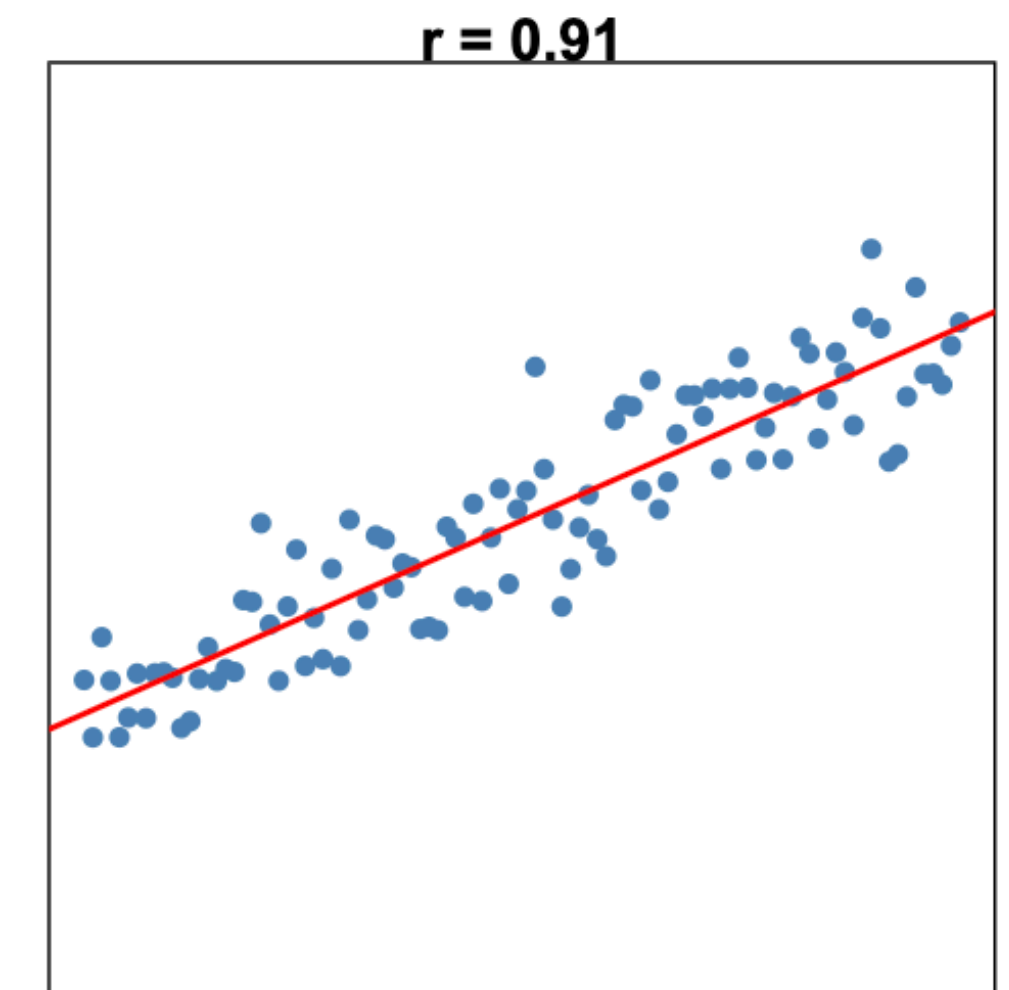
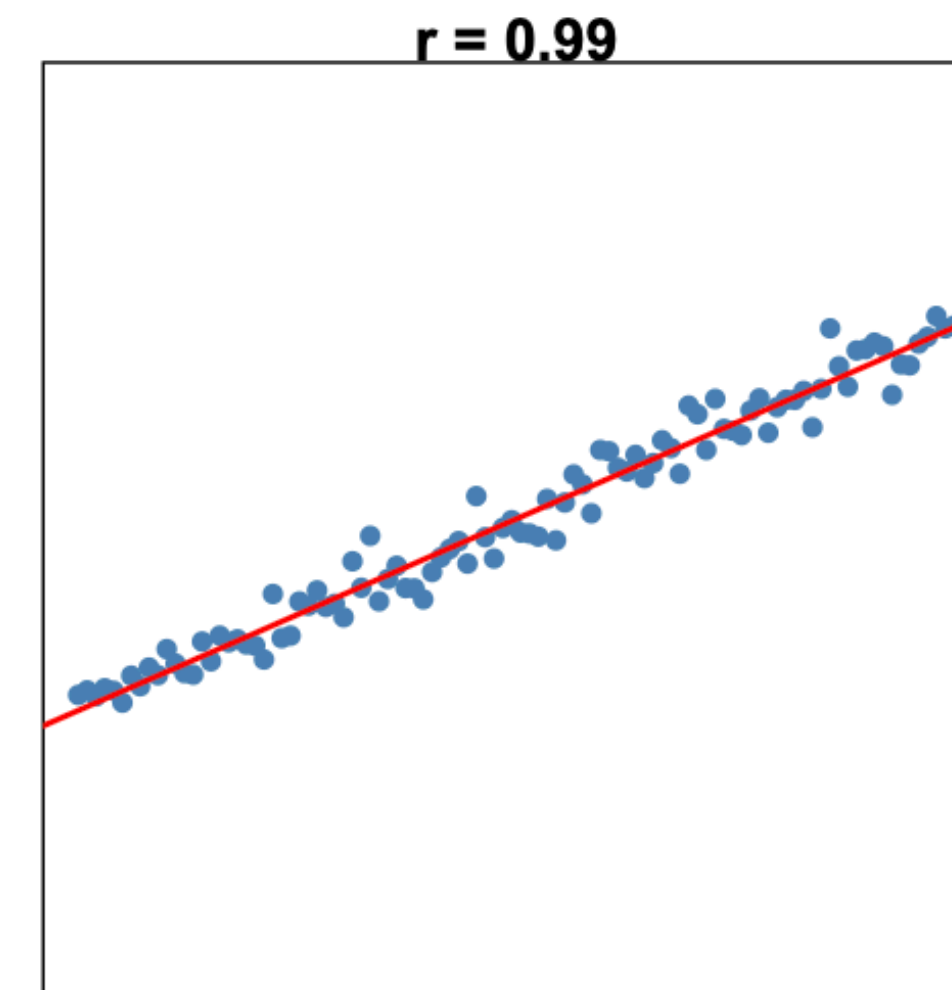
$$r = \frac{1}{n-1} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right)$$

number of observational units (points to n)

sum over all observational units (indexed by i); recall from Topic 2 (points to the summation symbol $\sum$)

mean of observed values for each variable (points to $\bar{x}$ and $\bar{y}$)

standard deviation of observed values for each variable (points to s_x and s_y)

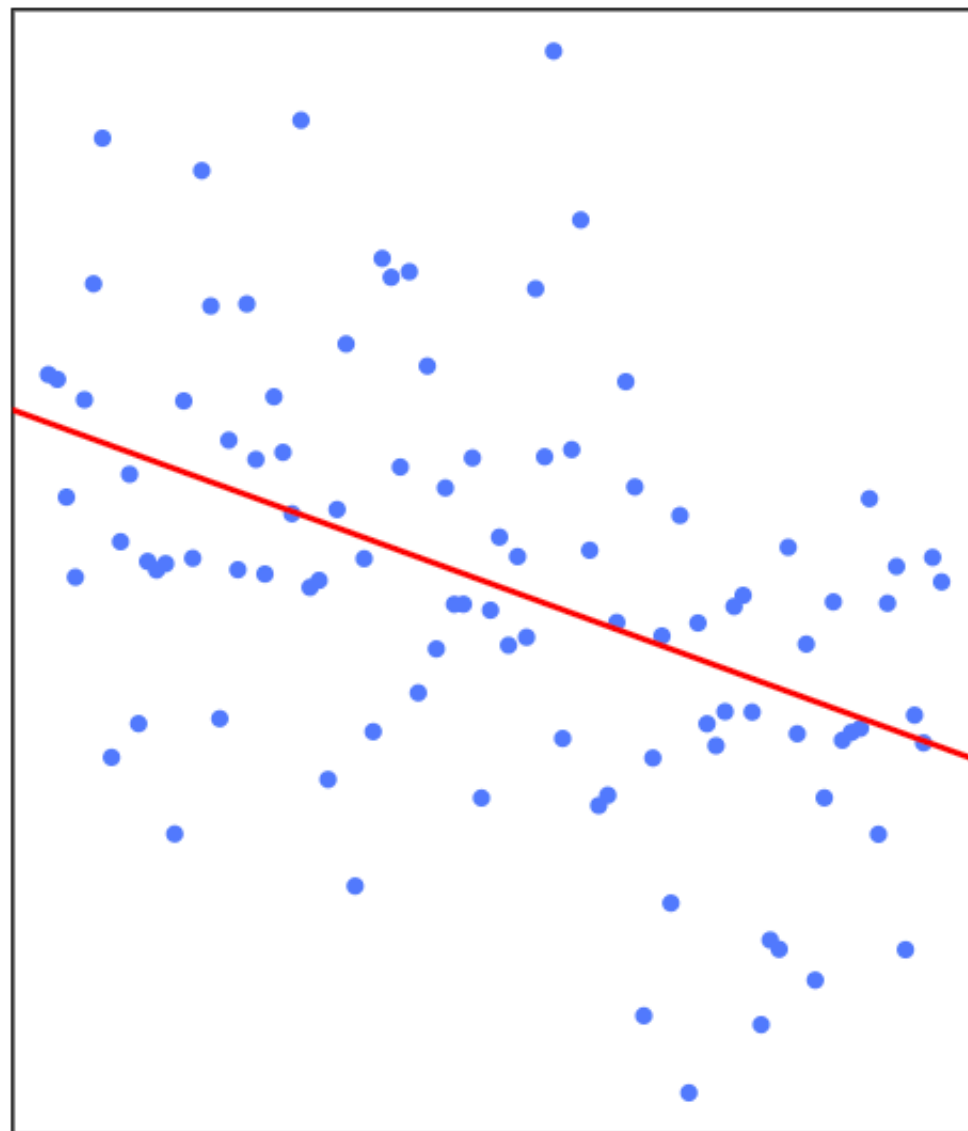


PollEverywhere

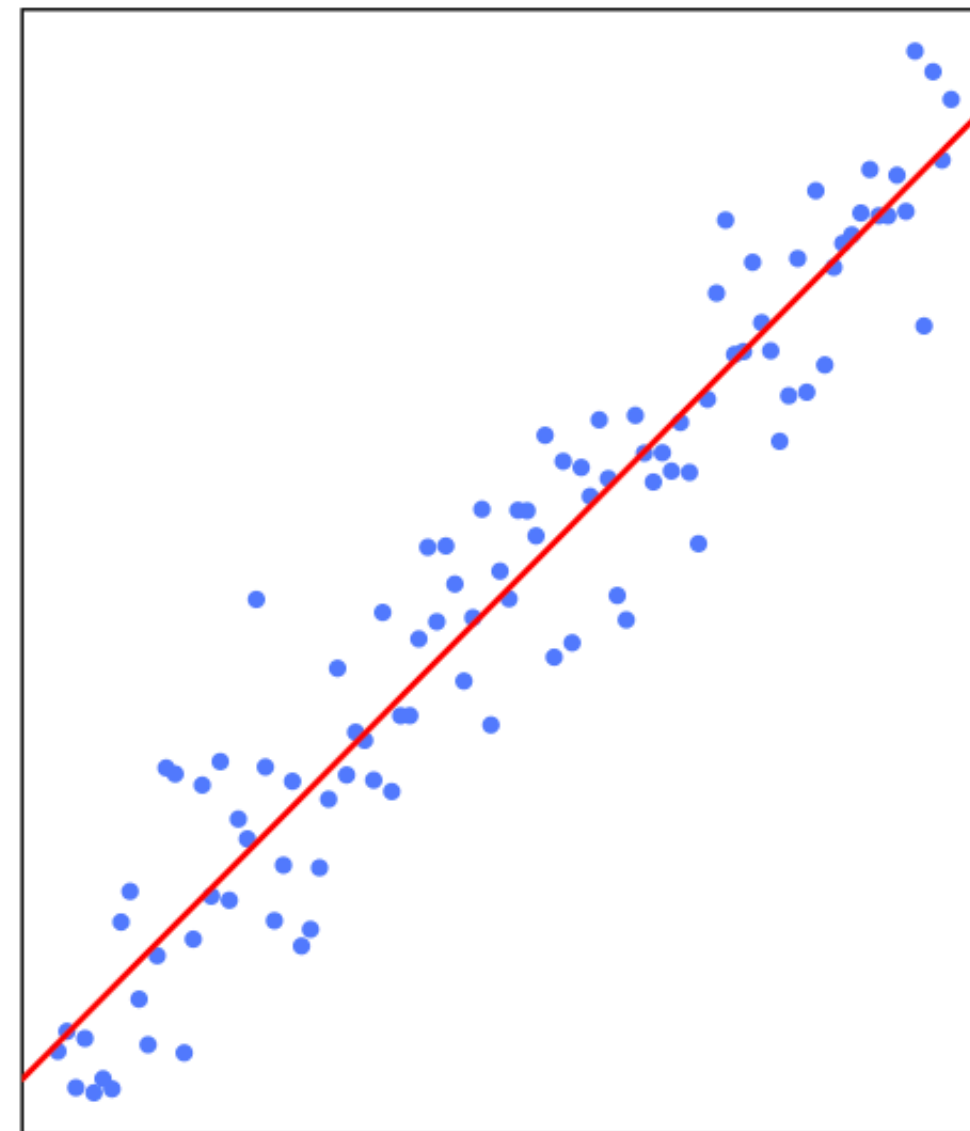
PollEv.com/stats8

- Which of the following scatterplots has a correlation coefficient r closest to 1?

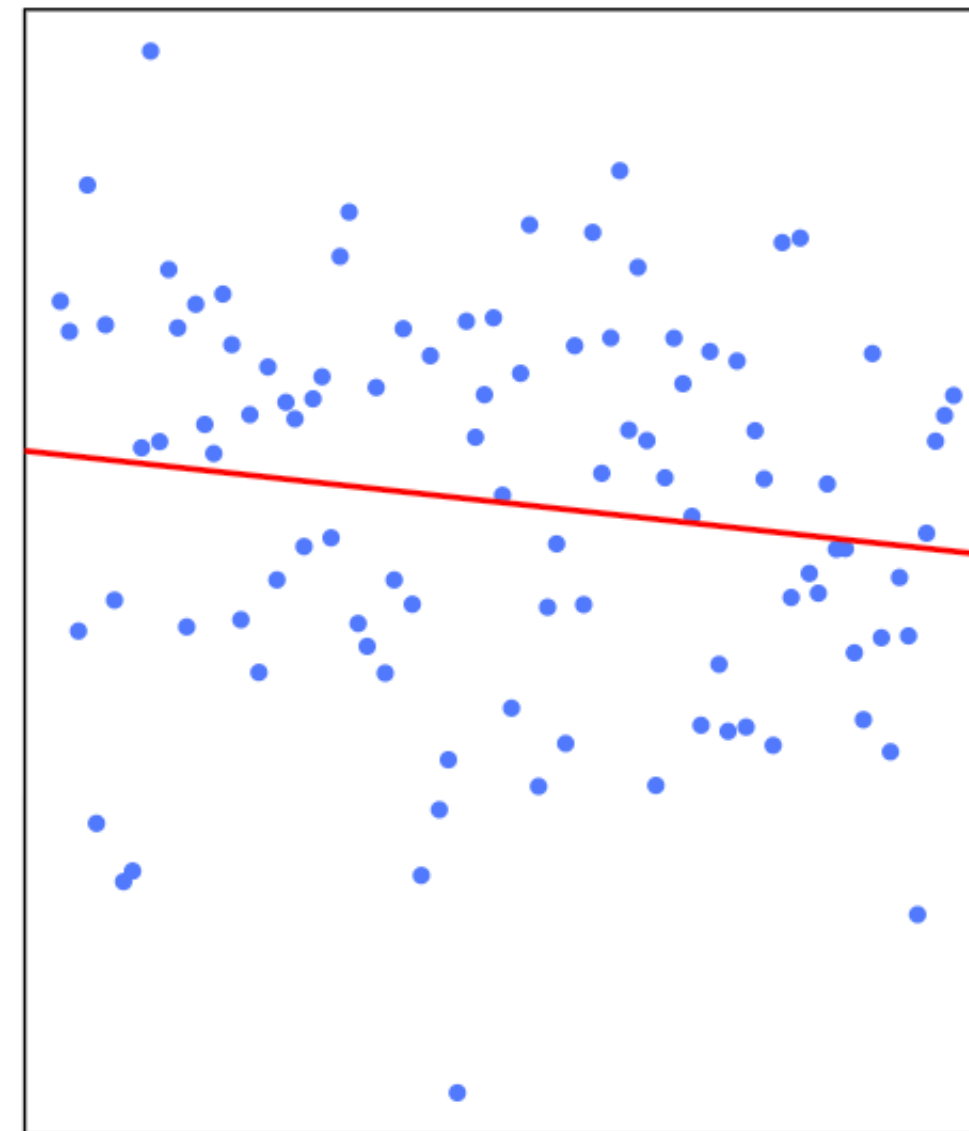
Plot A



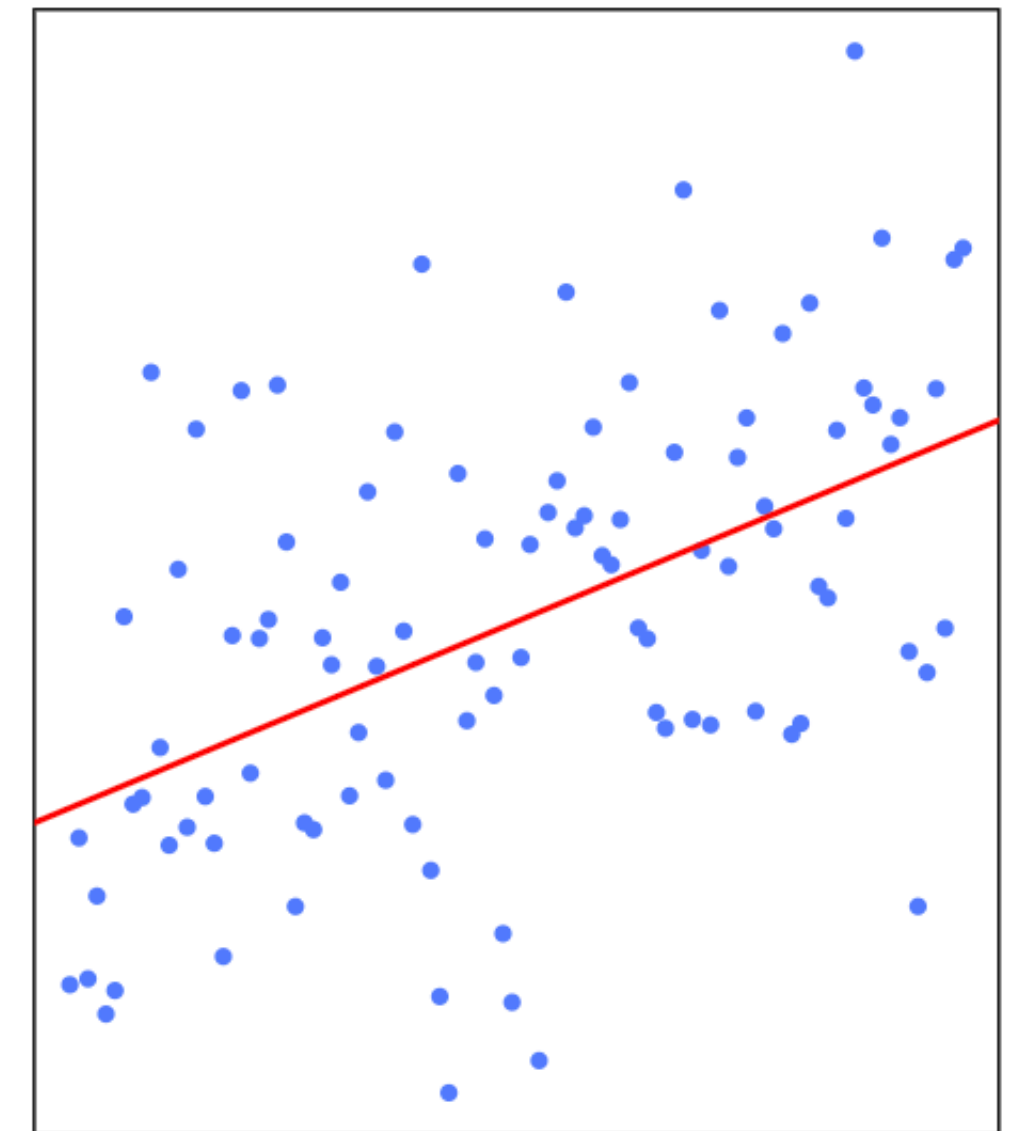
Plot B



Plot C



Plot D

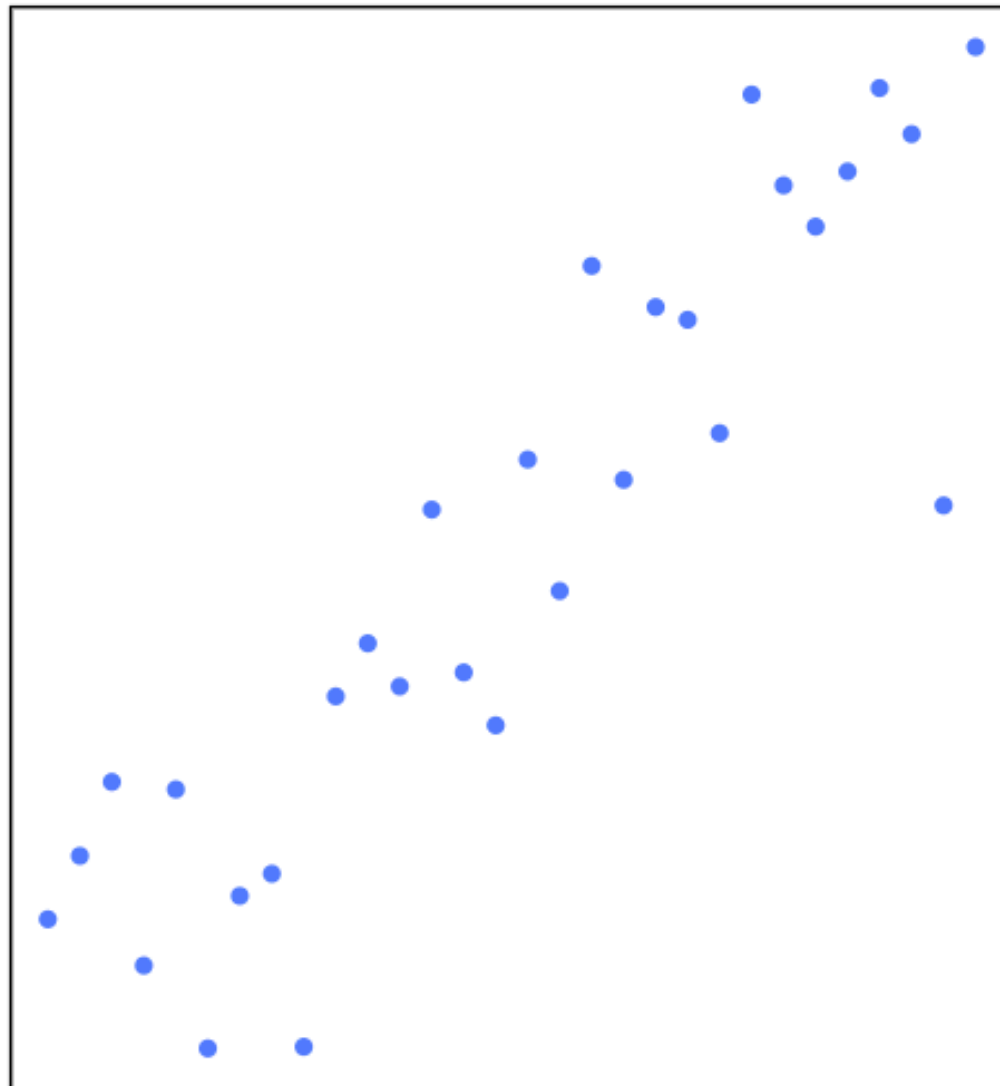


PollEverywhere

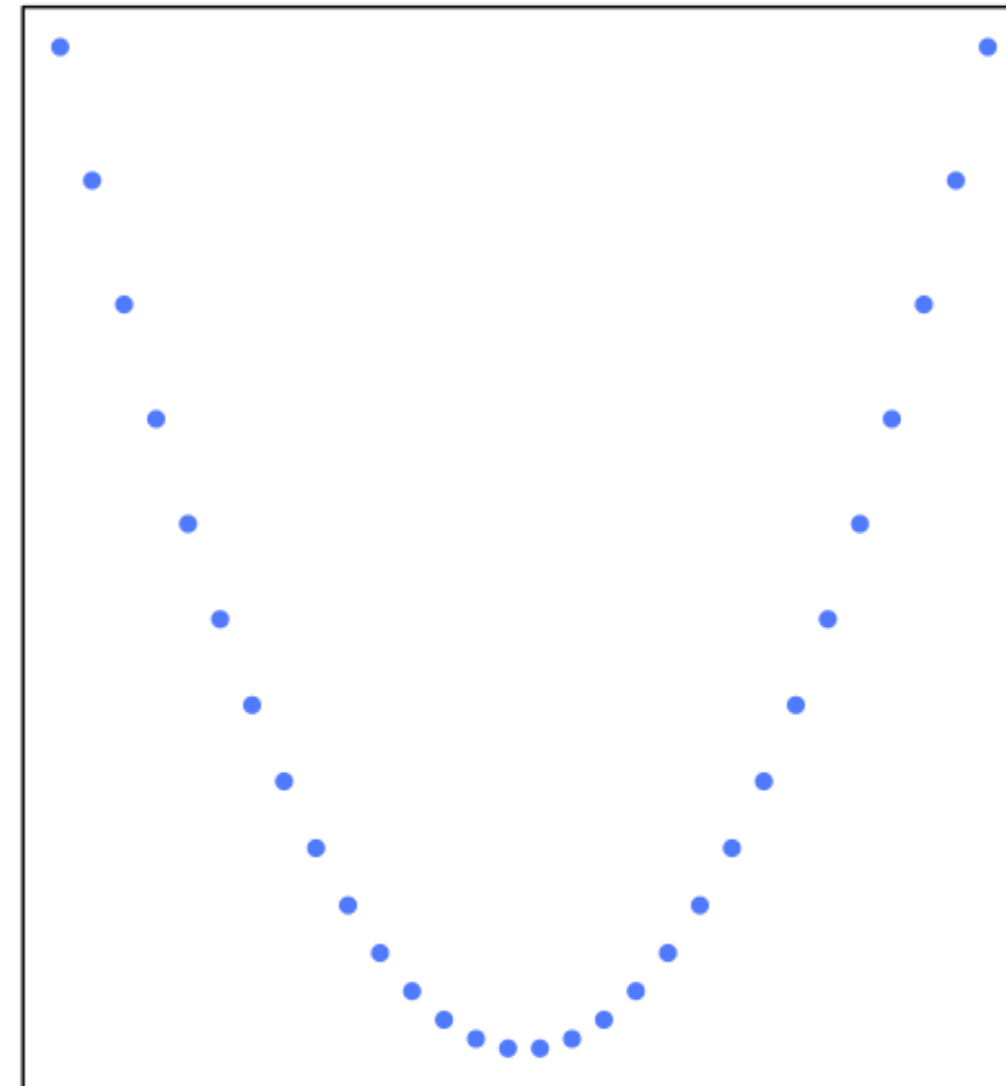
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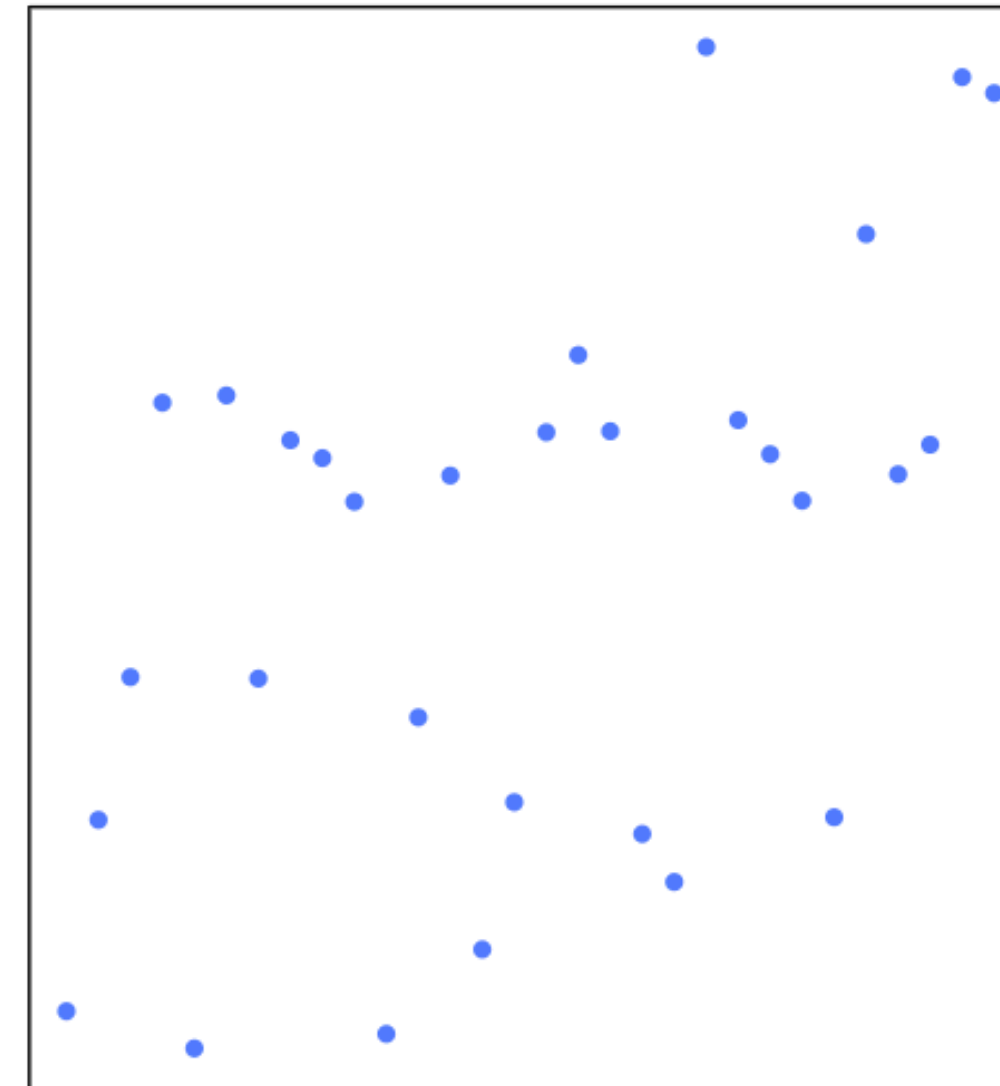
Plot A



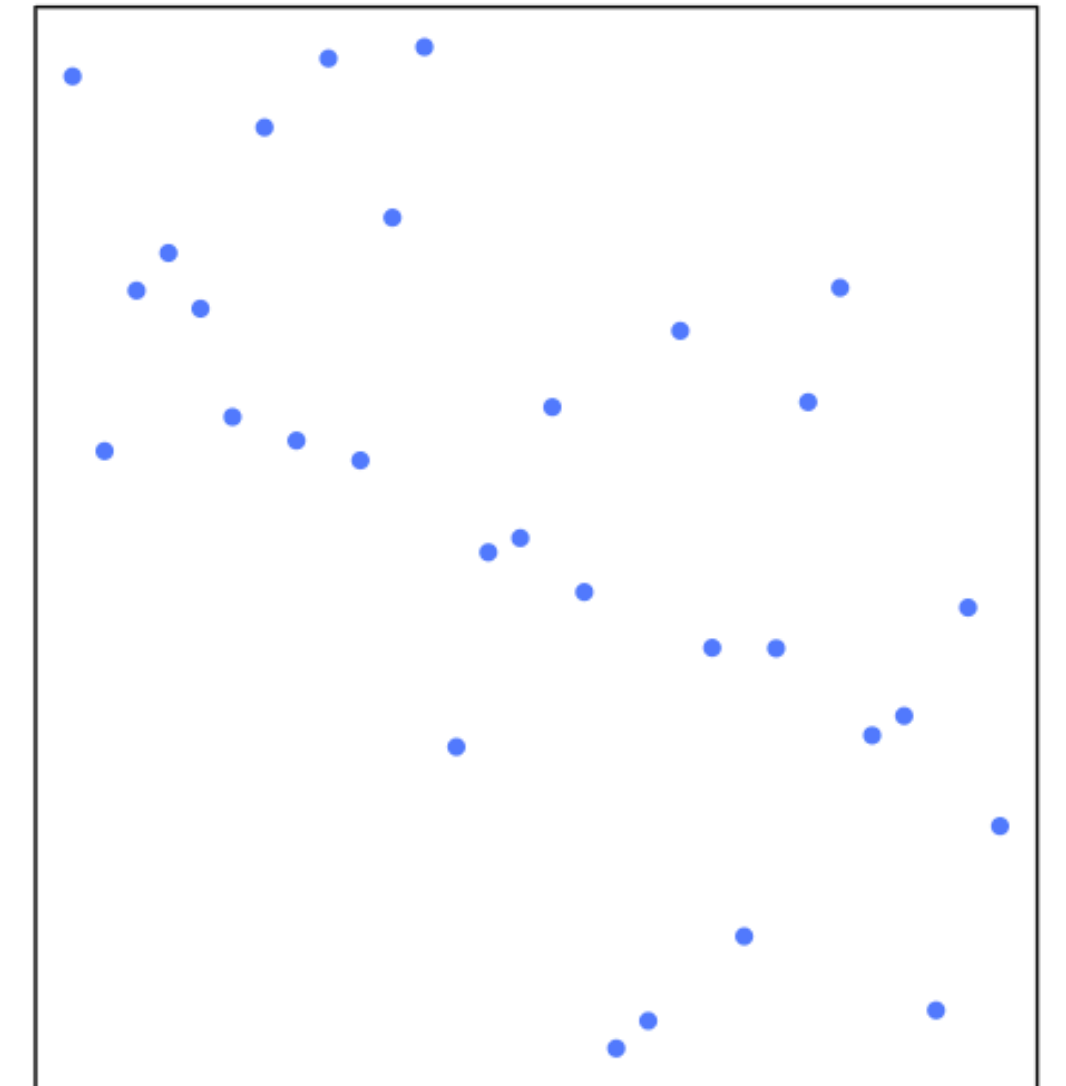
Plot B



Plot C



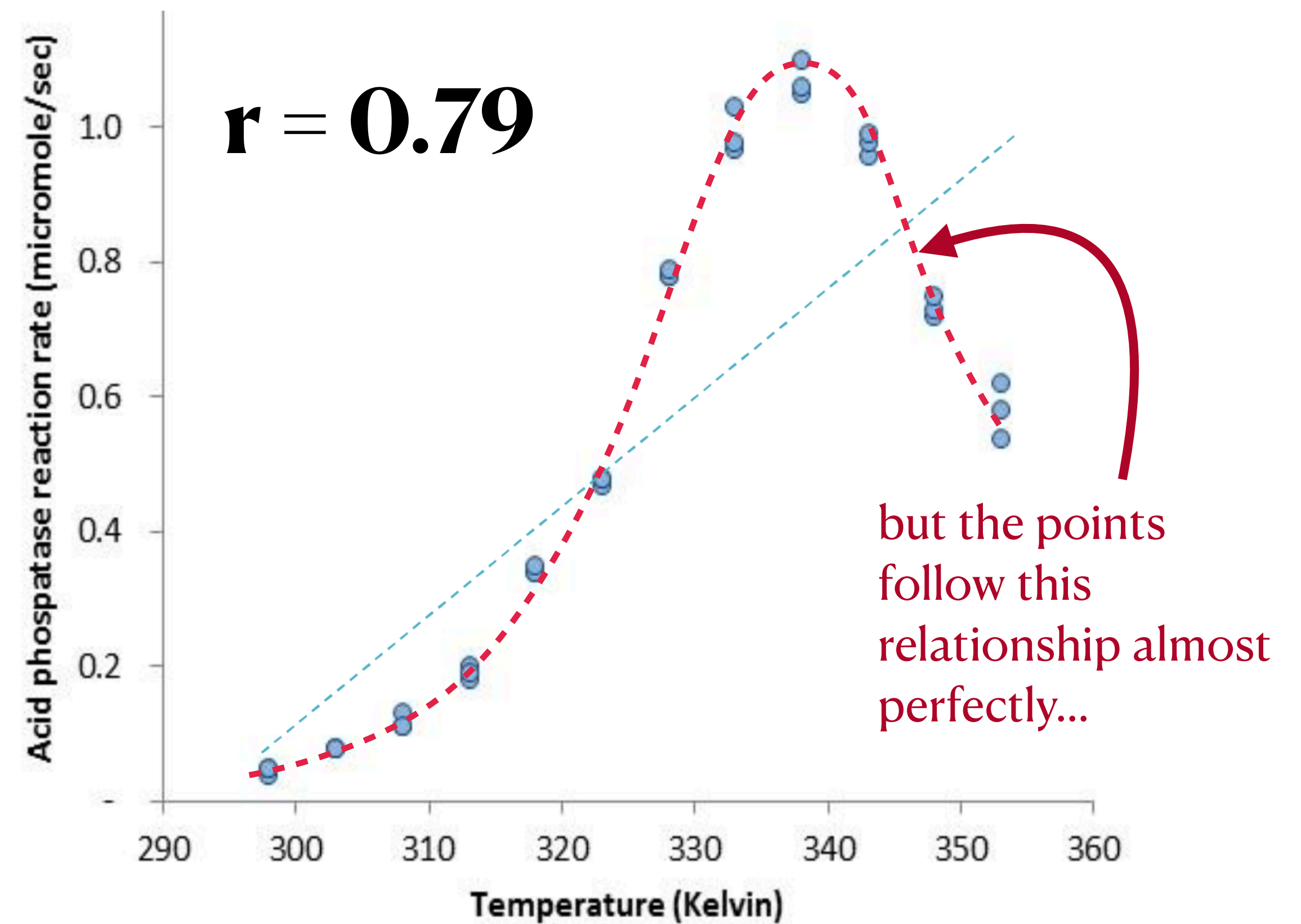
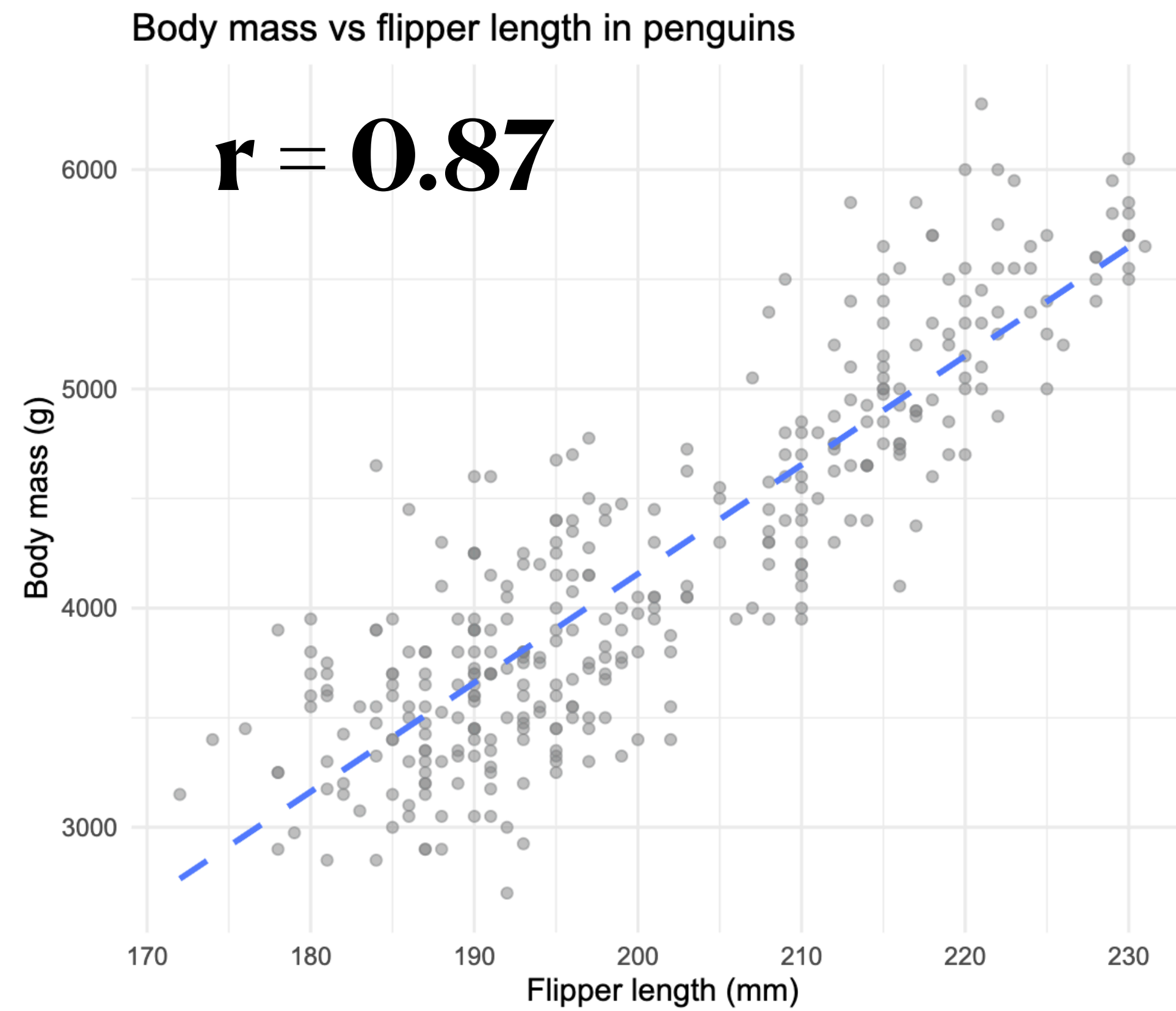
Plot D



Describing Associations

Strength

- 🌟 The correlation coefficient is only meaningful for strength of **linear relationships**!

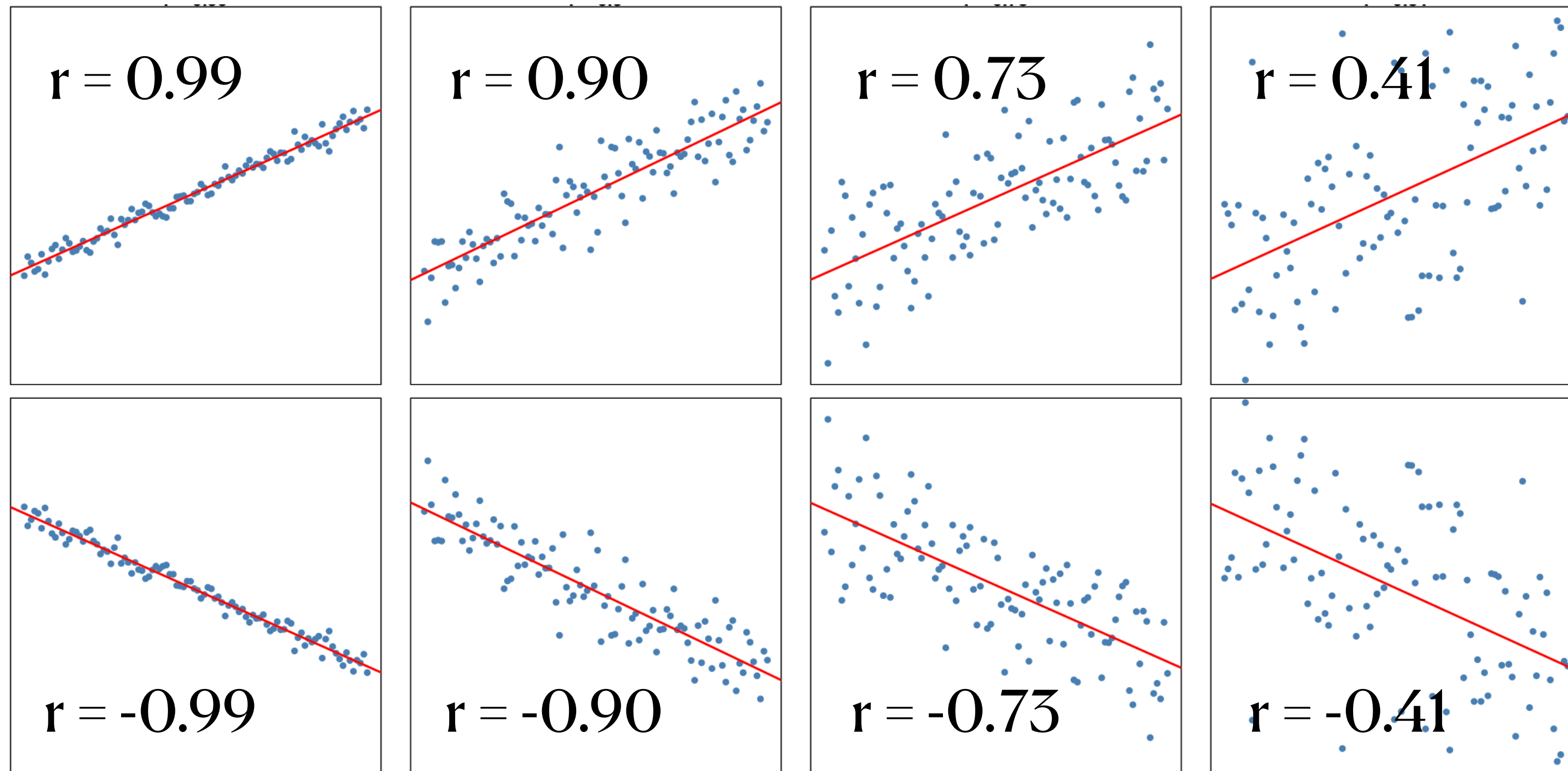


- As a general rule, blindly doing computations on data is not recommended — plot it too!

Describing Associations

Direction

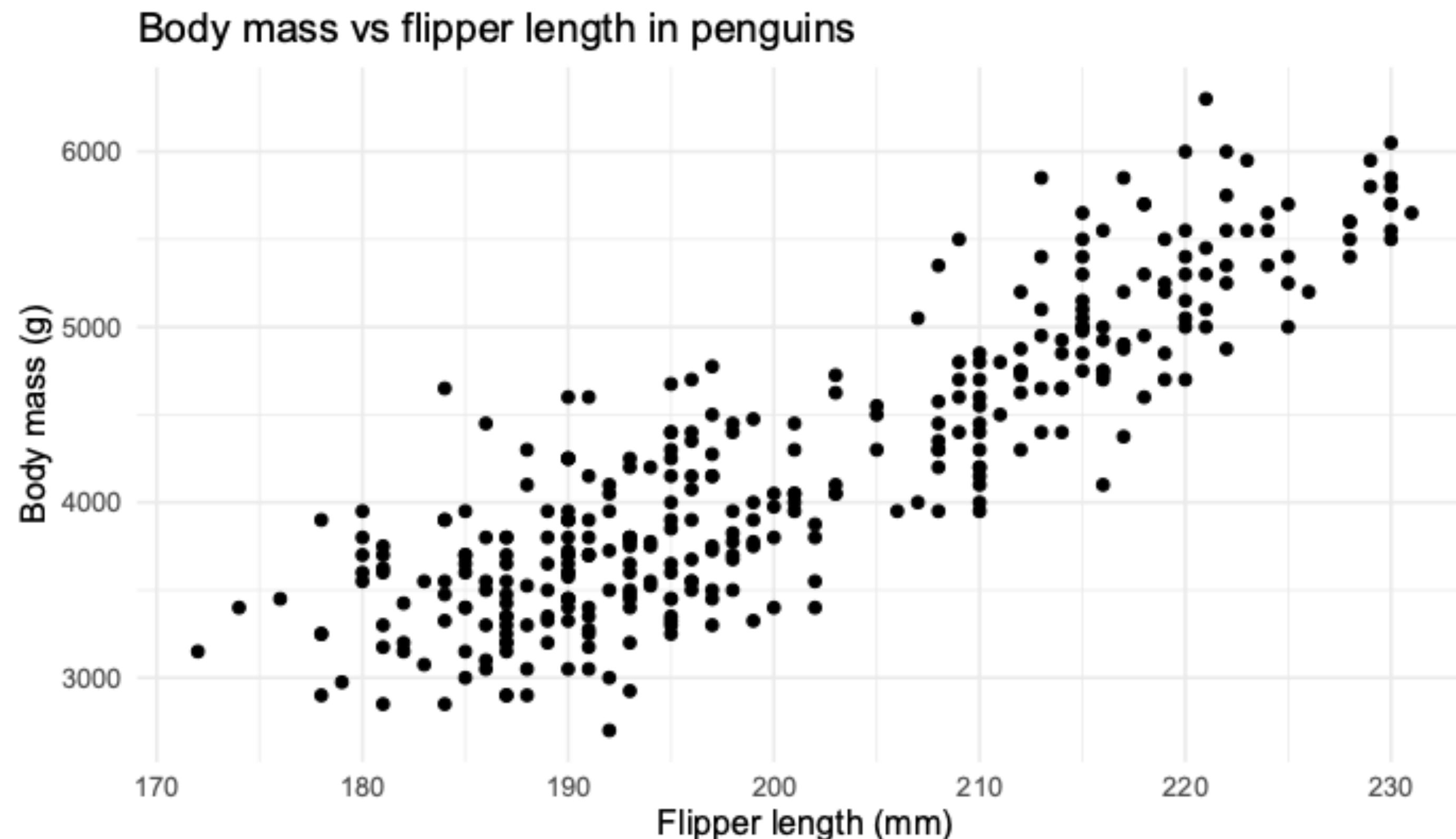
- The correlation coefficient r also gives us information on the **direction** of the relationship
 - Do higher values of one variable correspond to higher values of the other?



Describing Associations

Direction

- The correlation coefficient r also gives us information on the **direction** of the relationship
 - Do higher values of one variable correspond to higher values of the other?



- Penguin flipper length and body mass are **positively correlated** because penguins with longer flippers tend to have higher body mass
- The correlation coefficient $r=0.87$ is **positive**

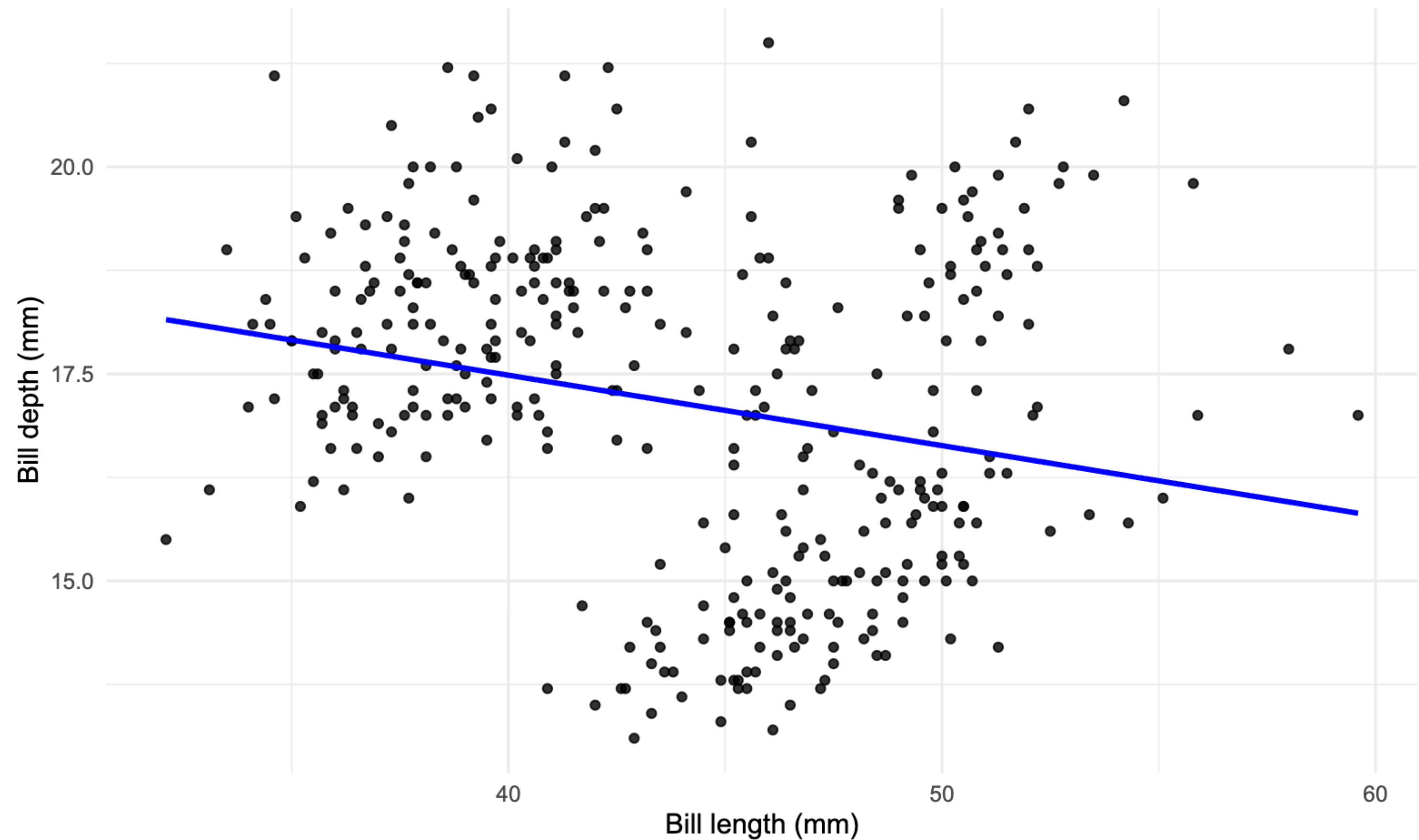
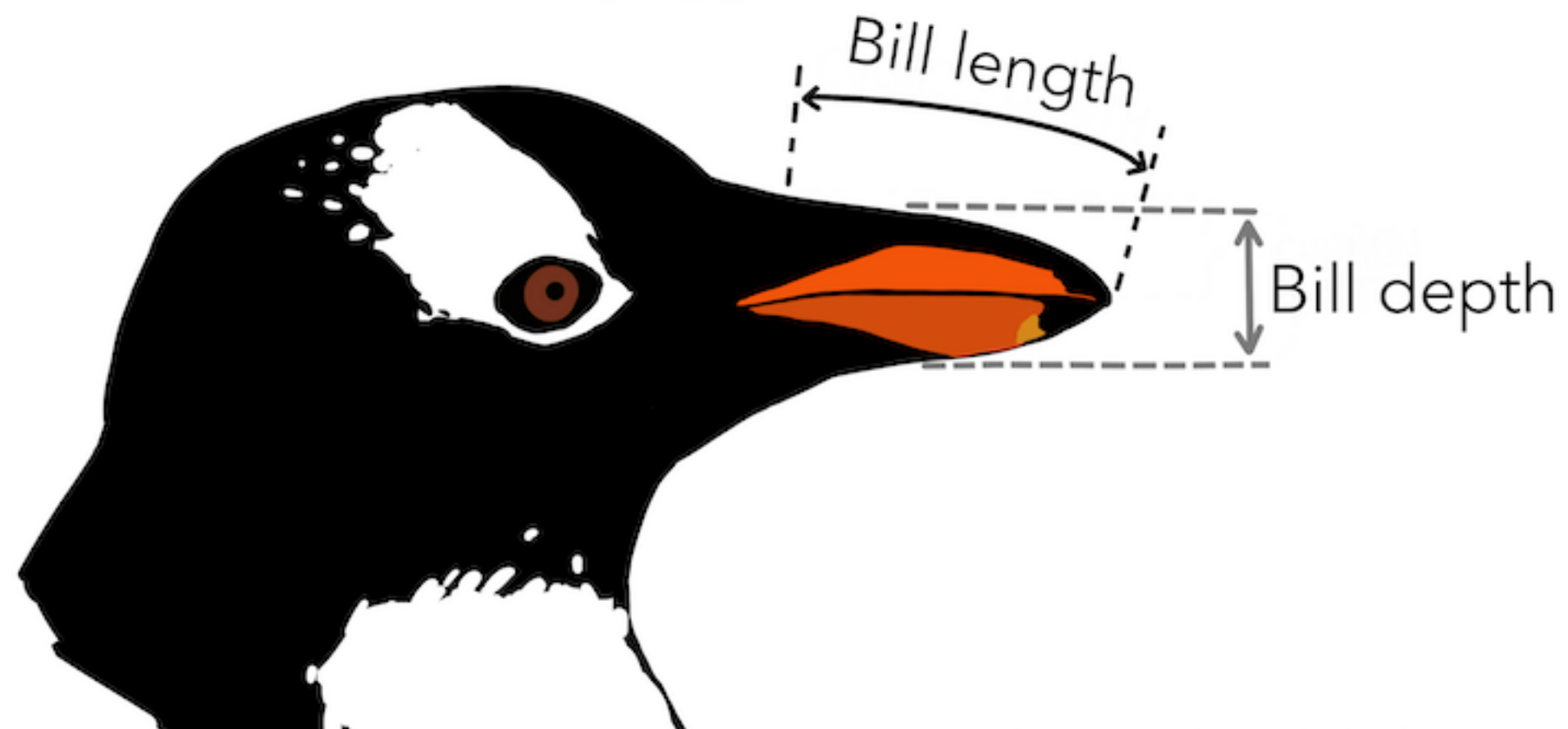
Describing Associations

Direction

Correlation coefficient is...?

Negative.

Penguins with longer bills tend to have thinner bills

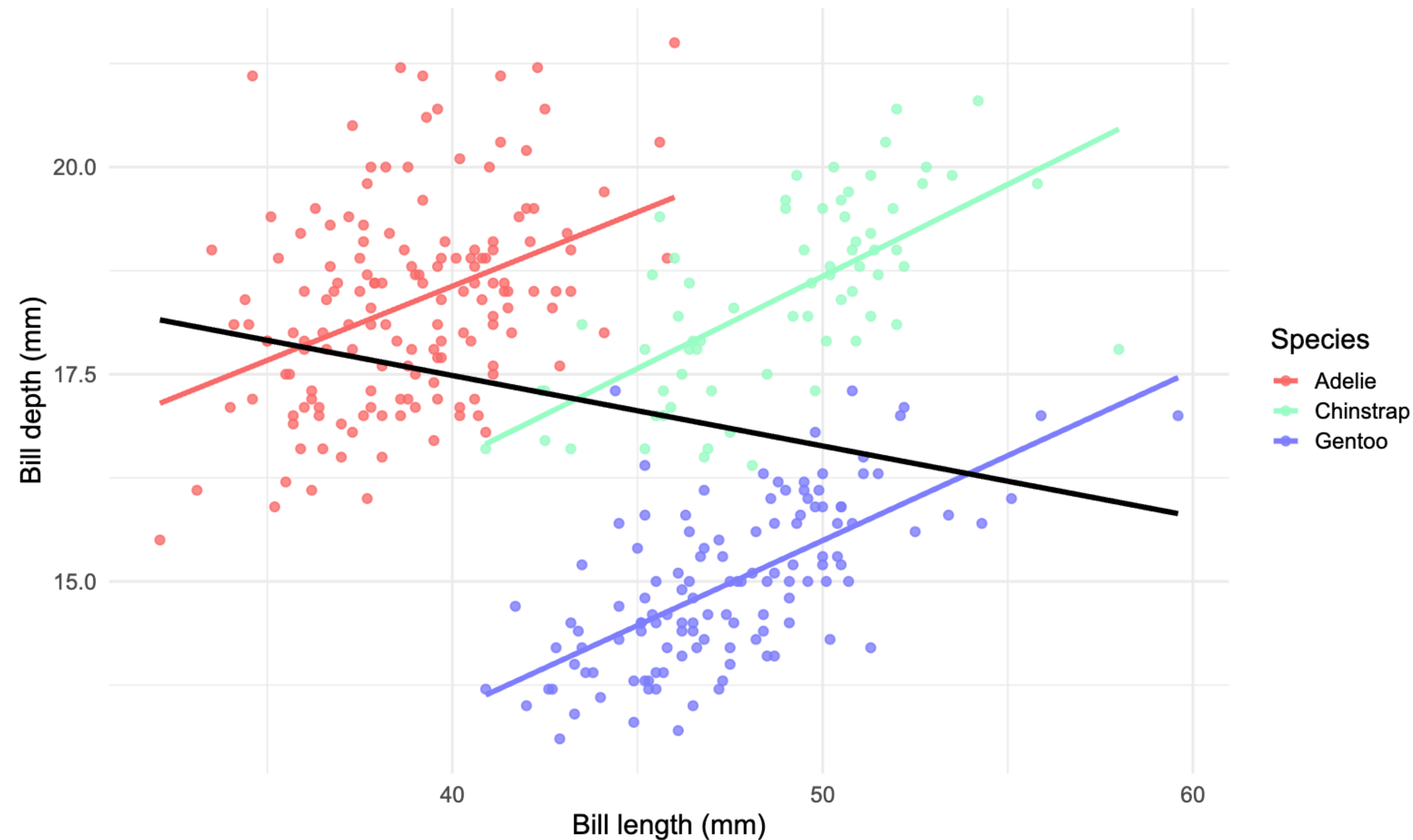
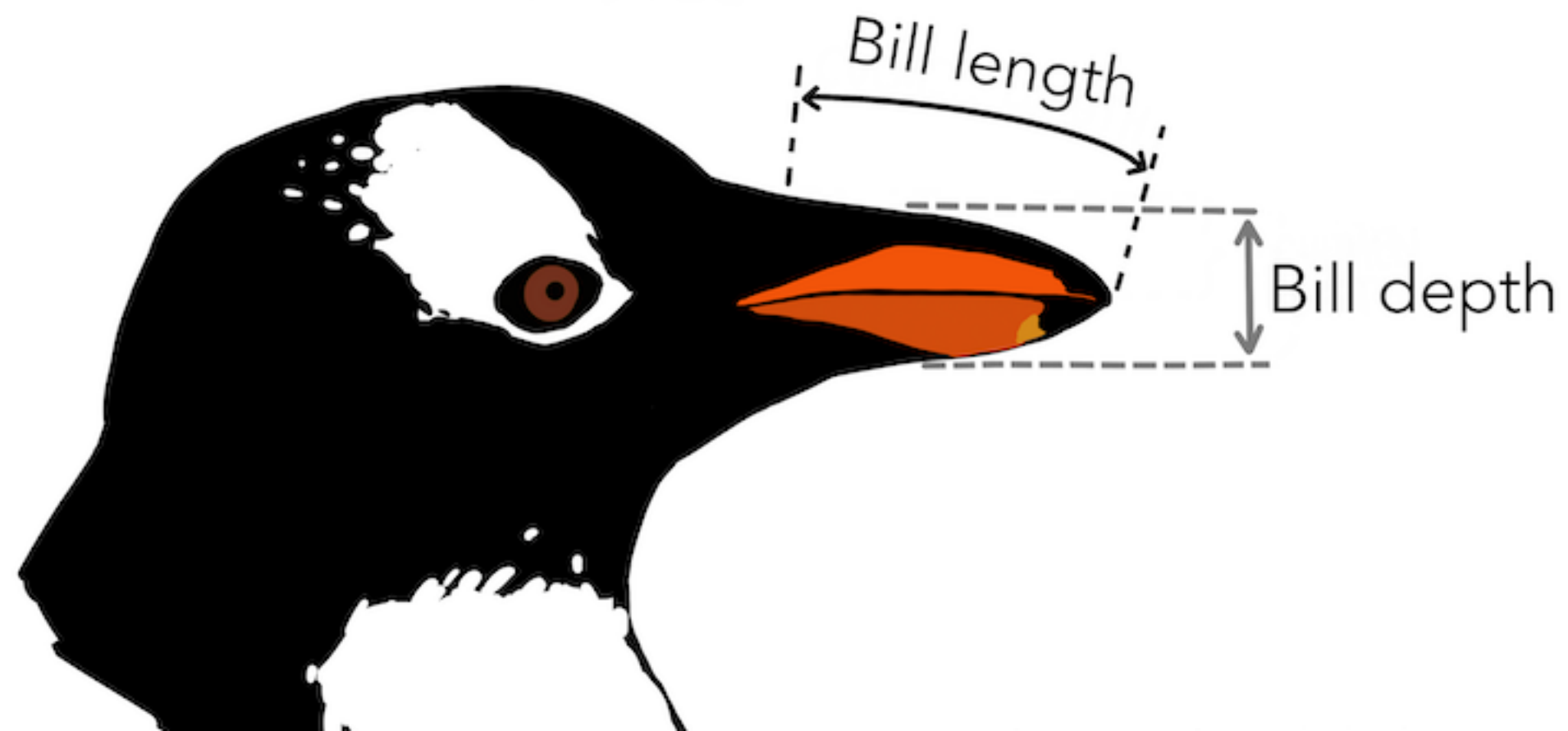


Describing Associations

Direction

But, when splitting by species, the separate correlations are positive!

This is called **Simpson's paradox**: An association that is true for each of several groups can appear to change direction when the data are combined to form a single group

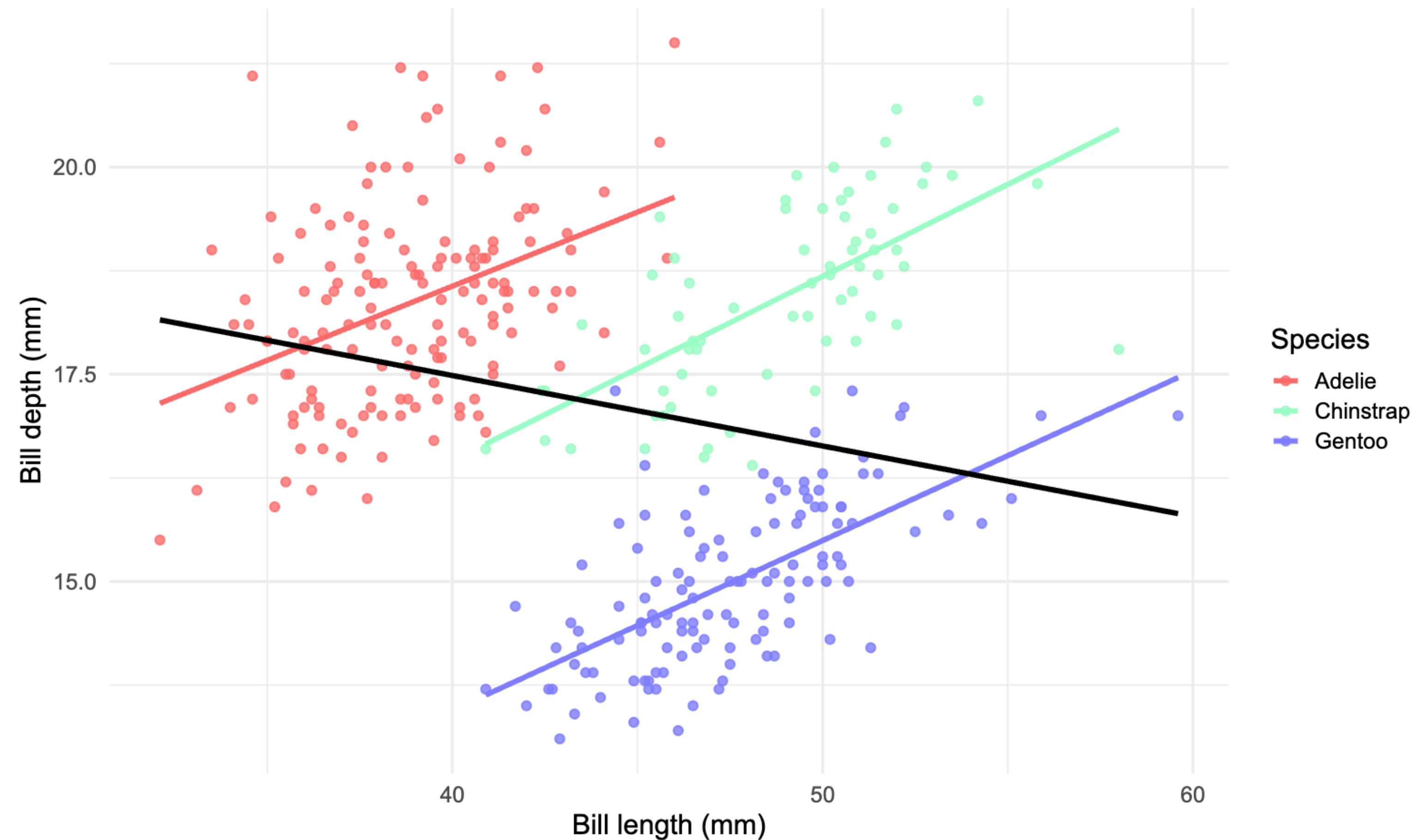
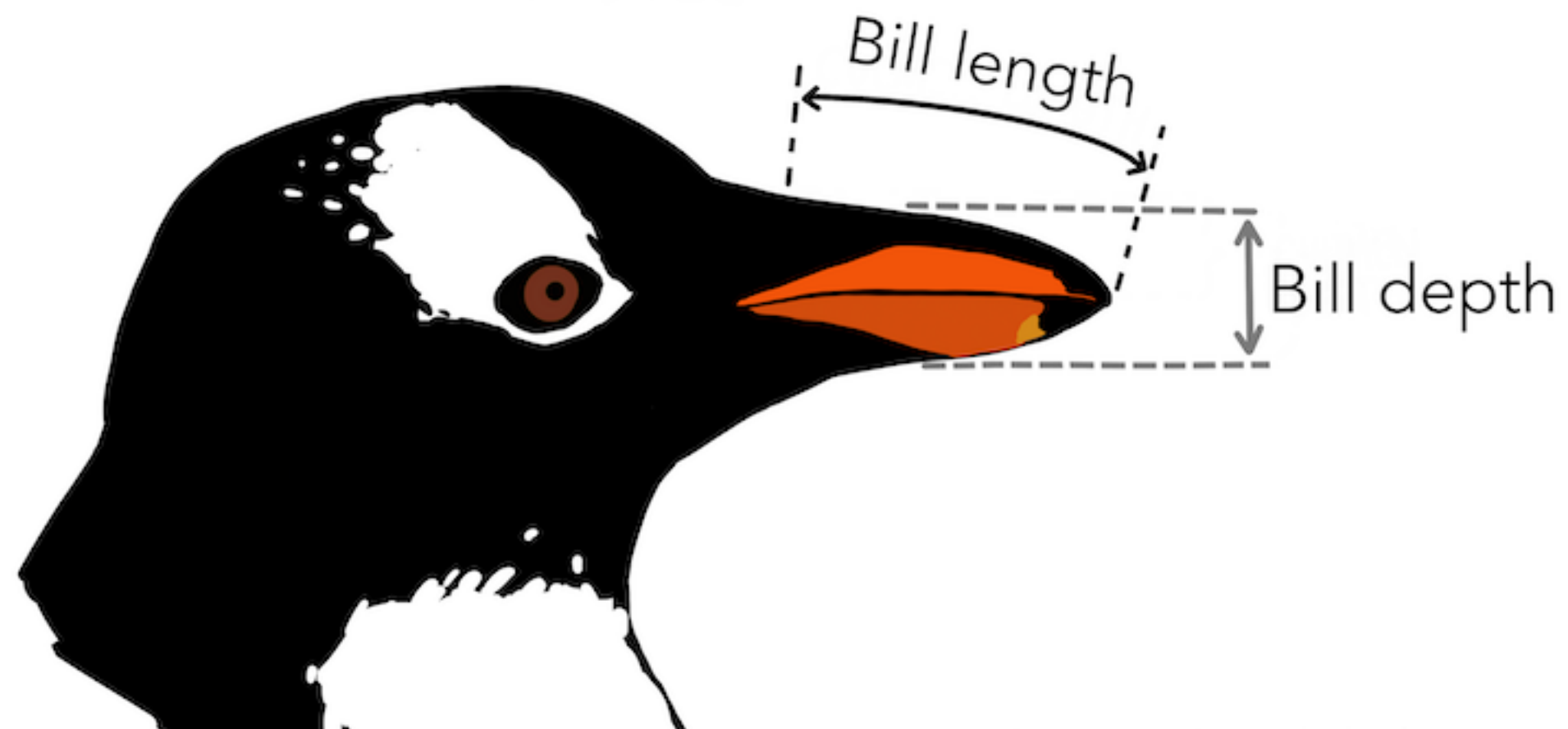


Describing Associations

Direction

But, when splitting by species, the separate correlations are positive!

Within each species, penguins with longer bills tend to also have thicker bills

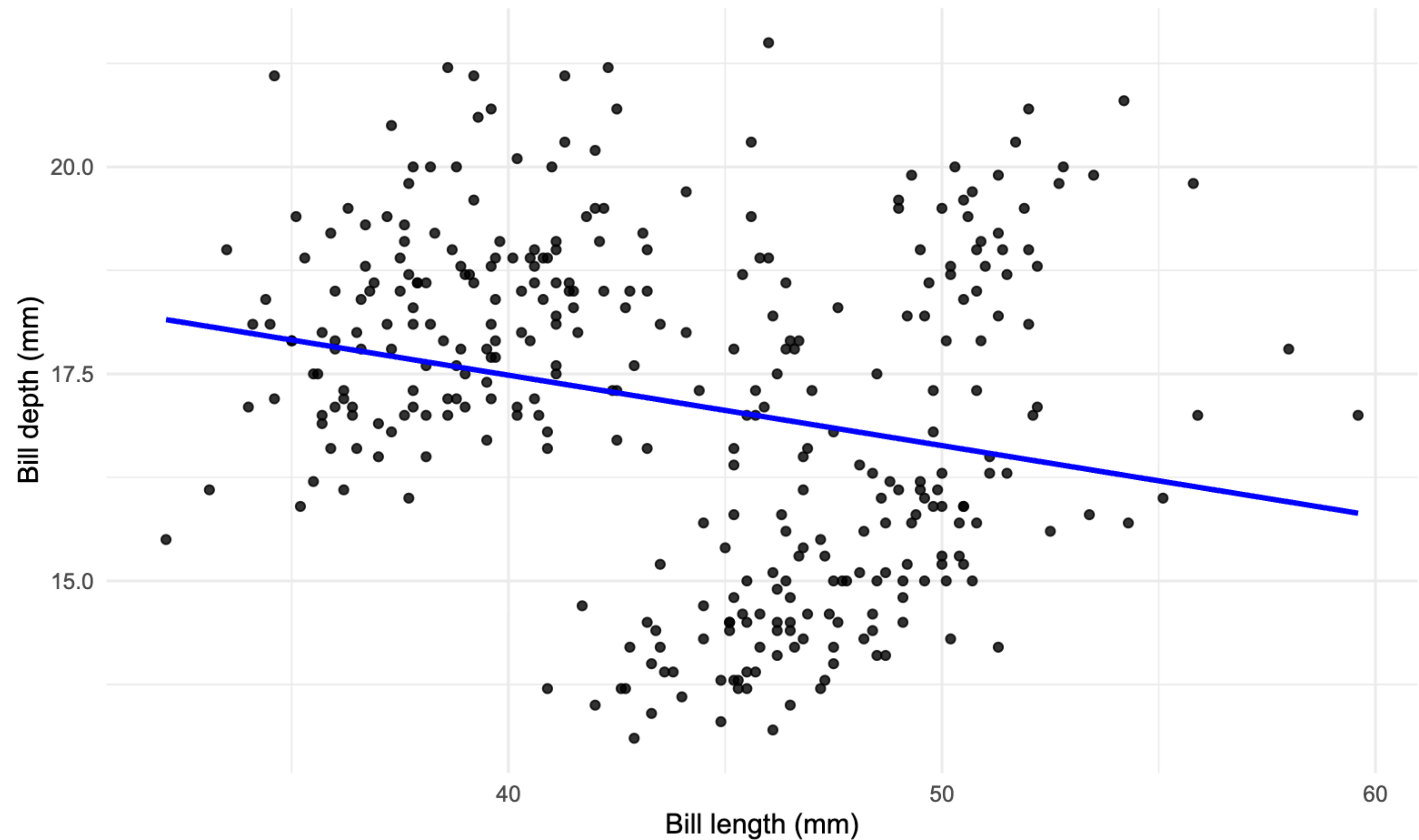
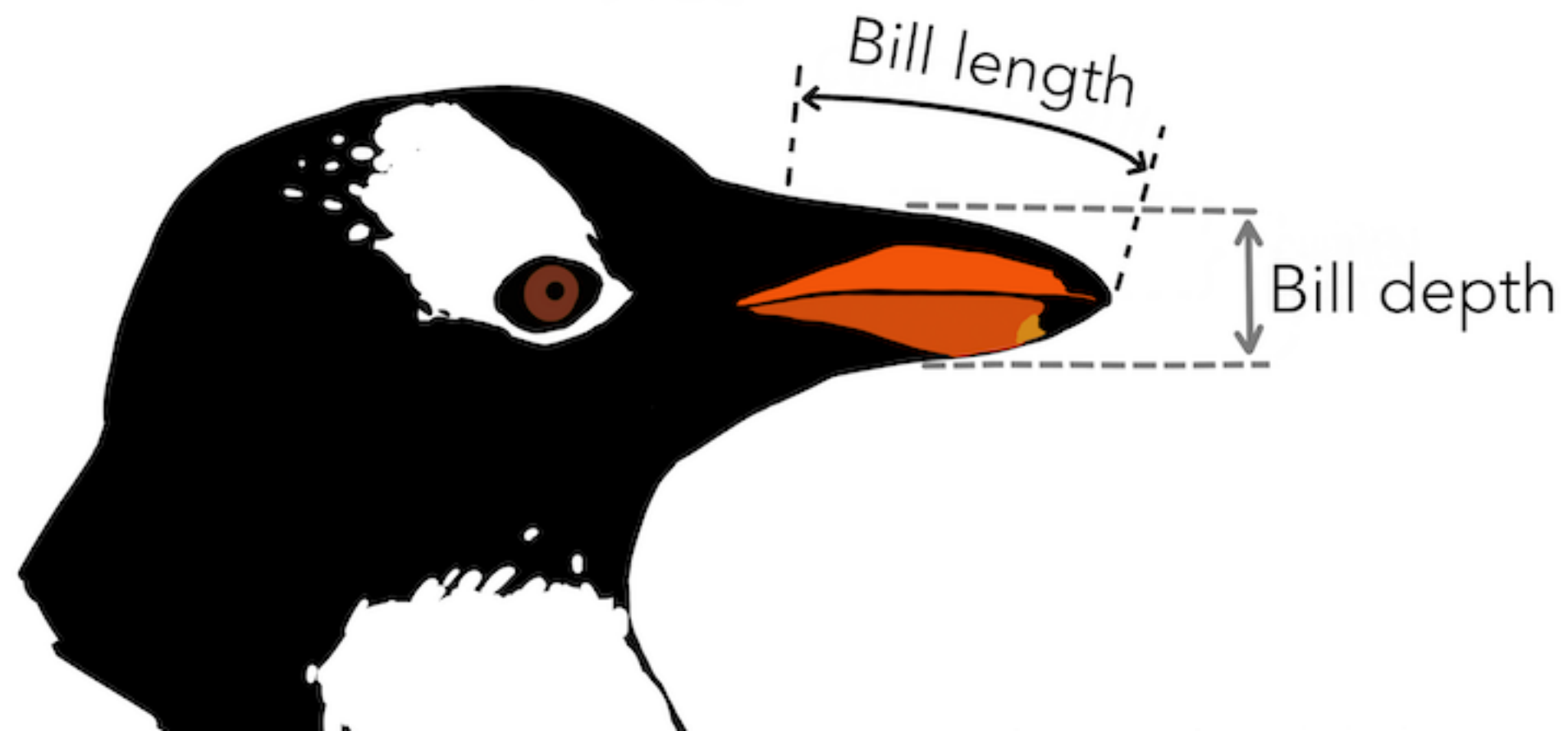


Describing Associations

$$R^2$$

The square of the correlation coefficient, r , also has an interesting/useful interpretation

r^2 is the **fraction of variation in the explanatory variable** that can be **explained by the response variable**

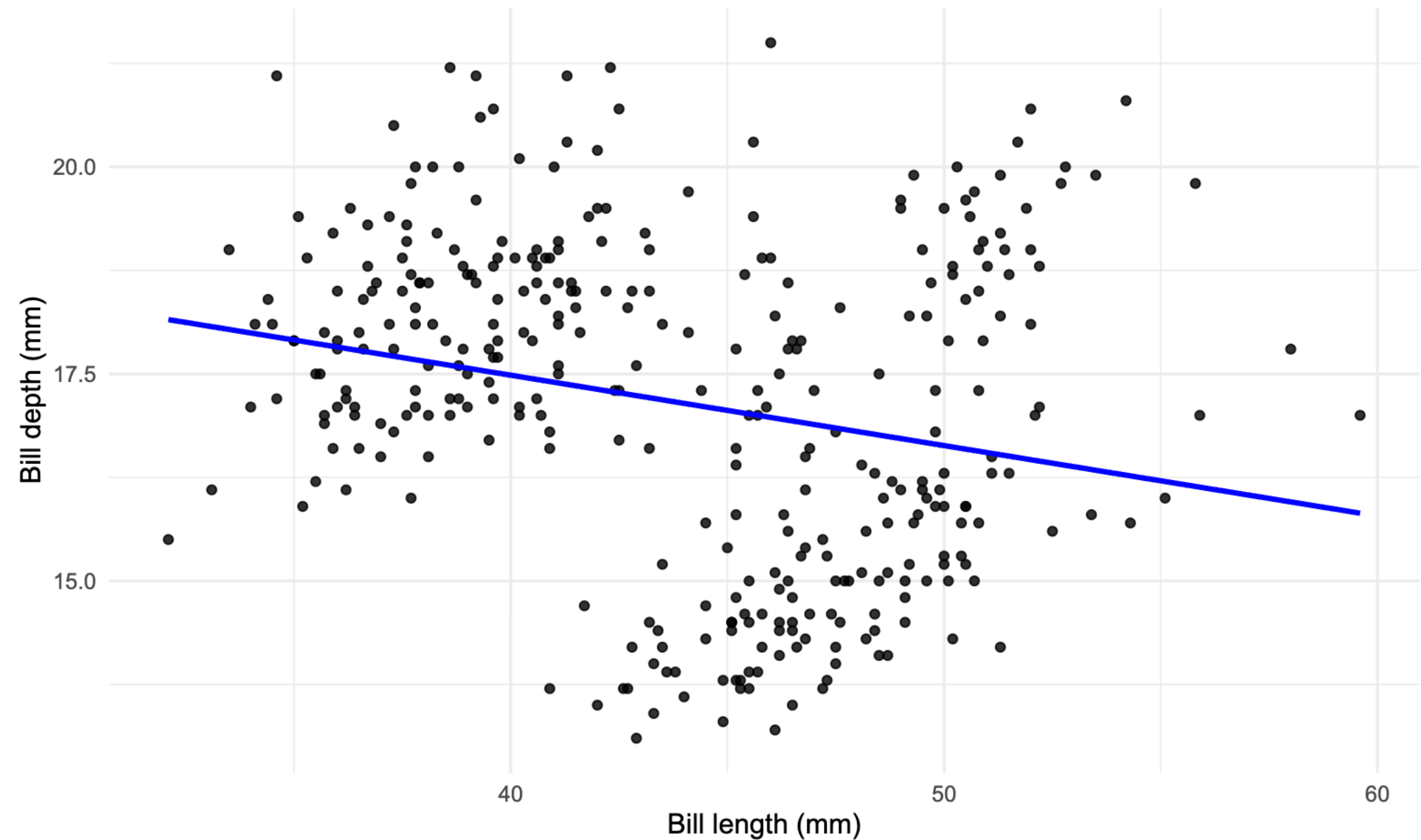
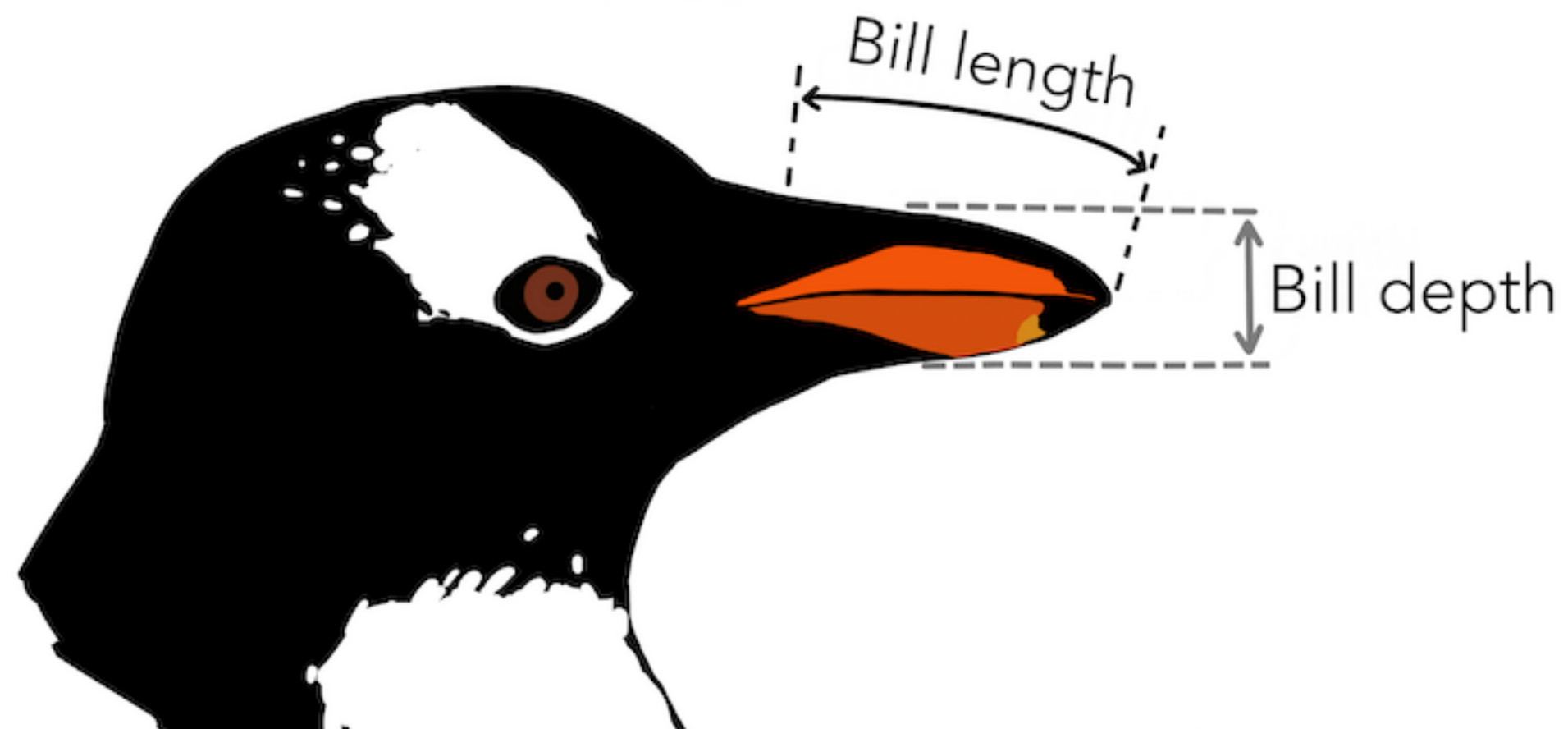


Describing Associations

$$R^2$$

Suppose $r = -0.3$.

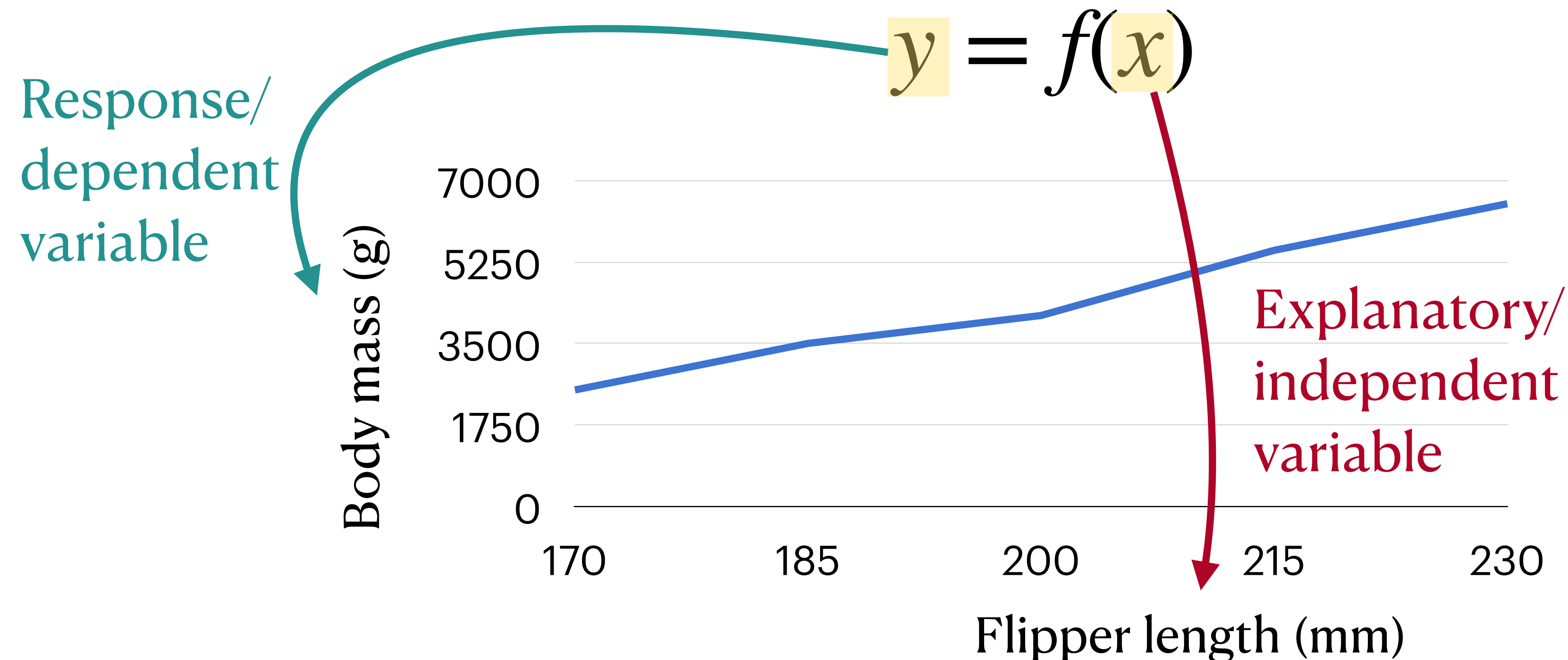
9% of the variability in bill depth can be explained by bill length in penguins.



Least-Squares Regression

Quantitative + Quantitative

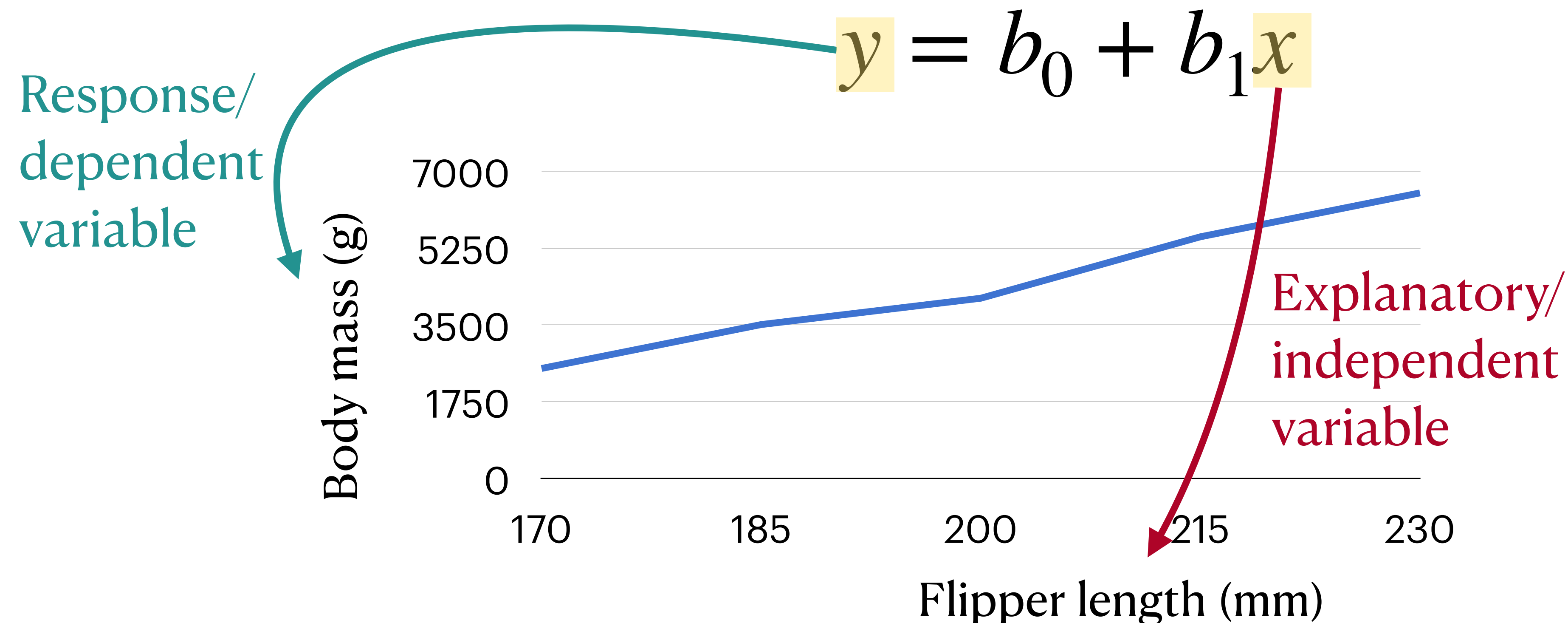
- We want to model one quantitative variable as a function of the other one
 - Mentioned earlier that we will focus on **linear** relationships, so...



Least-Squares Regression

Quantitative + Quantitative

- We want to model one quantitative variable as a function of the other one
- Mentioned earlier that we will focus on **linear** relationships, so let $f(x)$ be a line!



Least-Squares Regression

Equation of a line (refresher)

- We want to model one quantitative variable as a function of the other one
- Mentioned earlier that we will focus on **linear** relationships
- A **linear** relationship is a relationship with a constant rate of change (**slope**)

The diagram illustrates the equation of a line, $y = b_0 + b_1x$, with color-coded components and descriptive labels:

- response/dependent variable**: A red label with an arrow pointing to the variable y , which is enclosed in a pink square.
- intercept**: A purple label with an arrow pointing to the coefficient b_0 , which is enclosed in a purple square.
- slope**: An orange label with an arrow pointing to the coefficient b_1 , which is enclosed in an orange square.
- explanatory/independent variable**: A green label with an arrow pointing to the variable x , which is enclosed in a green square.

The equation is presented as $y = b_0 + b_1x$, where the plus sign and the variable x are not enclosed in colored boxes.

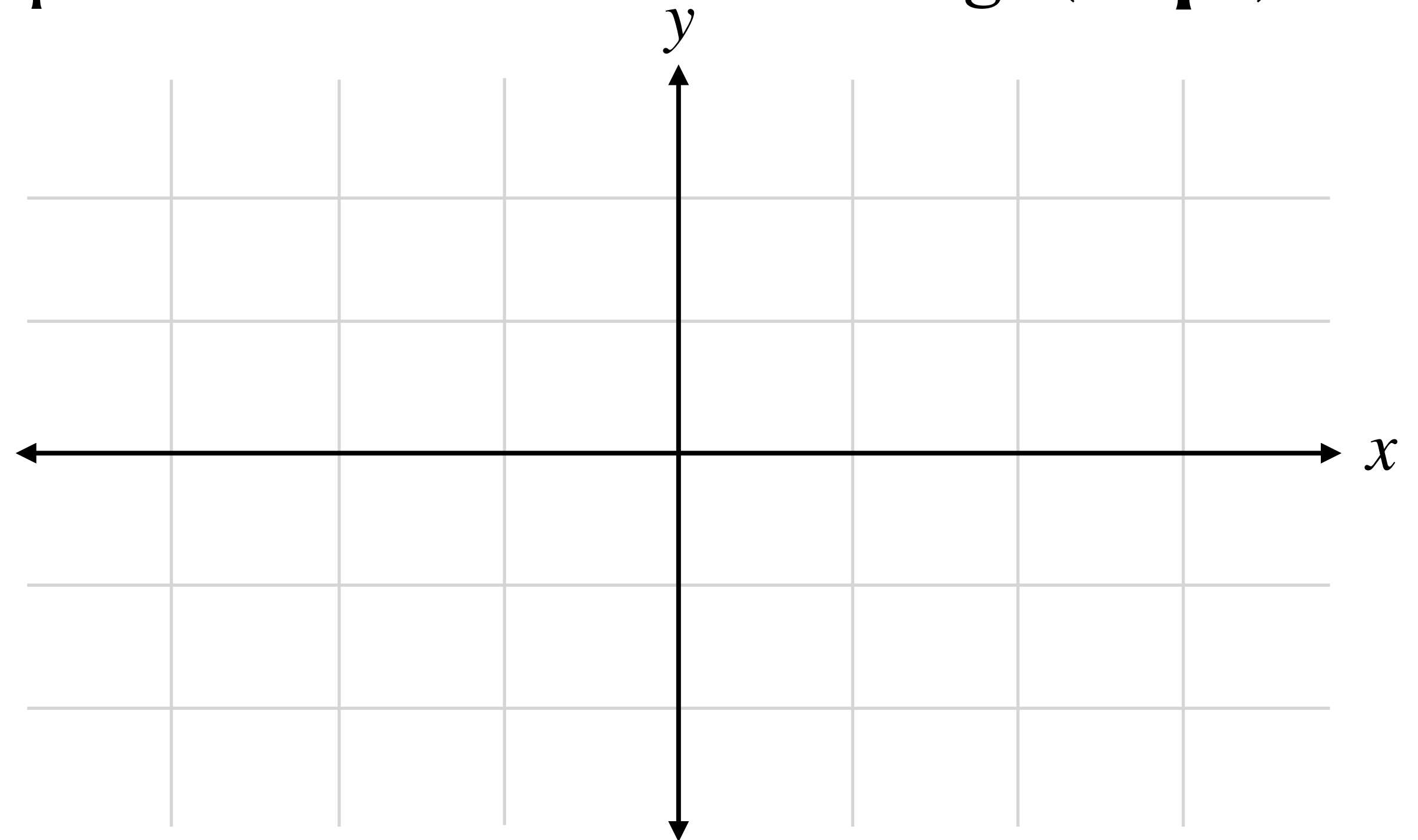
Least-Squares Regression

Equation of a line (refresher)

- We want to model one quantitative variable as a function of the other one
 - Mentioned earlier that we will focus on **linear** relationships
 - A **linear** relationship is a relationship with a constant rate of change (**slope**)

$$y = 1.2 + 0.5x$$

Diagram illustrating the equation of a line: $y = 1.2 + 0.5x$. The constant term 1.2 is highlighted in a purple box and labeled "intercept" with a purple arrow. The coefficient 0.5 is highlighted in an orange box and labeled "slope" with an orange arrow.



Least-Squares Regression

Equation of a line (refresher)

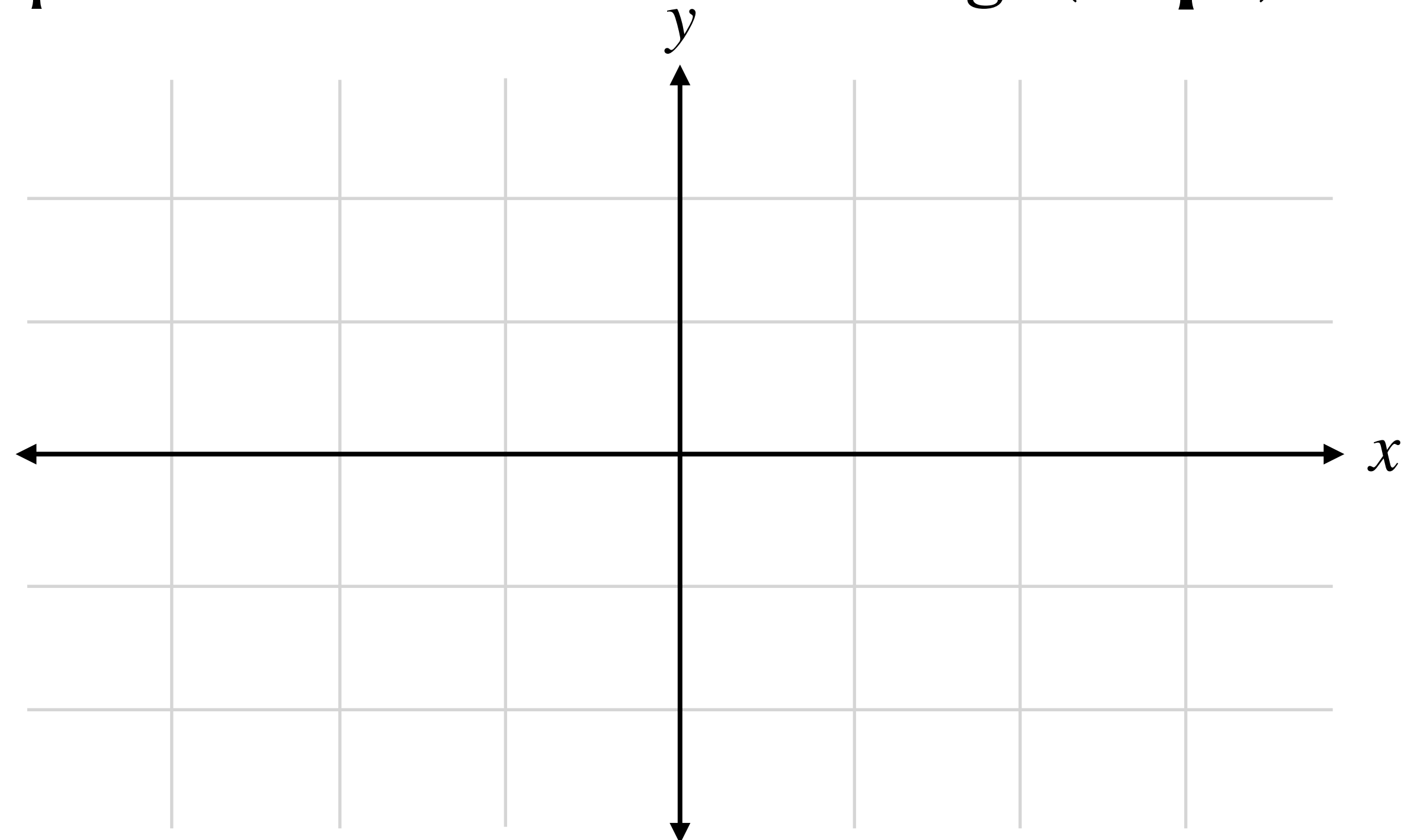
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$$y = 1.2 + 0.5x$$

The equation is annotated with colored boxes: a purple box around the constant term 1.2 and an orange box around the coefficient 0.5. An orange curved arrow points from the word "slope" to the orange box. A purple curved arrow points from the word "intercept" to the purple box.

$$x = -3$$

$$y = ?$$



Least-Squares Regression

Equation of a line (refresher)

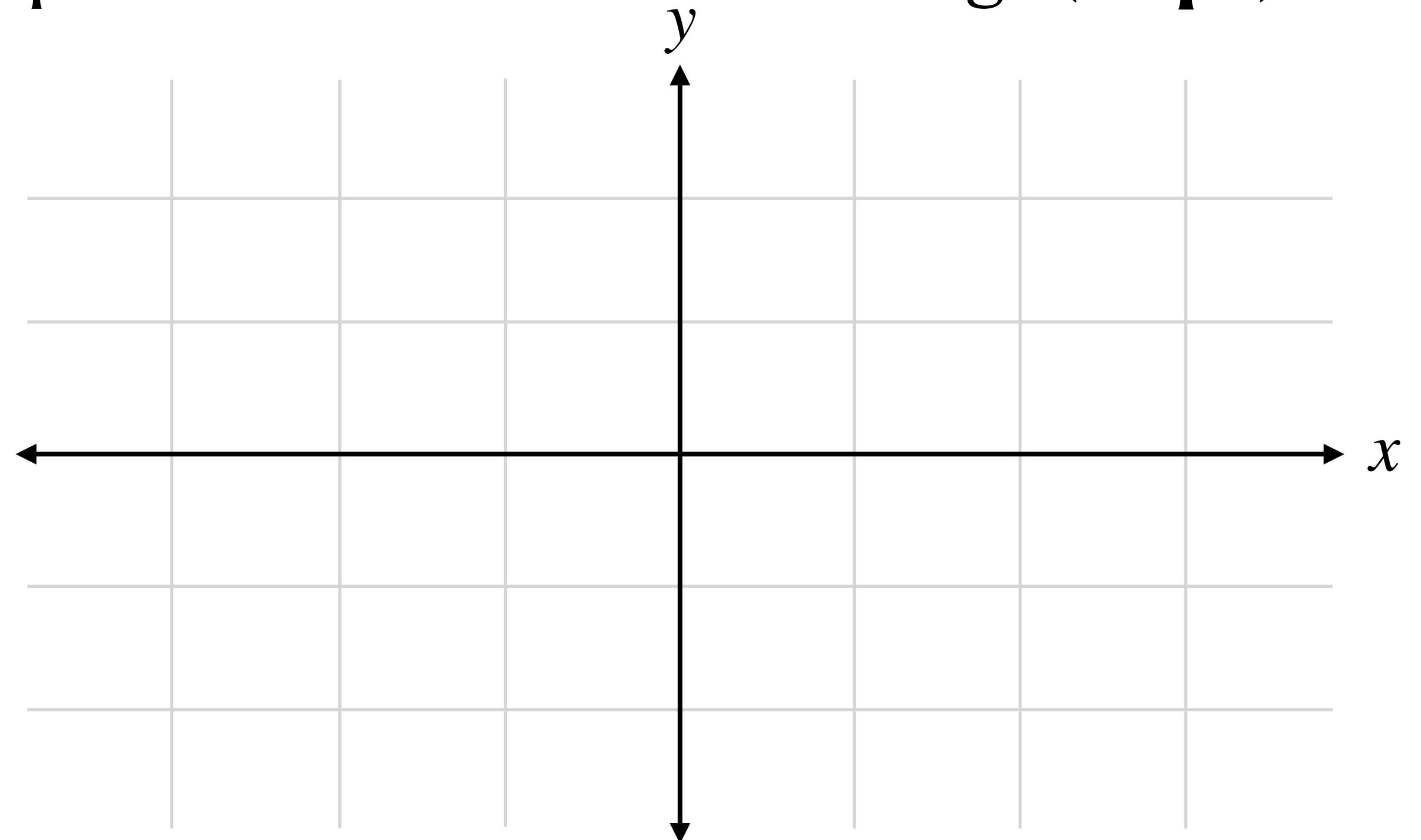
- We want to model one quantitative variable as a function of the other one
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$$y = 1.2 + 0.5x$$

intercept

slope

$x = -3$
 $y = 1.2 + 0.5x$



Least-Squares Regression

Equation of a line (refresher)

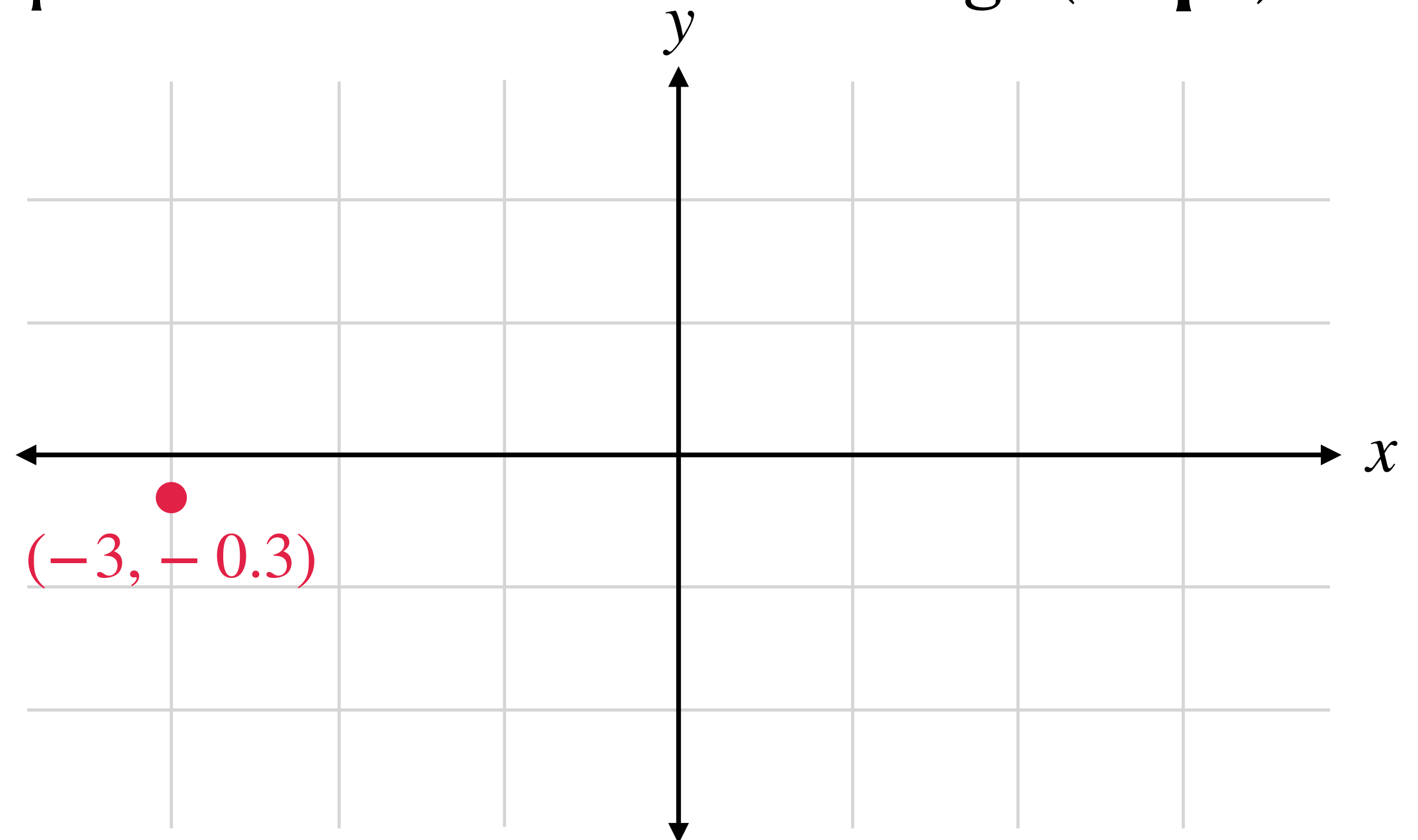
- We want to model one quantitative variable as a function of the other one
- Mentioned earlier that we will focus on **linear** relationships
- A **linear** relationship is a relationship with a constant rate of change (**slope**)

$$y = 1.2 + 0.5x$$

intercept

slope

$$x = -3$$
$$y = 1.2 + 0.5 \times -3$$
$$= 1.2 - 1.5 = -0.3$$



Least-Squares Regression

Equation of a line (refresher)

- We want to model one quantitative variable as a function of the other one
- Mentioned earlier that we will focus on **linear** relationships
- A **linear** relationship is a relationship with a constant rate of change (**slope**)

$$y = 1.2 + 0.5x$$

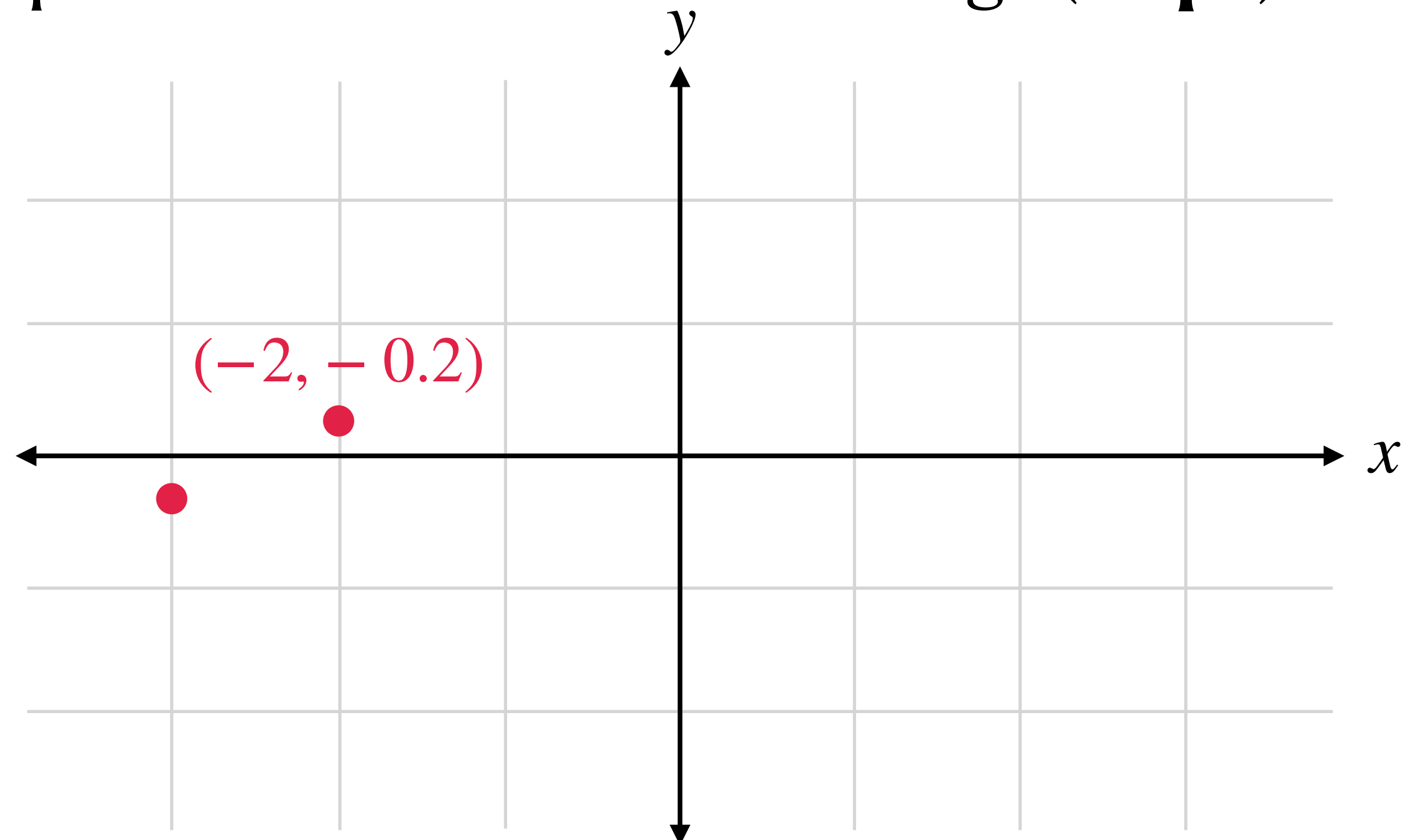
slope

$$x = -2$$

$$y = 1.2 + 0.5 \times -2$$

$$= 1.2 - 1 = 0.2$$

intercept



Least-Squares Regression

Equation of a line (refresher)

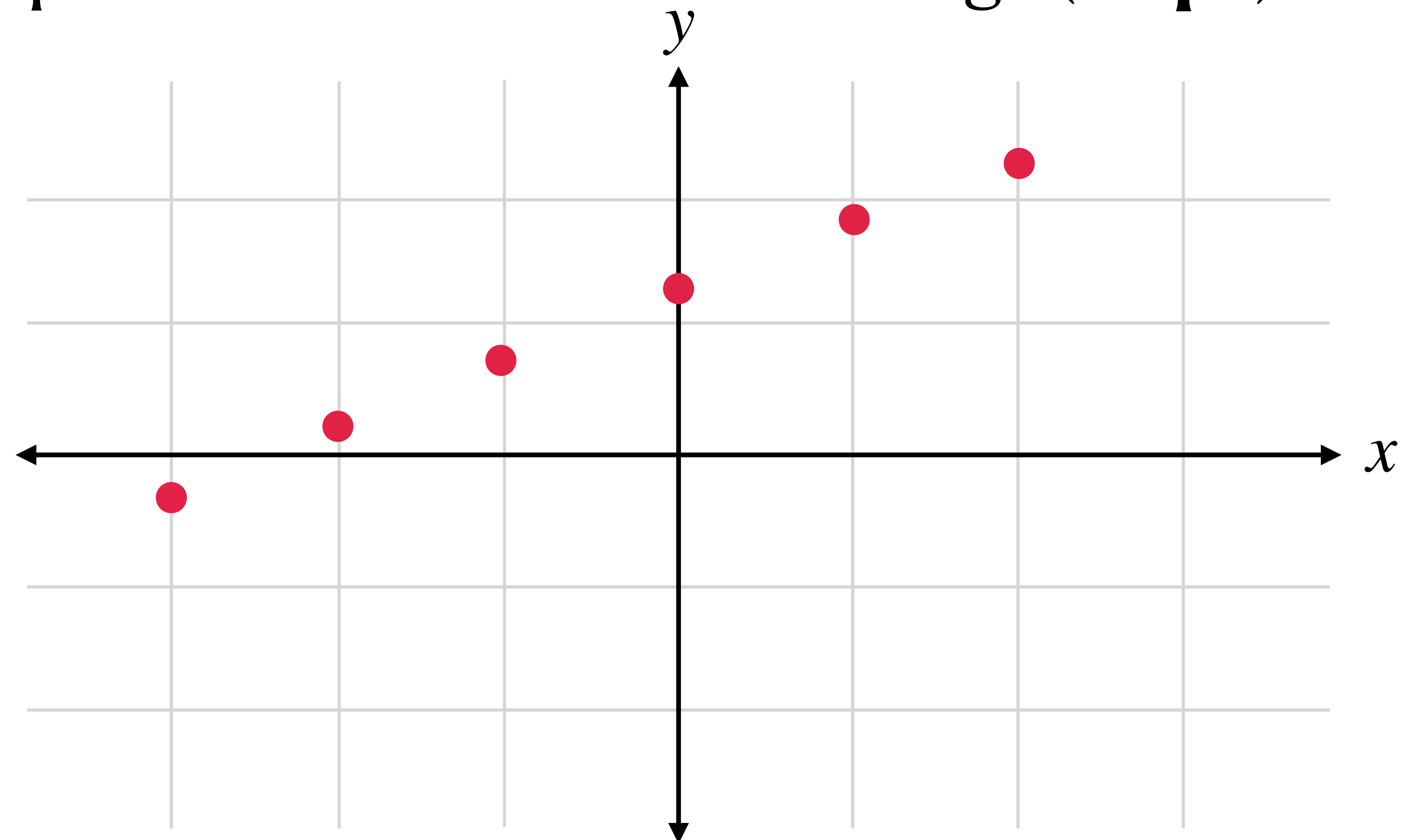
- We want to model one quantitative variable as a function of the other one
 - Mentioned earlier that we will focus on **linear** relationships
 - A **linear** relationship is a relationship with a constant rate of change (**slope**)

$$y = 1.2 + 0.5x$$

The equation is annotated with a purple box around 1.2 and an orange box around 0.5. A purple arrow points from the word "intercept" to the 1.2 box. An orange arrow points from the word "slope" to the 0.5 box.

$(-3, -0.3)$
 $(-2, -0.2)$

And so on...



Least-Squares Regression

Equation of a line (refresher)

- What is the value of y when $x = 0$?
- PollEv.com/stats8

$$y = 1.2 + 0.5x$$

intercept

slope

The diagram shows the equation $y = 1.2 + 0.5x$. The number 1.2 is enclosed in a light purple box, and the number 0.5 is enclosed in a light orange box. A purple curved arrow points from the word 'intercept' to the purple box. An orange curved arrow points from the word 'slope' to the orange box.

Least-Squares Regression

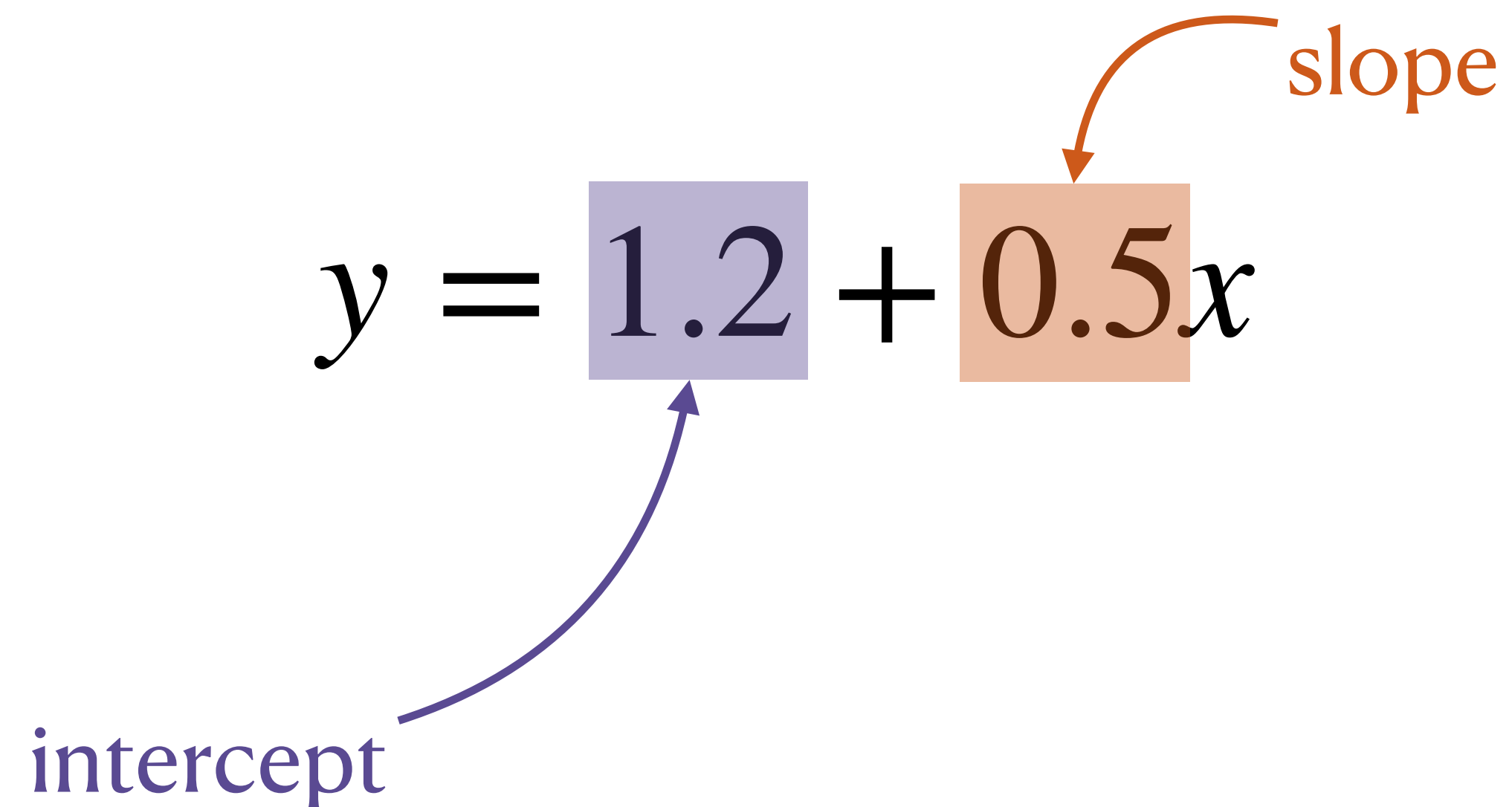
Equation of a line (refresher)

- Suppose we take two points on the line that are one unit apart in x . How far apart are they in y ?
- PollEv.com/stats8

$$y = 1.2 + 0.5x$$

intercept

slope



Least-Squares Regression

Equation of a line (refresher)

- Suppose we take two points on the line that are one unit apart in x . How far apart are they in y ?
- Say one point has an x -value of “ c ”. The other point has x -value “ $c + 1$ ”.

$x = c$	$x = c + 1$
$y_1 = 1.2 + 0.5 \times c$	$y_2 = 1.2 + 0.5 \times (c + 1)$

Least-Squares Regression

Equation of a line (refresher)

- Suppose we take two points on the line that are one unit apart in x . How far apart are they in y ?
- Say one point has an x -value of “ c ”. The other point has x -value “ $c + 1$ ”.

$x = c$	$x = c + 1$
$y_1 = 1.2 + 0.5 \times c$	$y_2 = 1.2 + 0.5 \times (c + 1)$ $= 1.2 + 0.5 \times c + 0.5$

Least-Squares Regression

Equation of a line (refresher)

- Suppose we take two points on the line that are one unit apart in x . How far apart are they in y ?
- Say one point has an x -value of “ c ”. The other point has x -value “ $c + 1$ ”.

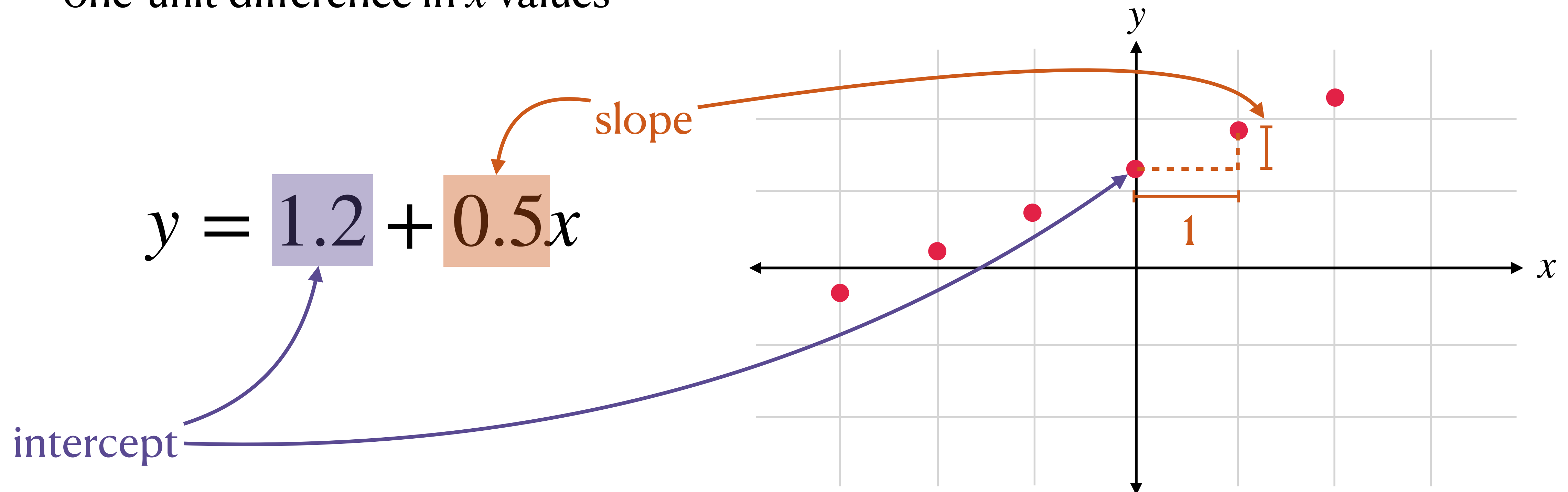
$x = c$	$x = c + 1$
$y_1 = 1.2 + 0.5 \times c$	$y_2 = 1.2 + 0.5 \times (c + 1)$ $= 1.2 + 0.5 \times c + 0.5$

- So, two points on the line that are **1** unit apart in x are **0.5** units apart in y
- This is the slope of the line!

Least-Squares Regression

Equation of a line (refresher)

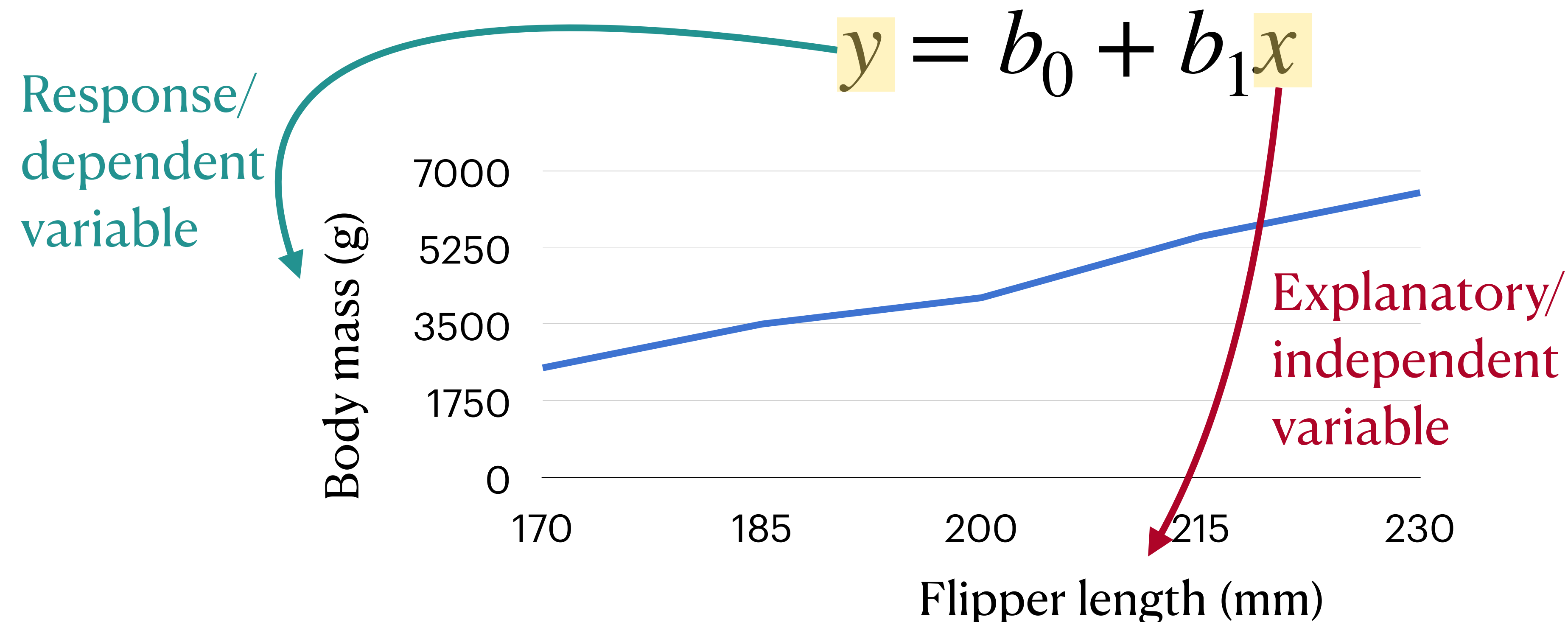
- So the **intercept** of the line represents the y where $x = 0$
- And the **slope** of the line represents the difference in y values corresponding to a one-unit difference in x values



Least-Squares Regression

Quantitative + Quantitative

- Back to statistics...
 - Model the relationship between explanatory and response as a line:

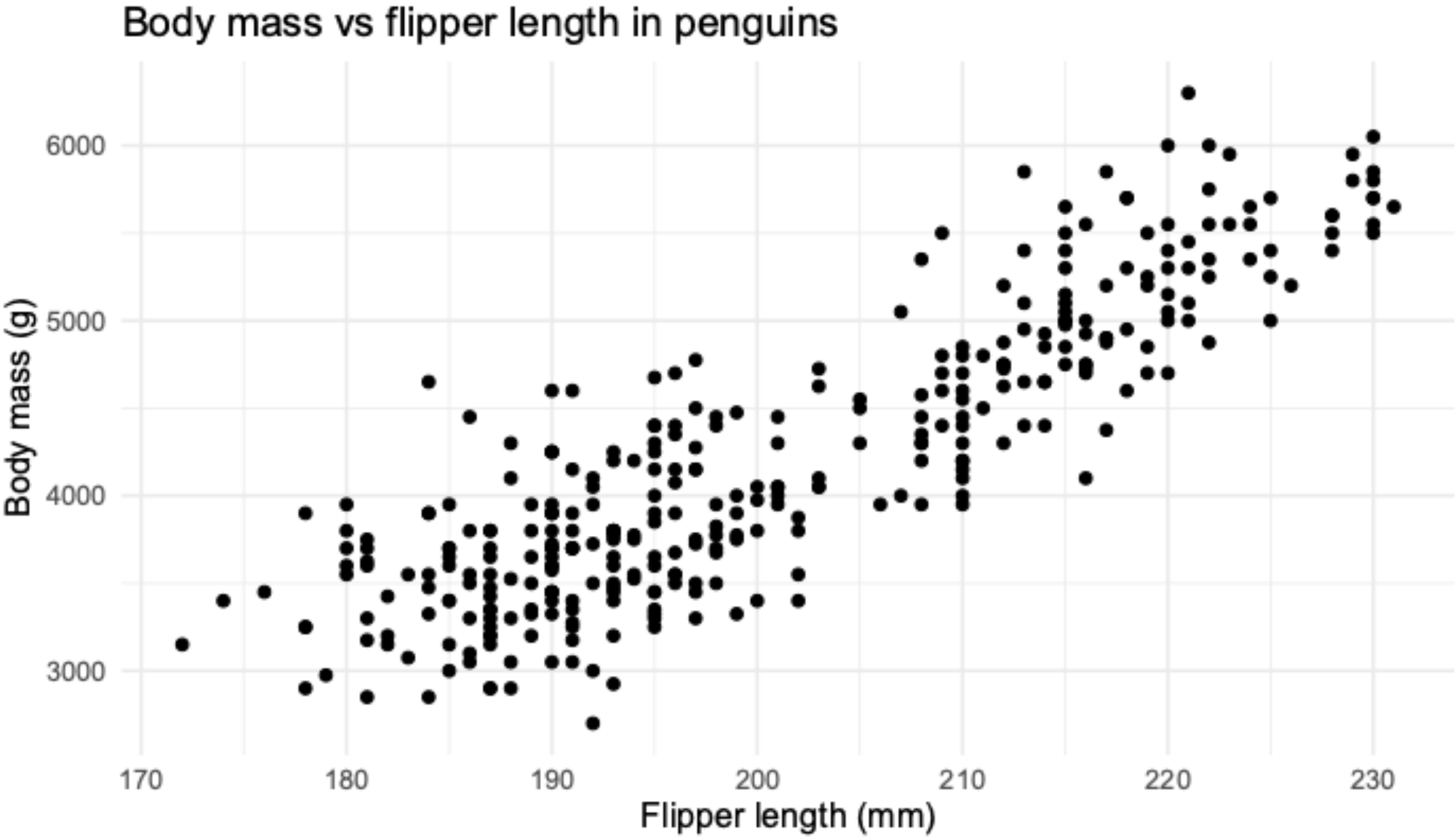


Least-Squares Regression

Quantitative + Quantitative

- Model the relationship between explanatory and response as a line:

$$y = b_0 + b_1x$$

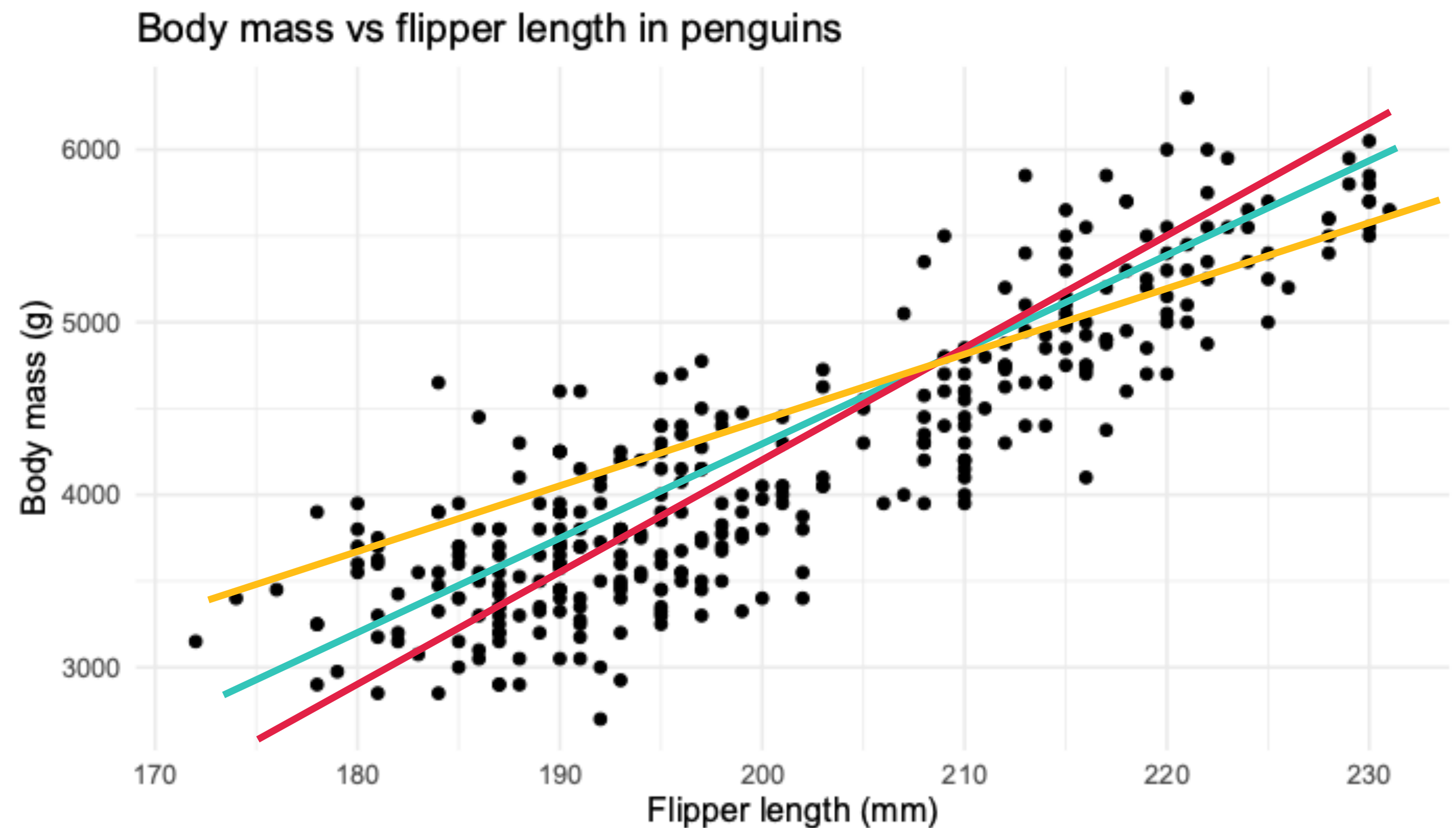


Least-Squares Regression

Quantitative + Quantitative

- Great, but how do we choose a specific line?

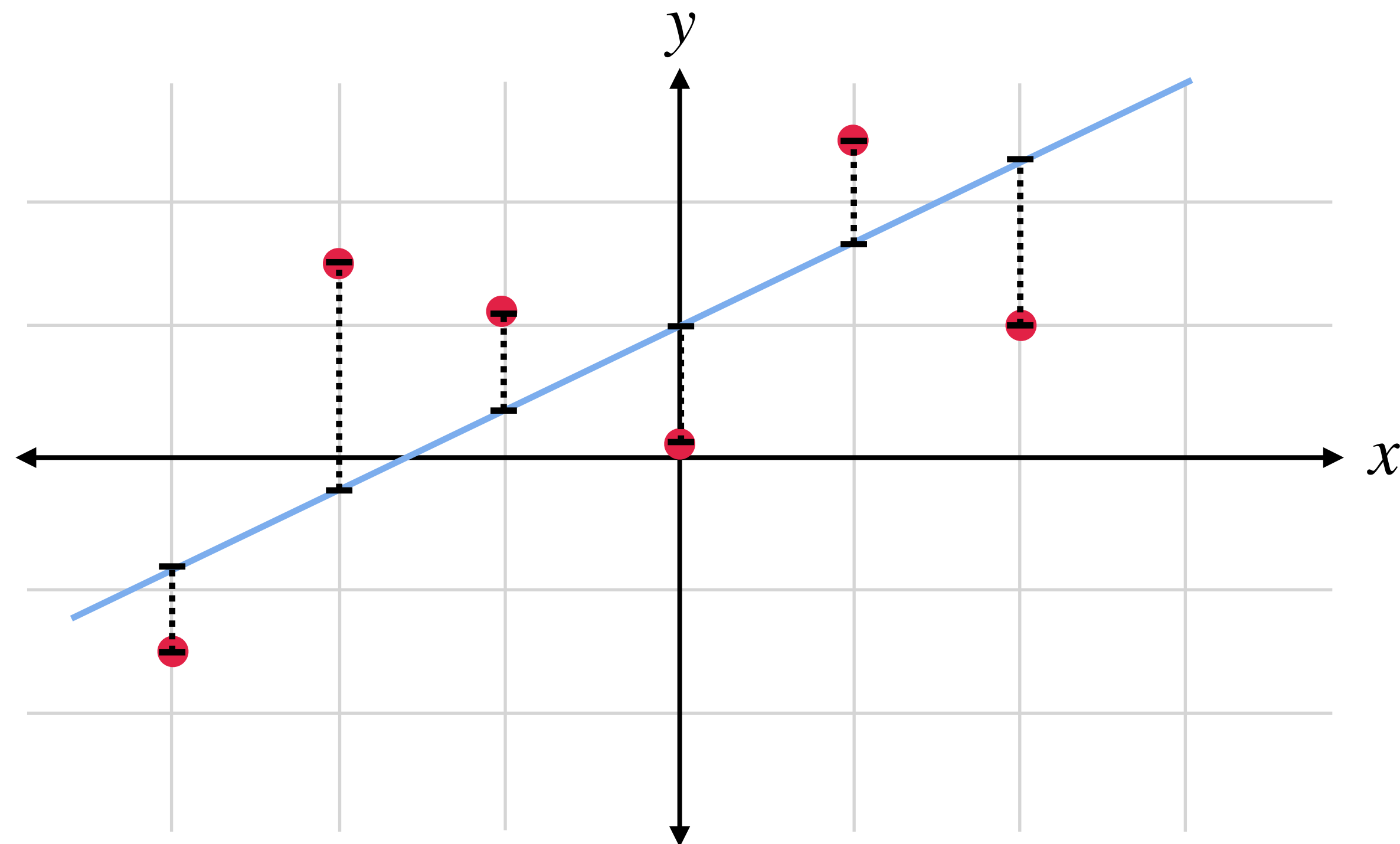
$$y = b_0 + b_1x$$



Least-Squares Regression

Quantitative + Quantitative

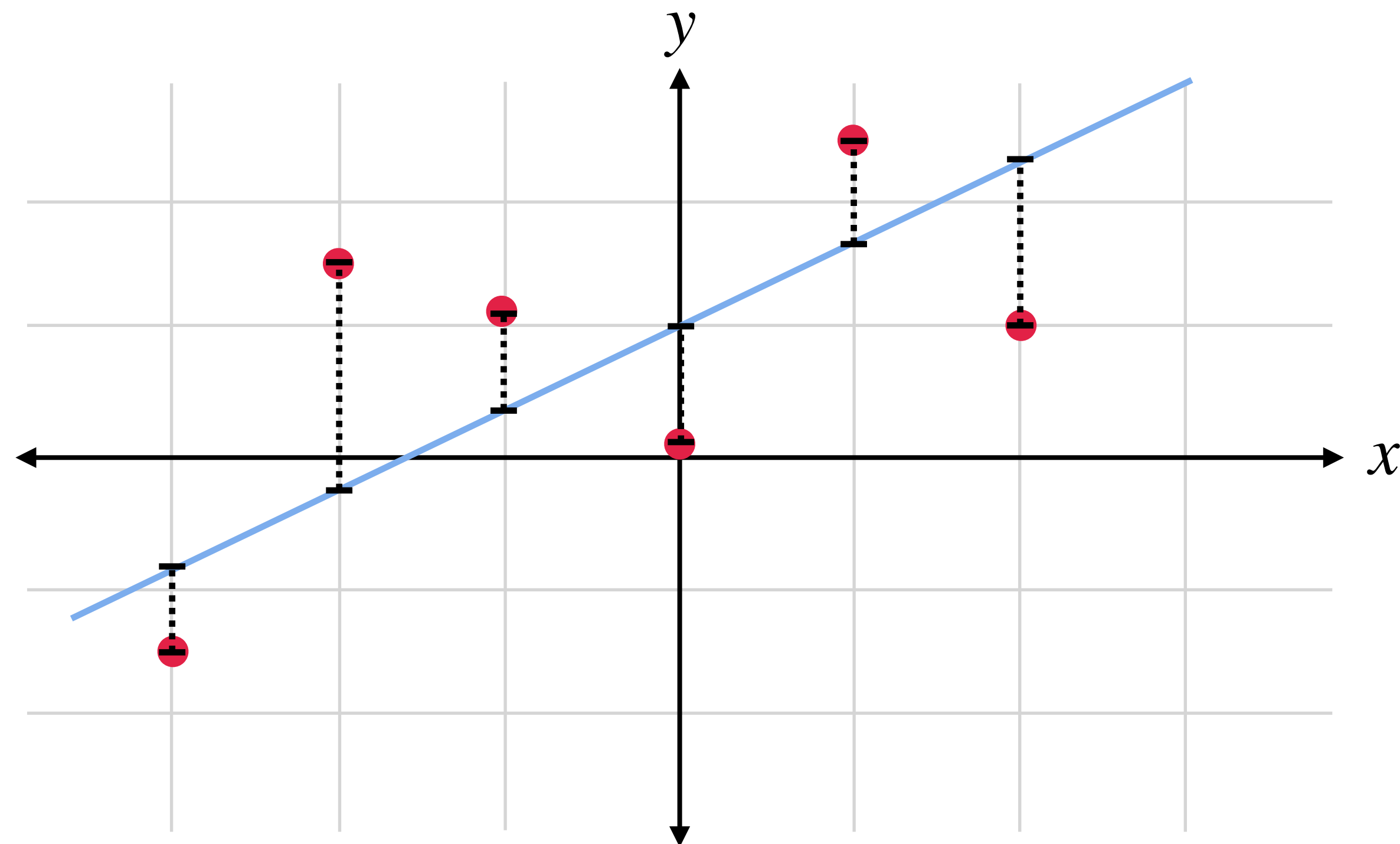
- This is why it's called “least-squares” regression:
 - Find the line that minimizes the sum of squared distances between each point and the line



Least-Squares Regression

Quantitative + Quantitative

- Find the line that minimizes the sum of squared distances between each point and the line
- These distances are called **residuals**

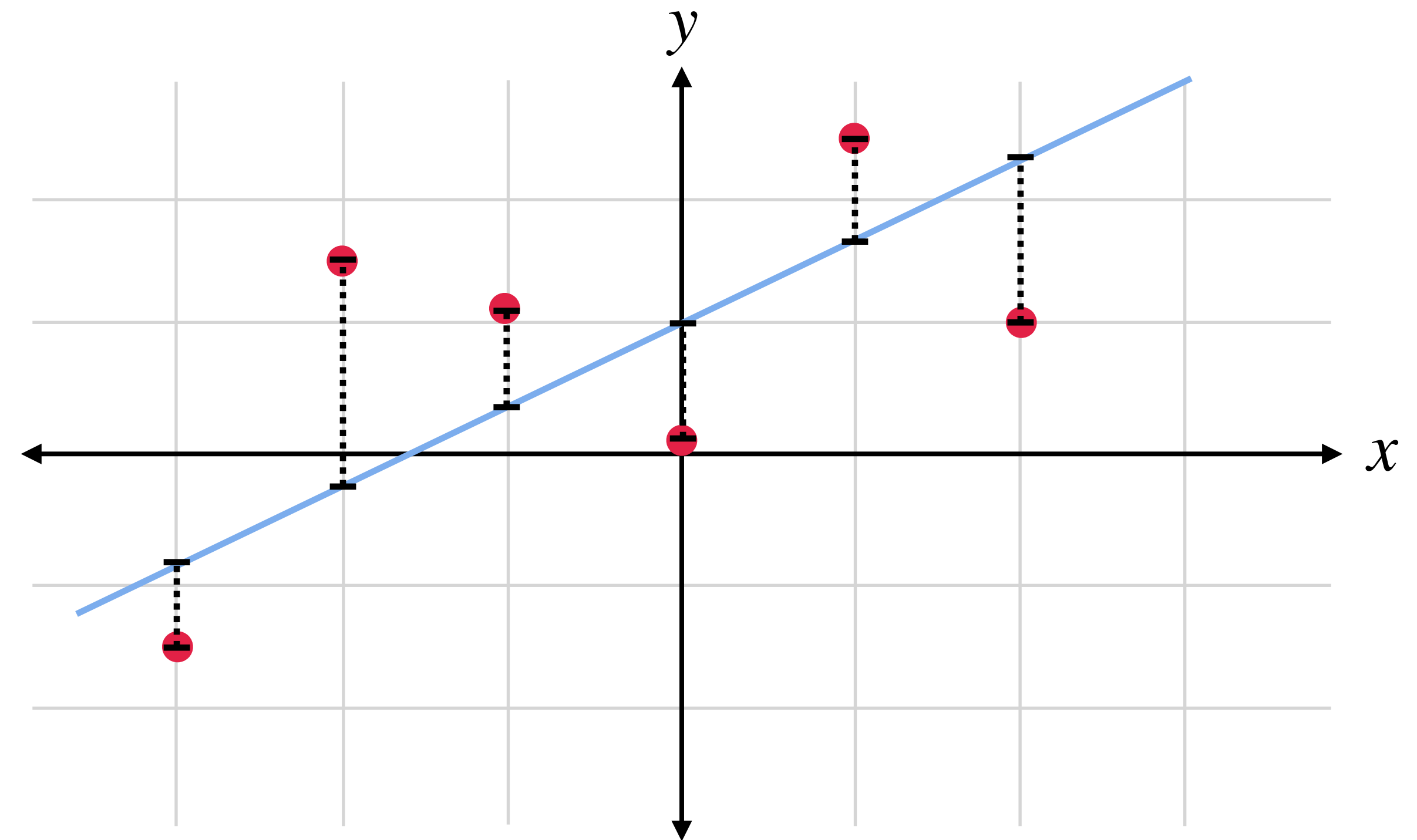


*** You don't need to be able to compute this yourself (computers can do that), but you should remember that we obtain the least-squares line by finding the **line that minimizes the sum of squared residuals**

Least-Squares Regression

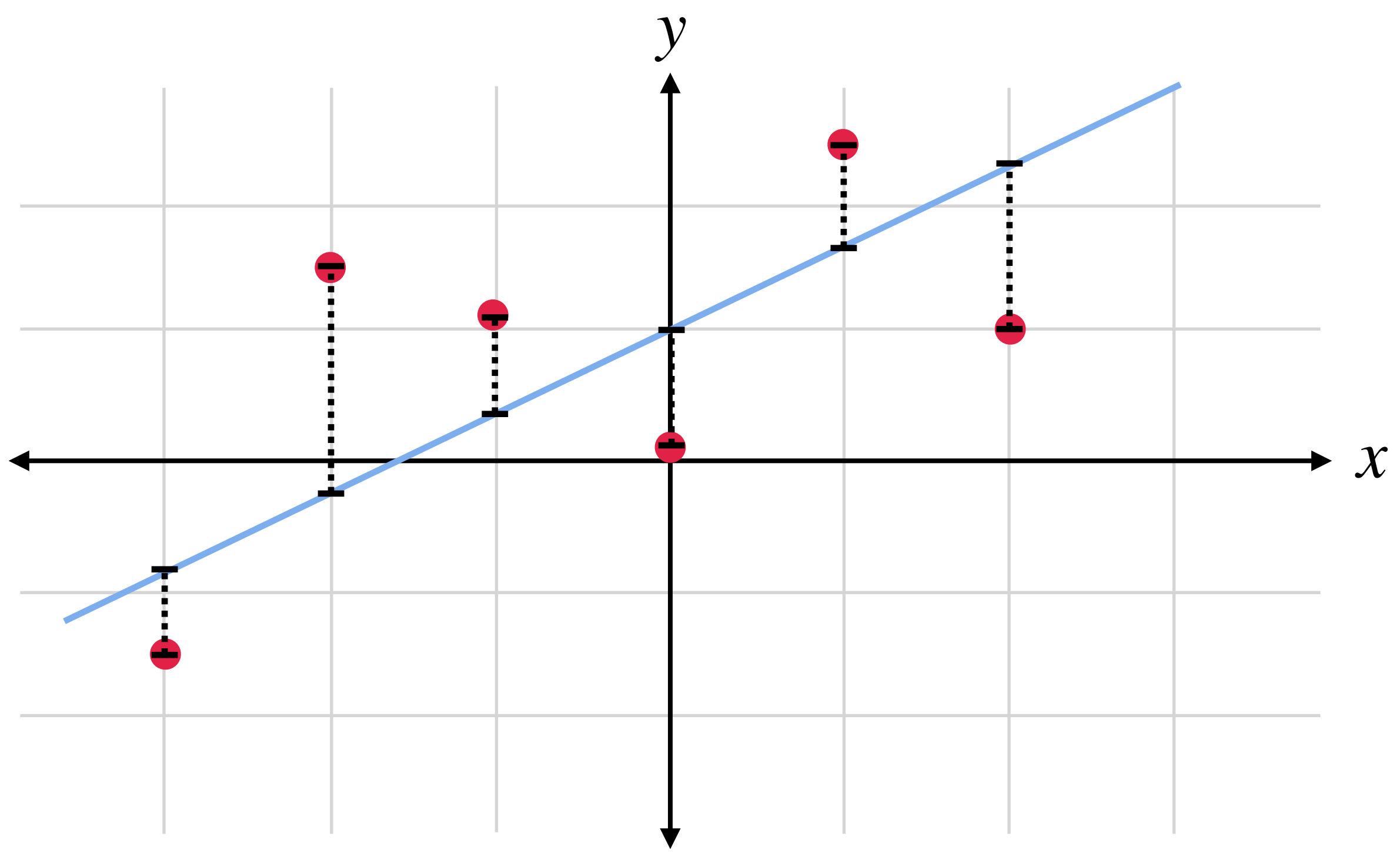
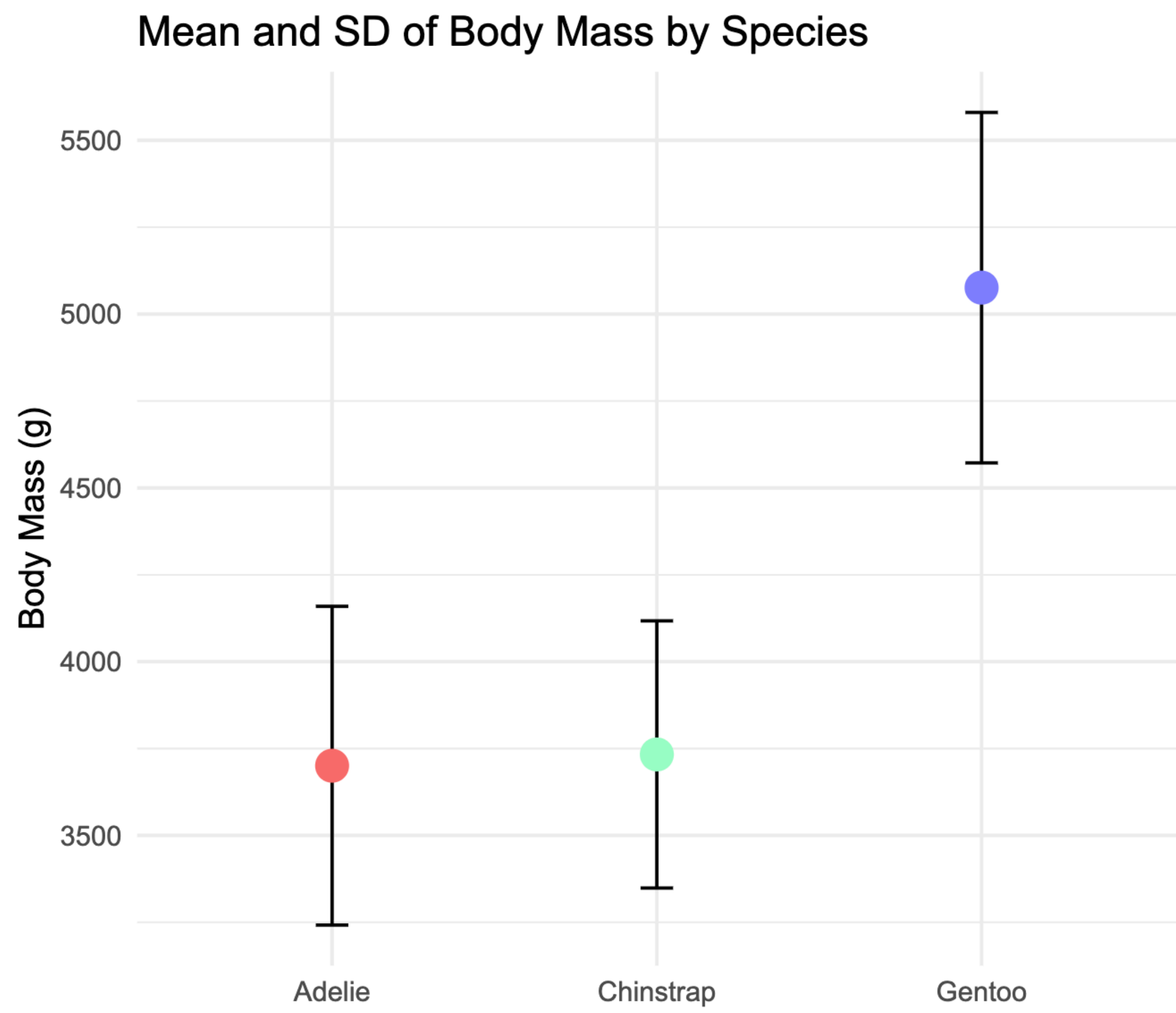
Intuition: Categorical + Quantitative vs. Quantitative + Quantitative

- With categorical + quantitative, we compared means between categories of the categorical variable
- With quantitative + quantitative, we are assuming that the **mean** of one quantitative variable changes **linearly** with the other quantitative variable



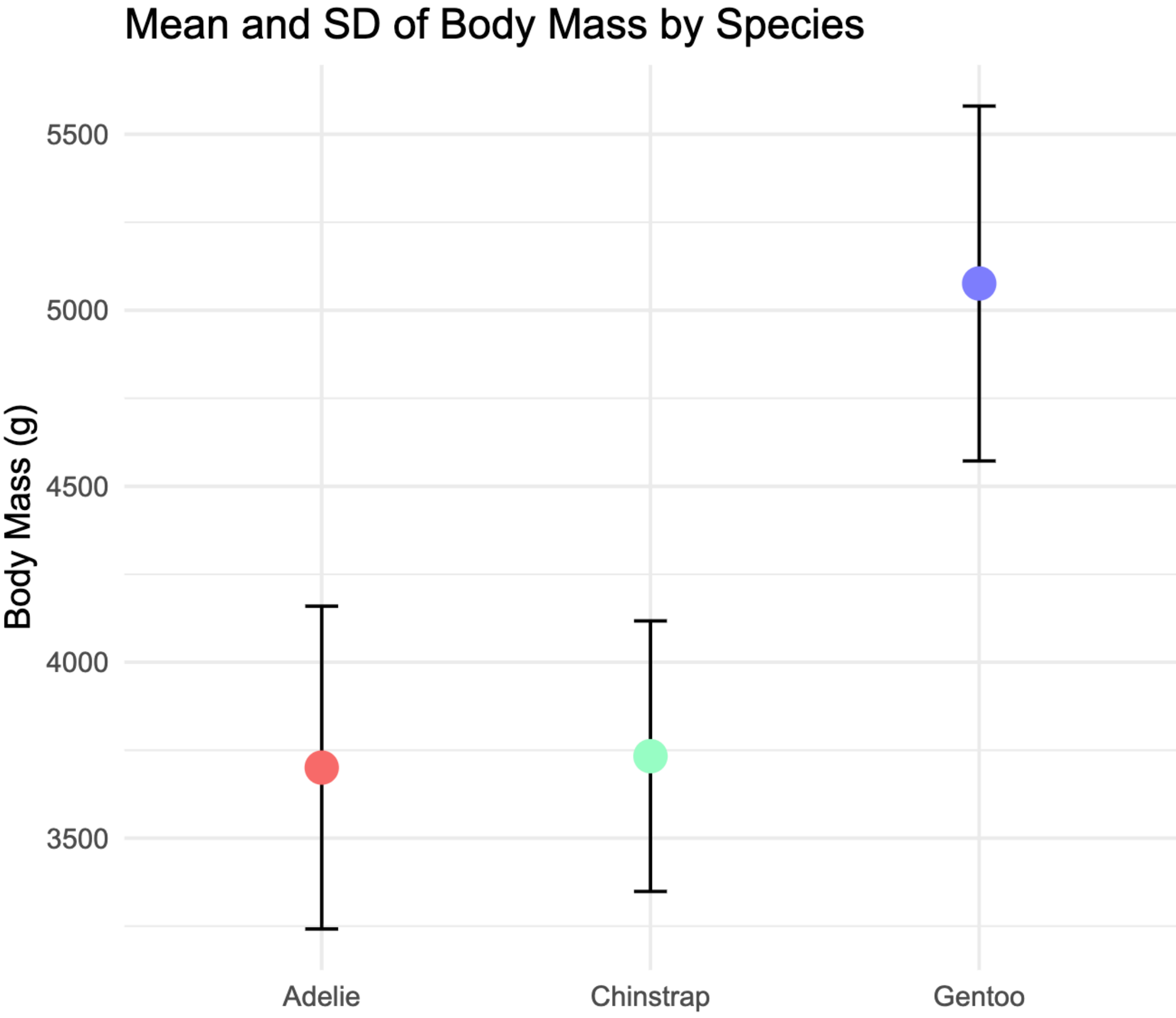
Least-Squares Regression

Intuition: Categorical + Quantitative vs. Quantitative + Quantitative



Least-Squares Regression

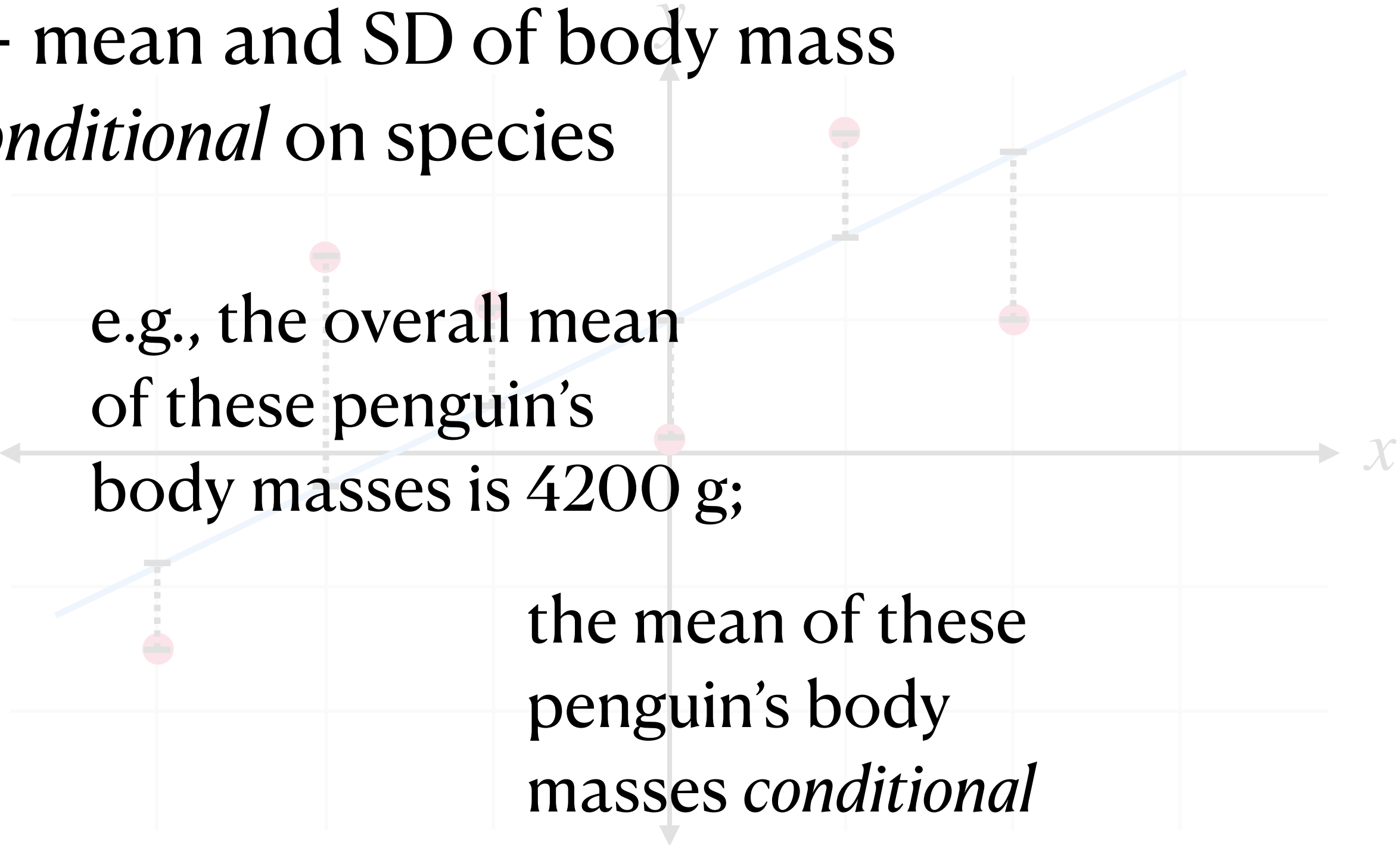
Some intuition



← mean and SD of body mass
conditional on species

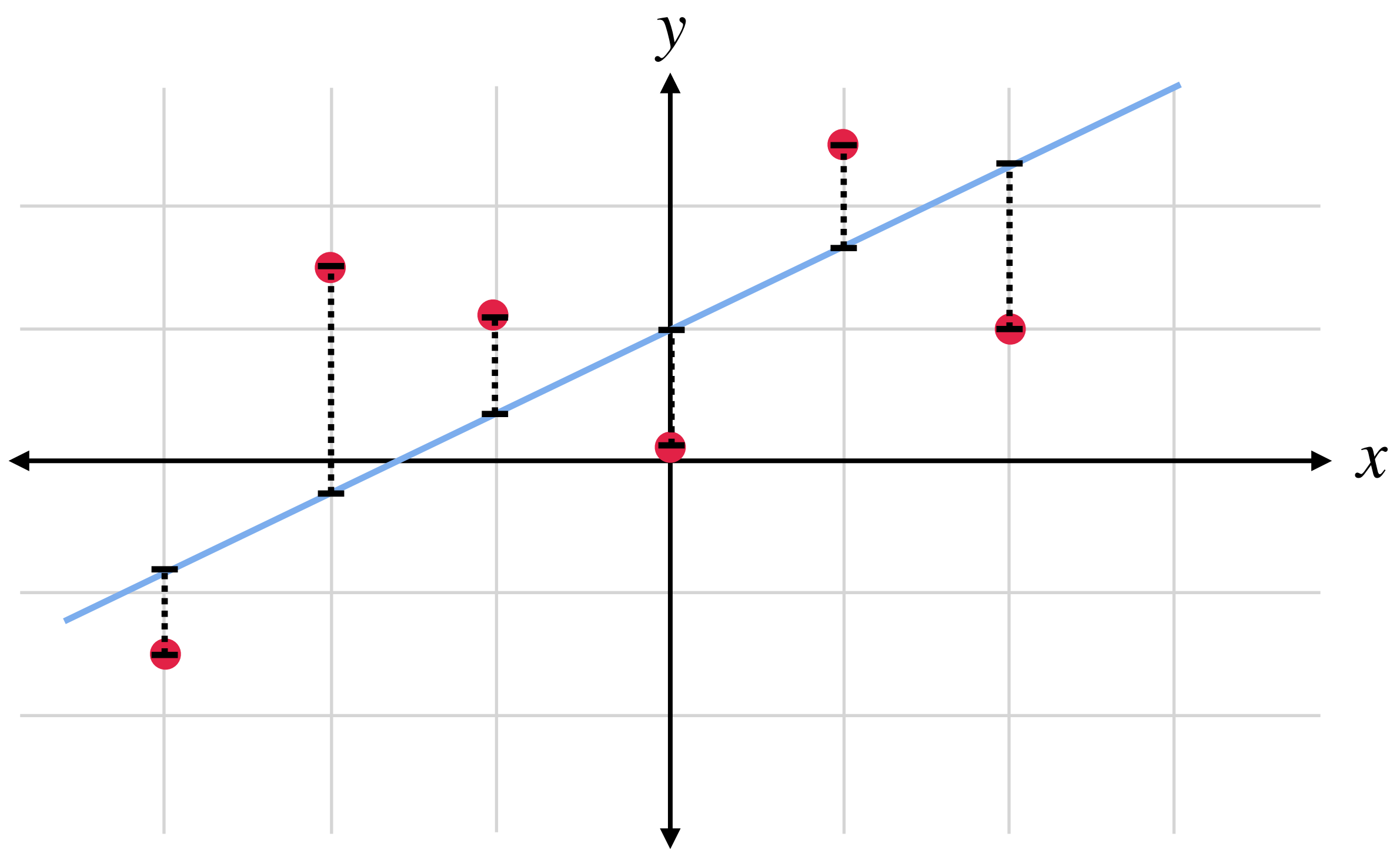
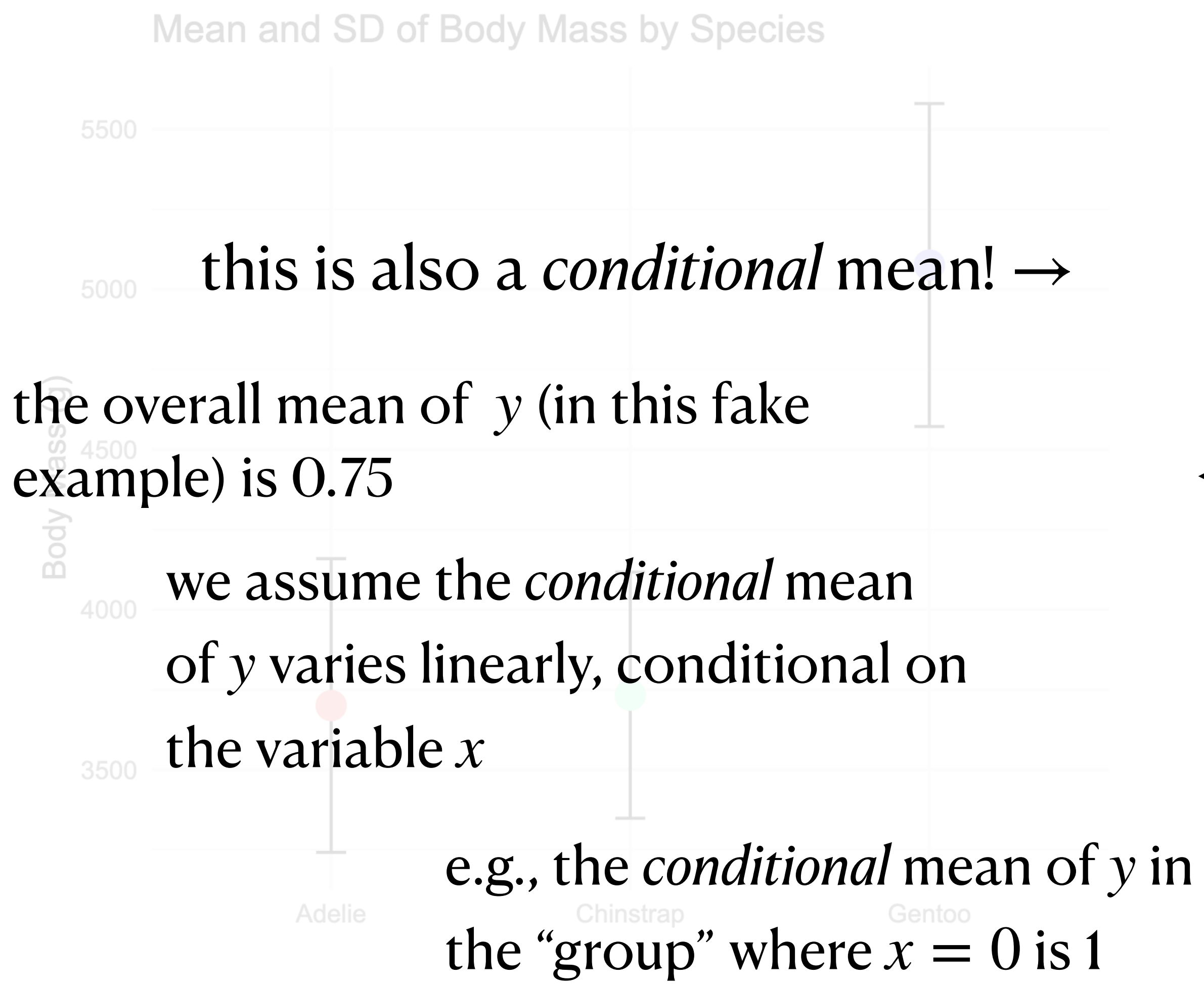
e.g., the overall mean
of these penguin's
body masses is 4200 g;

the mean of these
penguin's body
masses *conditional*
on being Gentoo is
5100 g



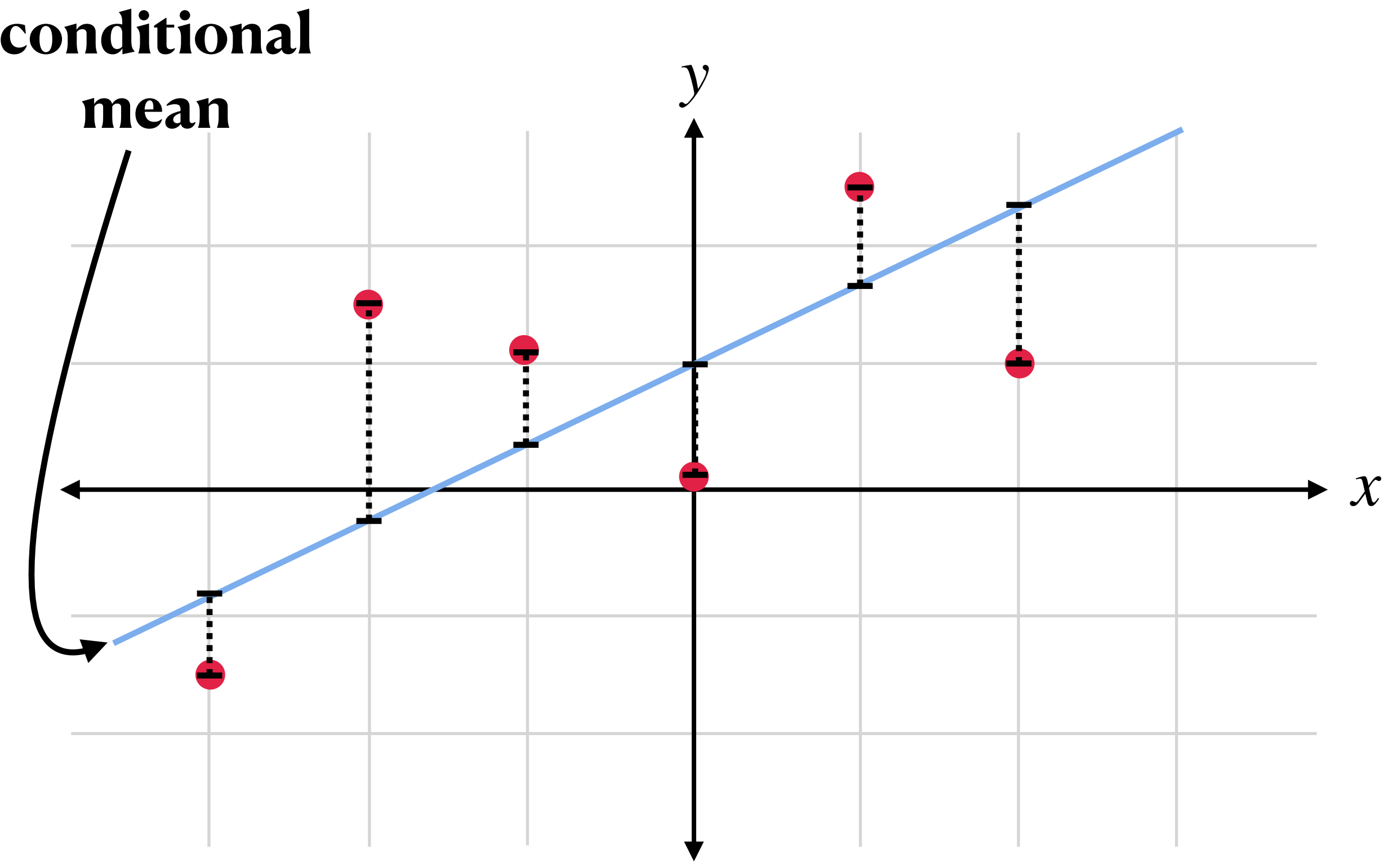
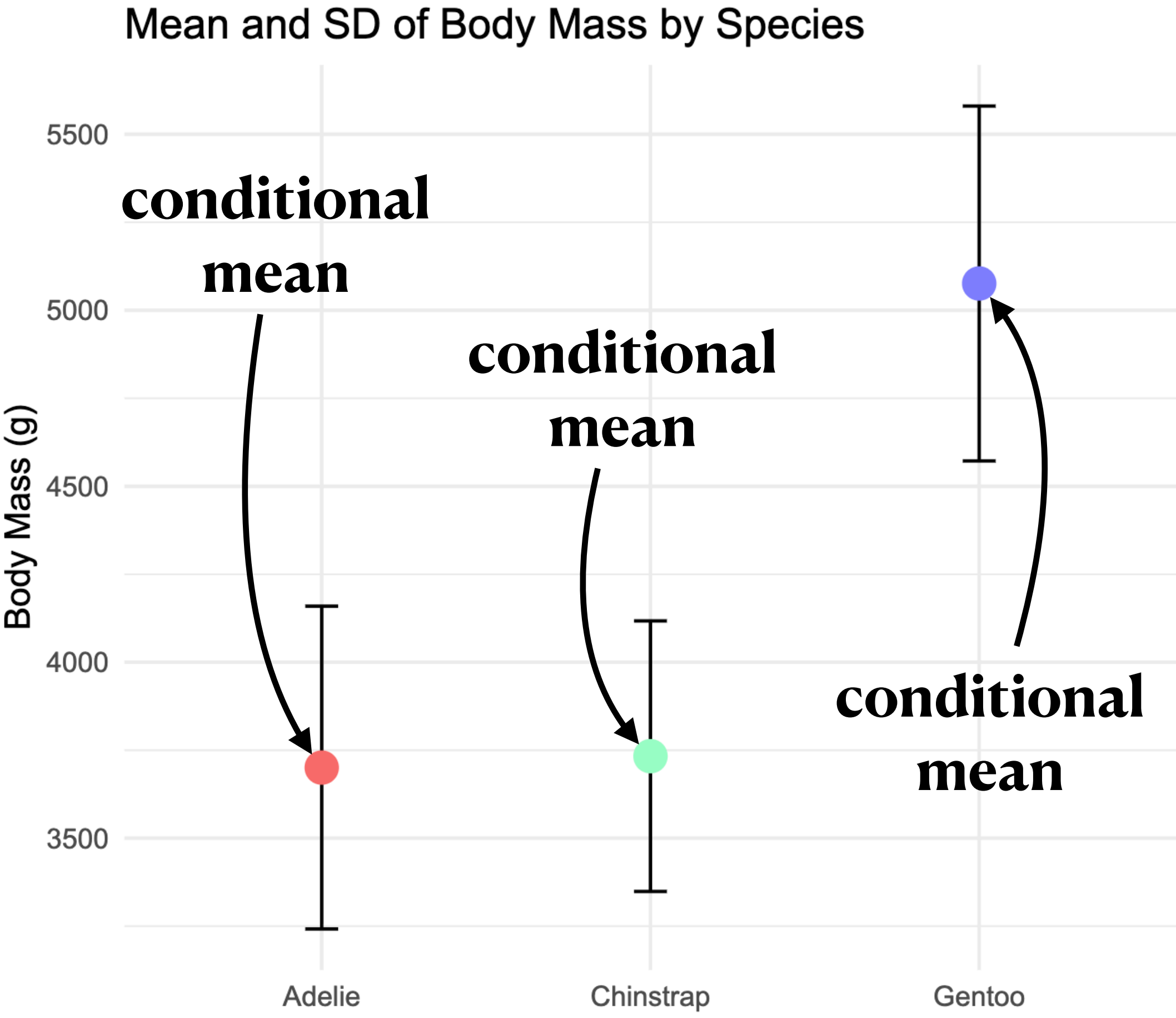
Least-Squares Regression

Some intuition



Least-Squares Regression

Some intuition



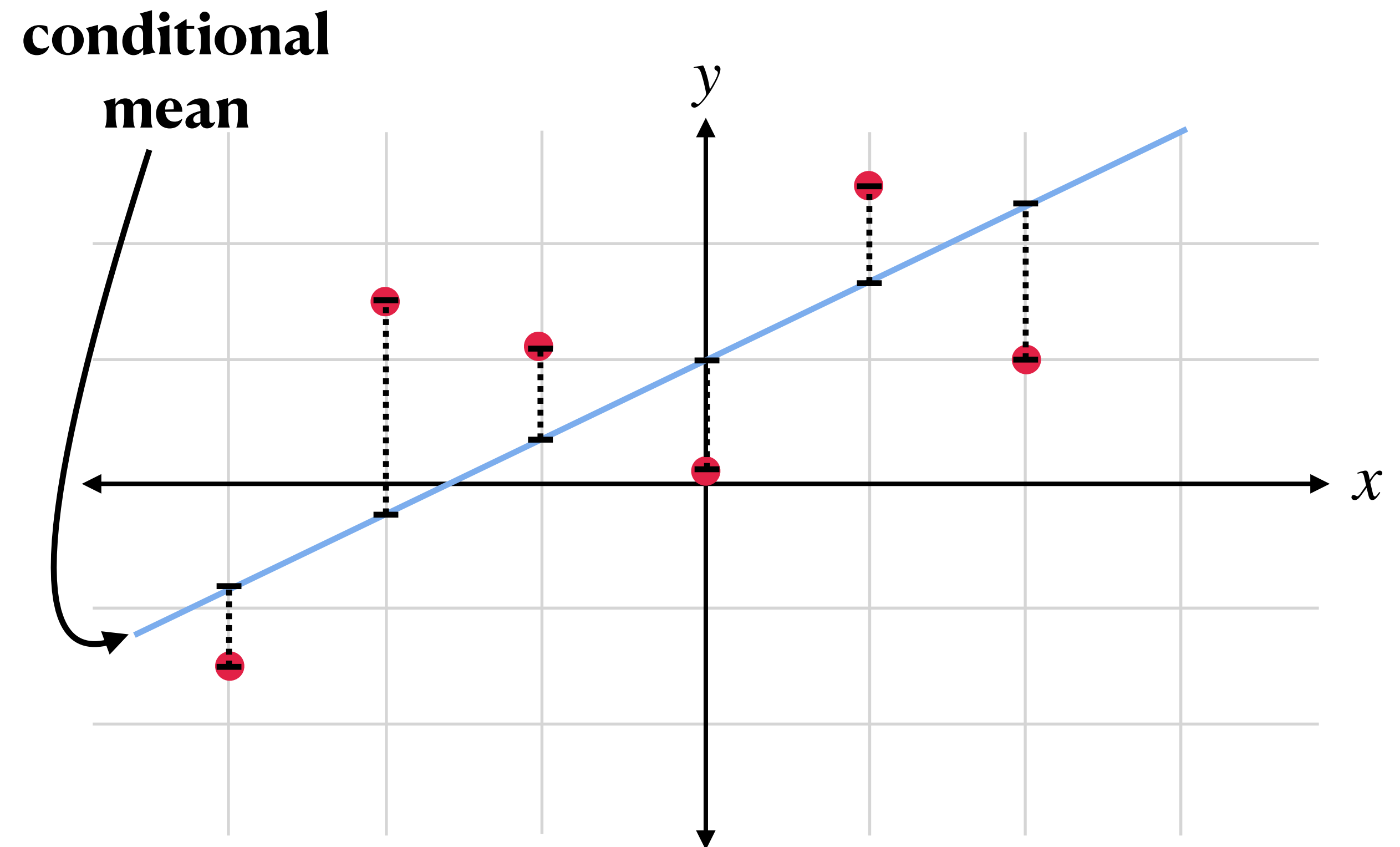
Least-Squares Regression

Notation

the observed data is usually denoted y , and the conditional mean is denoted $\hat{y}$ (“y hat”)

$$\hat{y} = b_0 + b_1x$$

residuals are $e = y - \hat{y}$

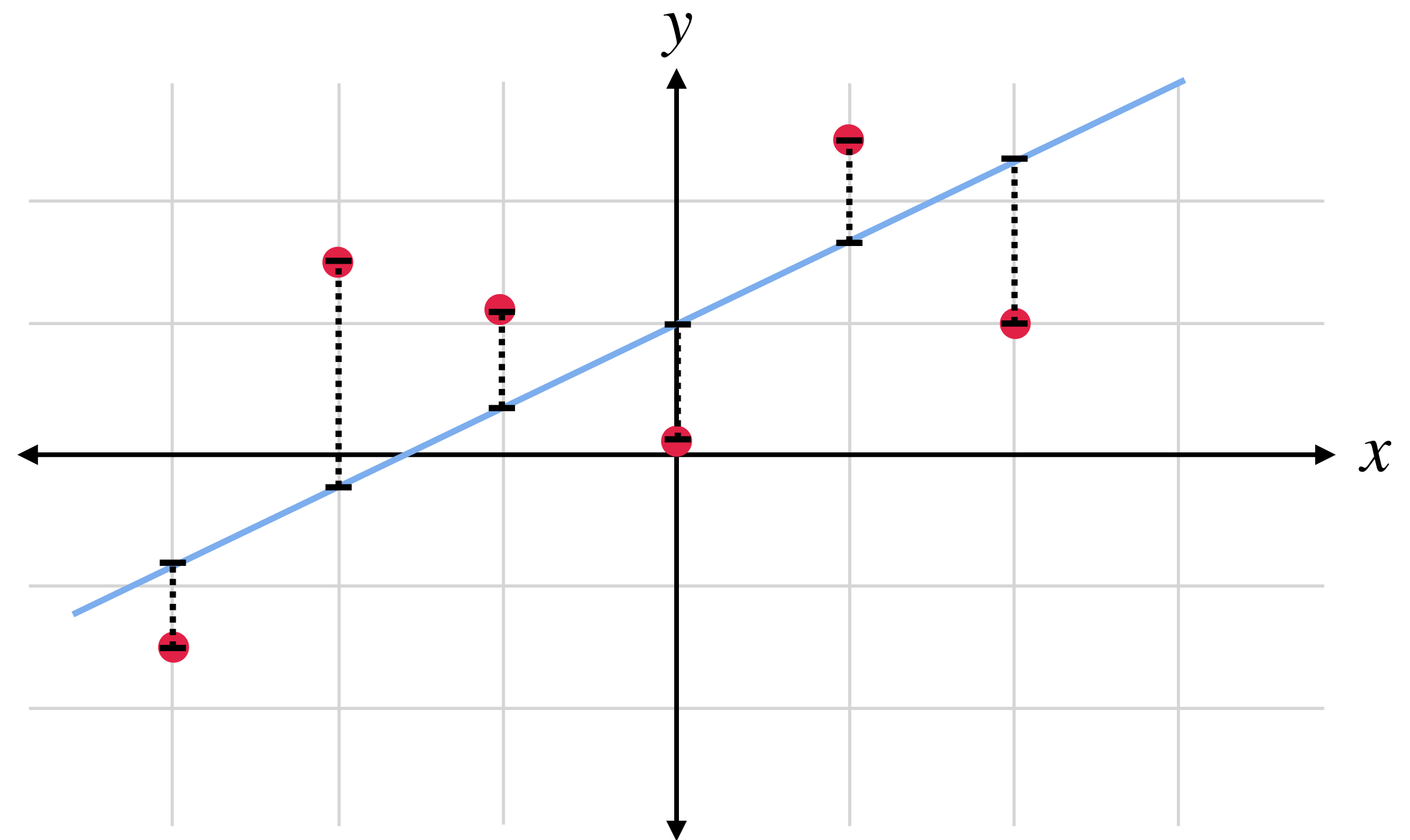


Least-Squares Regression

Interpretation

$$\hat{y} = b_0 + b_1x$$

- Intercept, b_0 , is the conditional mean of y in the group where $x = 0$
- Slope, b_1 , is the difference in conditional means of y for two groups that are 1 unit apart in x



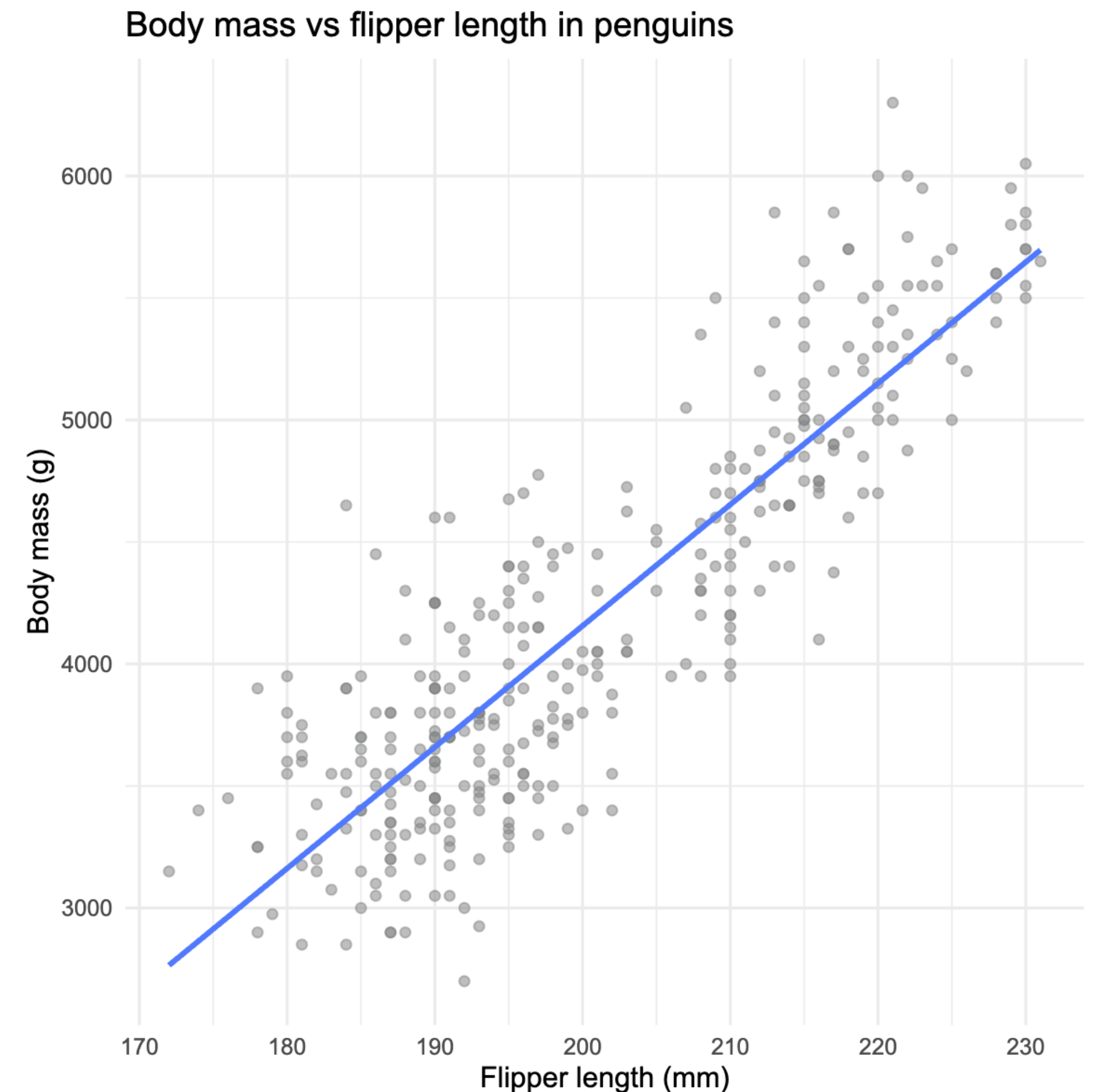
Least-Squares Regression

Interpretation

$$\hat{y} = b_0 + b_1x$$

(avg. body mass) = $-5781 + 49.7$ (flipper length)

Intercept interpretation: The average body mass for penguins with 0 mm flippers is -5781 g.



Least-Squares Regression

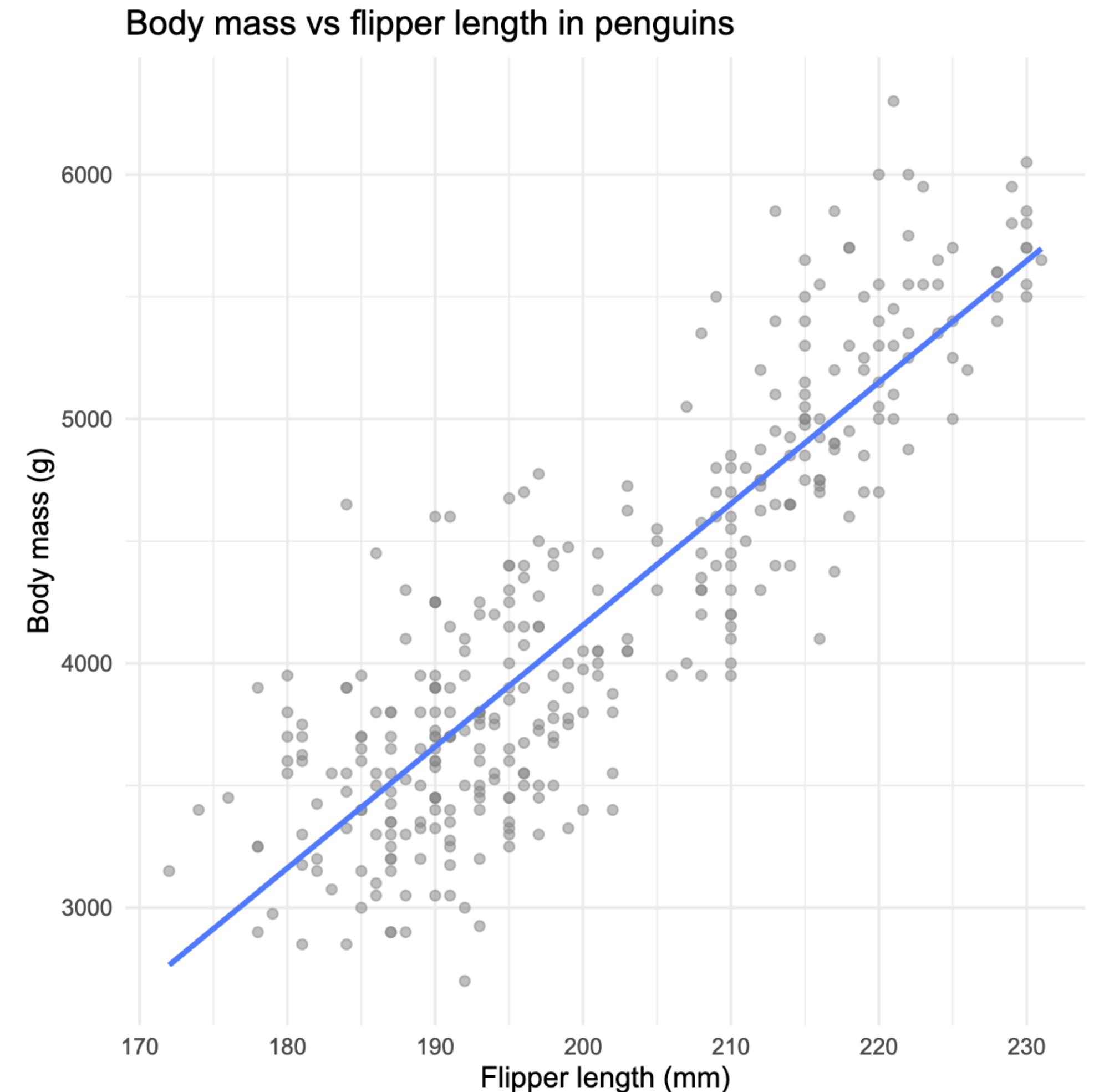
Interpretation

$$\hat{y} = b_0 + b_1x$$

(avg. body mass) = $-5781 + 49.7$ (flipper length)

Intercept interpretation: The average body mass for penguins with 0 mm flippers is -5781 g.

Wait a minute...



Least-Squares Regression

Interpretation

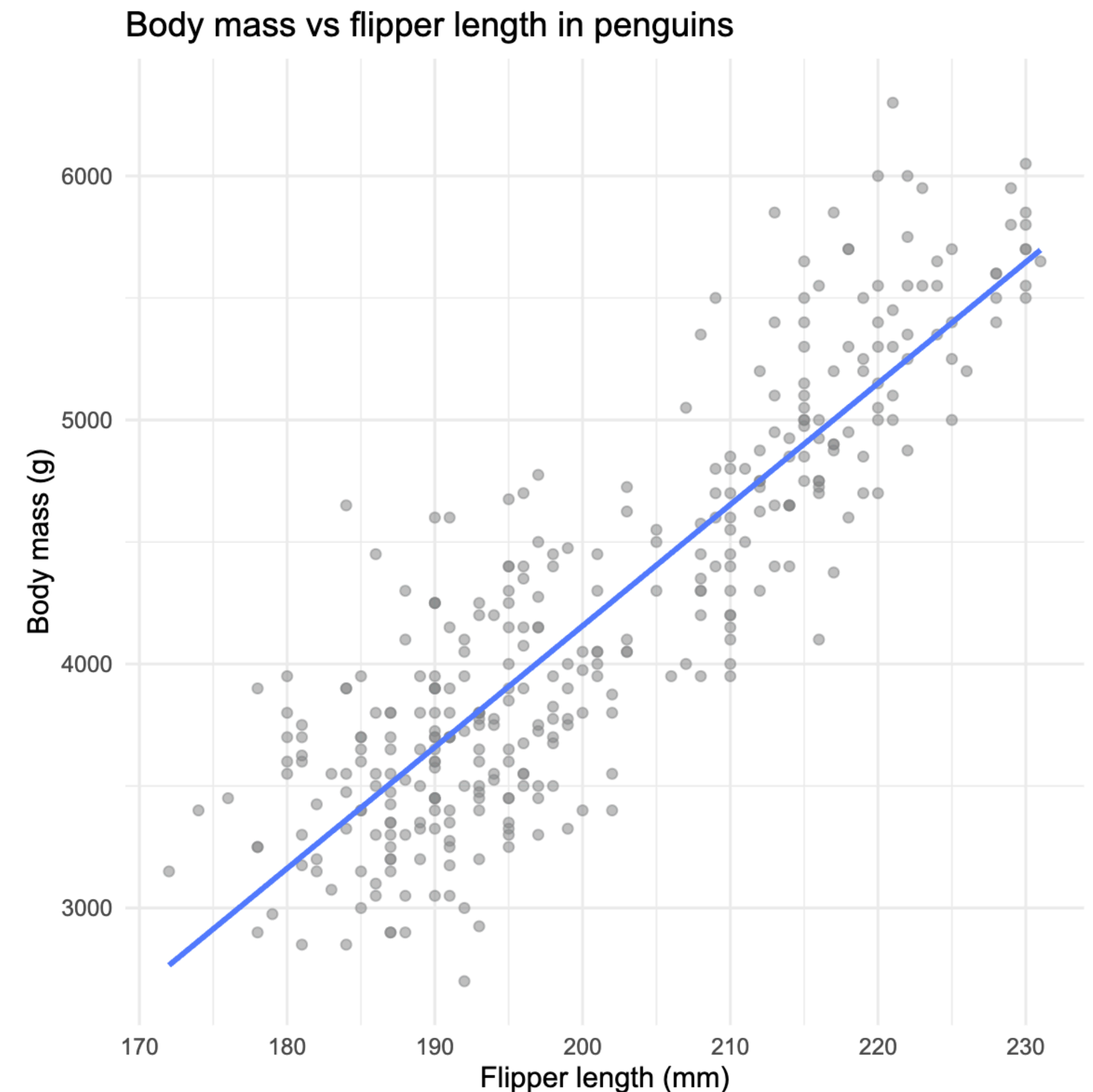
$$\hat{y} = b_0 + b_1x$$

(avg. body mass) = $-5781 + 49.7$ (flipper length)

Intercept interpretation: The average body mass for penguins with 0 mm flippers is -5781 g.

Using the conditional mean outside the range of the data is called **extrapolation** and it is not generally valid

(This means that the intercept is often not practically interpretable!)



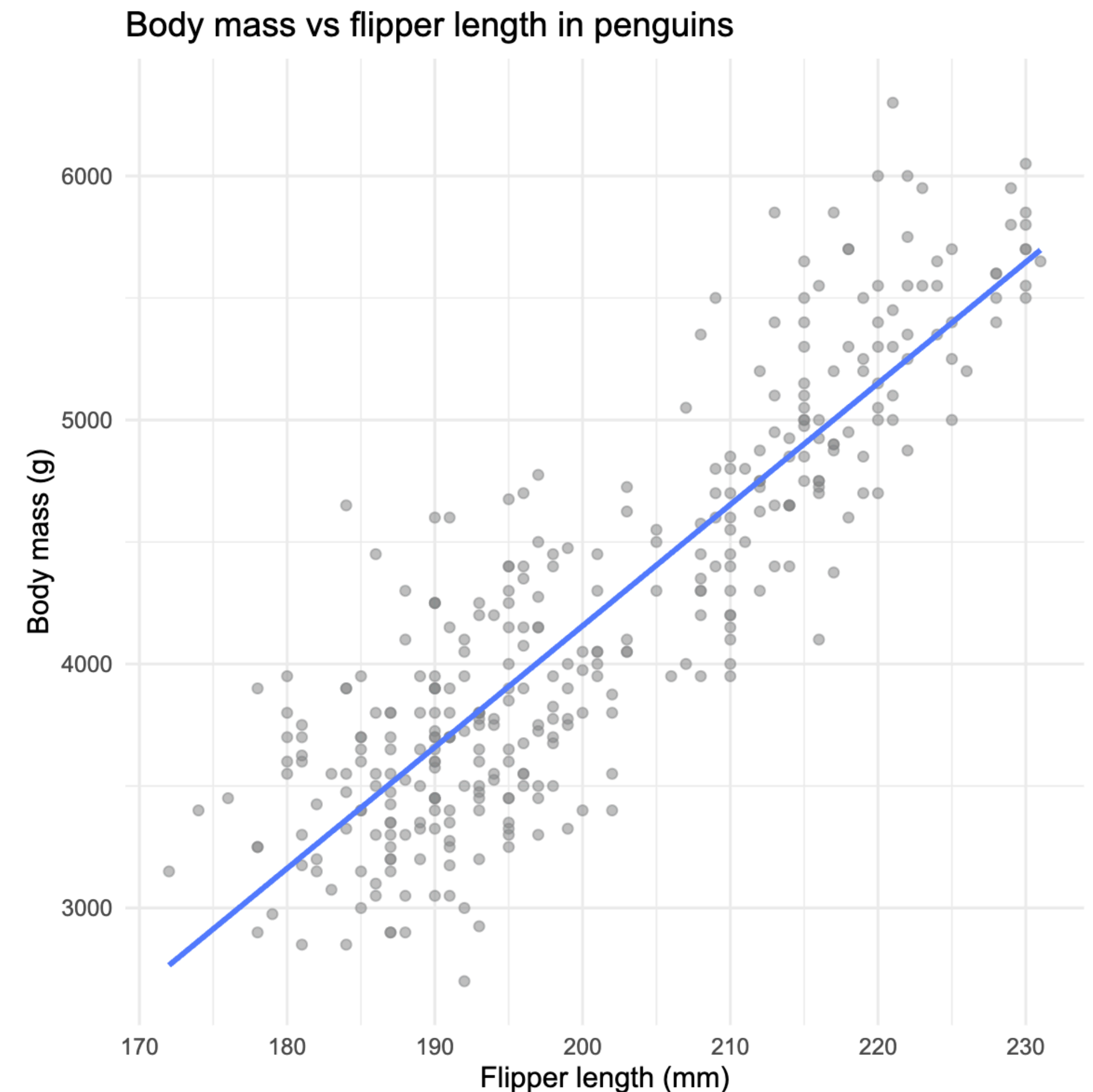
Least-Squares Regression

Interpretation

$$\hat{y} = b_0 + b_1x$$

(avg. body mass) = $-5781 + 49.7$ (flipper length)

Slope interpretation: Comparing two groups of penguins that differ by 1 mm in flipper length, the average body mass of the group with longer flippers is 49.7 g greater.



Least-Squares Regression

Interpretation

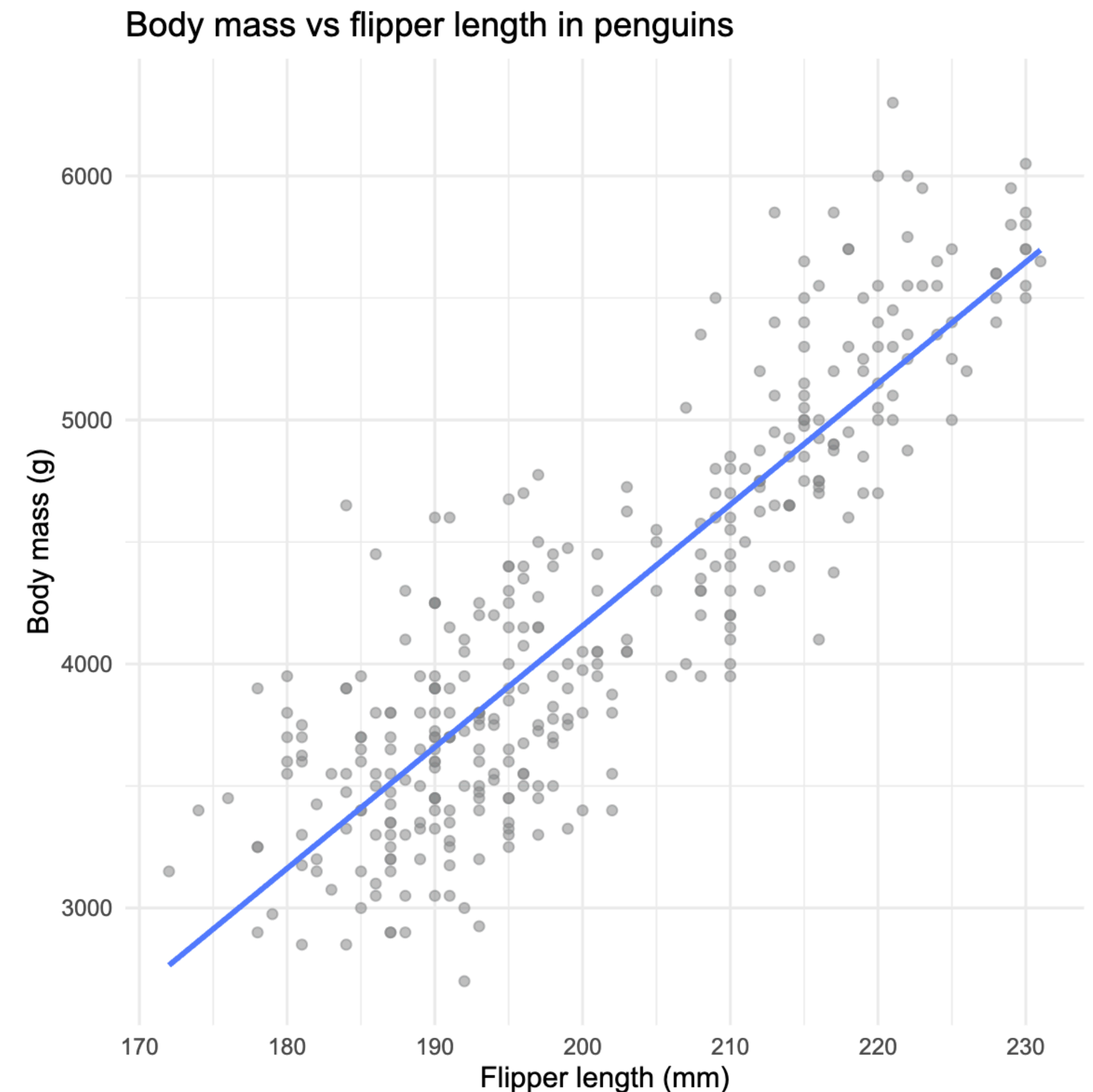
$$\hat{y} = b_0 + b_1x$$

(avg. body mass) = $-5781 + 49.7$ (flipper length)

A common alternative slope interpretation:

For every 1 mm increase in flipper length in penguins, the body mass increases by 49.7 g on average.

💡 This is one of the things I do differently from the textbook! 💡



Least-Squares Regression

Interpretation

$$\hat{y} = b_0 + b_1x$$

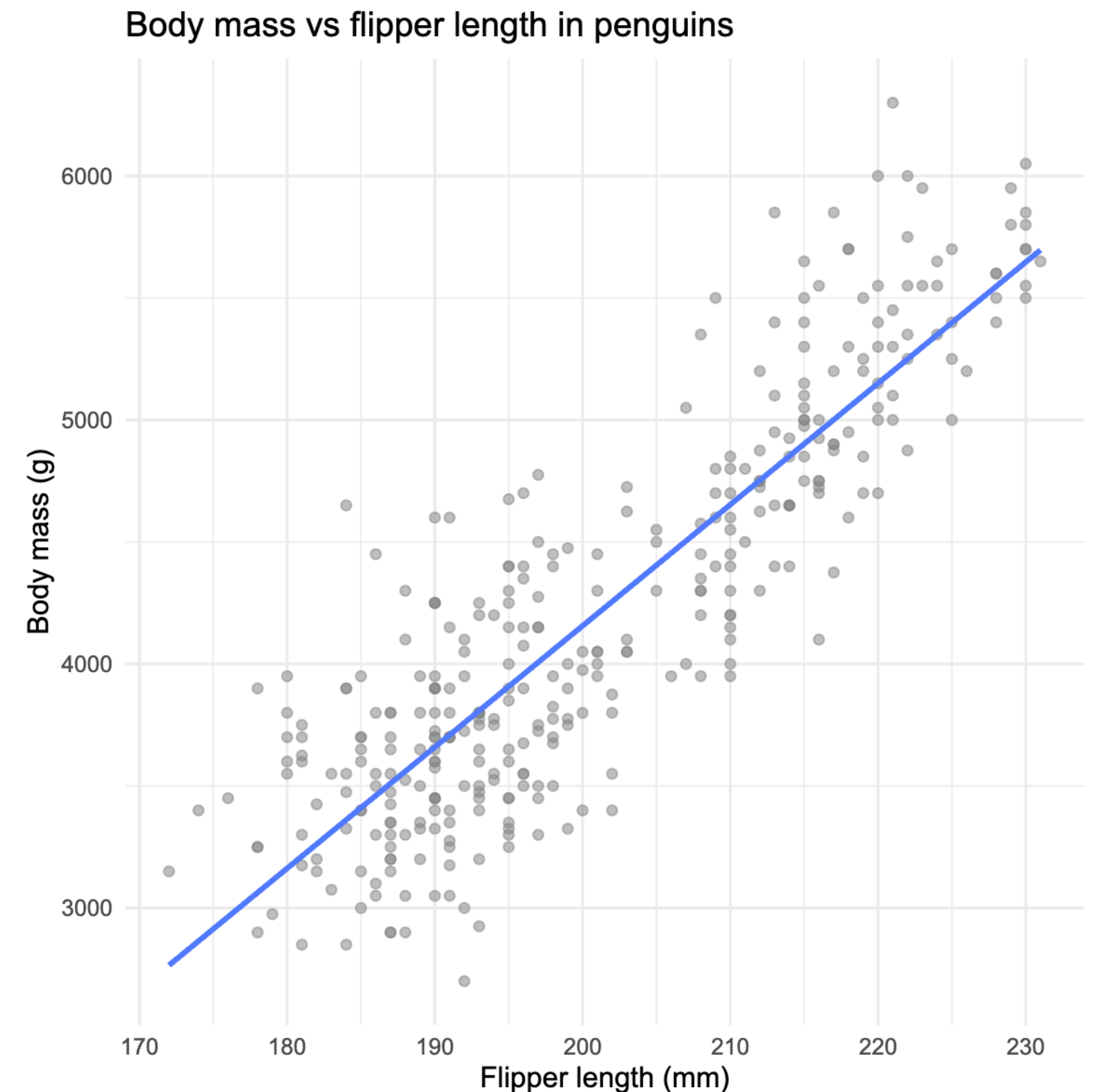
(avg. body mass) = $-5781 + 49.7$ (flipper length)

A common alternative slope interpretation:

For every 1 mm increase in flipper length in penguins, the body mass increases by 49.7 g on average.

Are we implying that if penguins' flippers grow 1 mm, they will gain 49.7 g of mass? Sounds very causal to me...

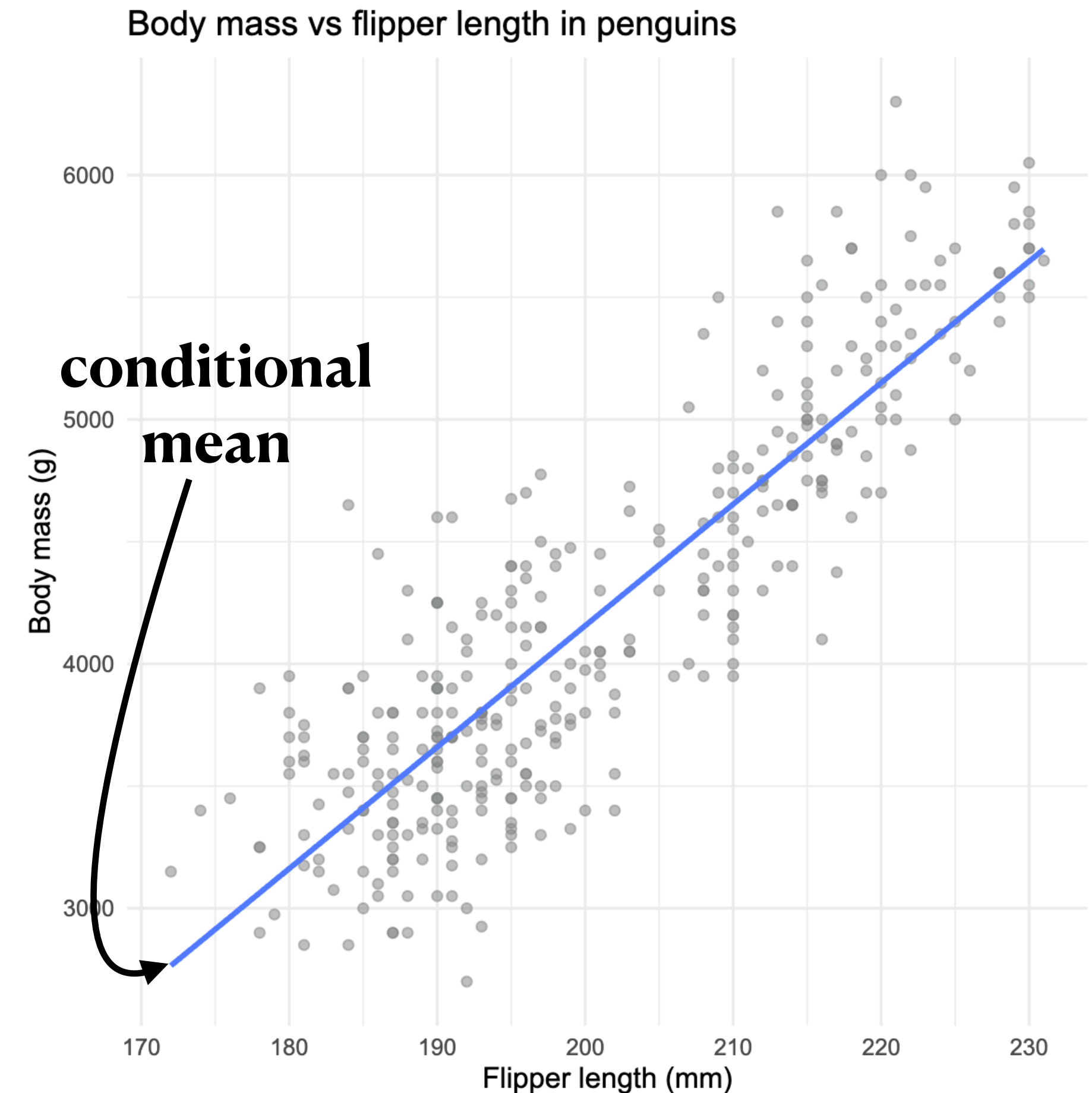
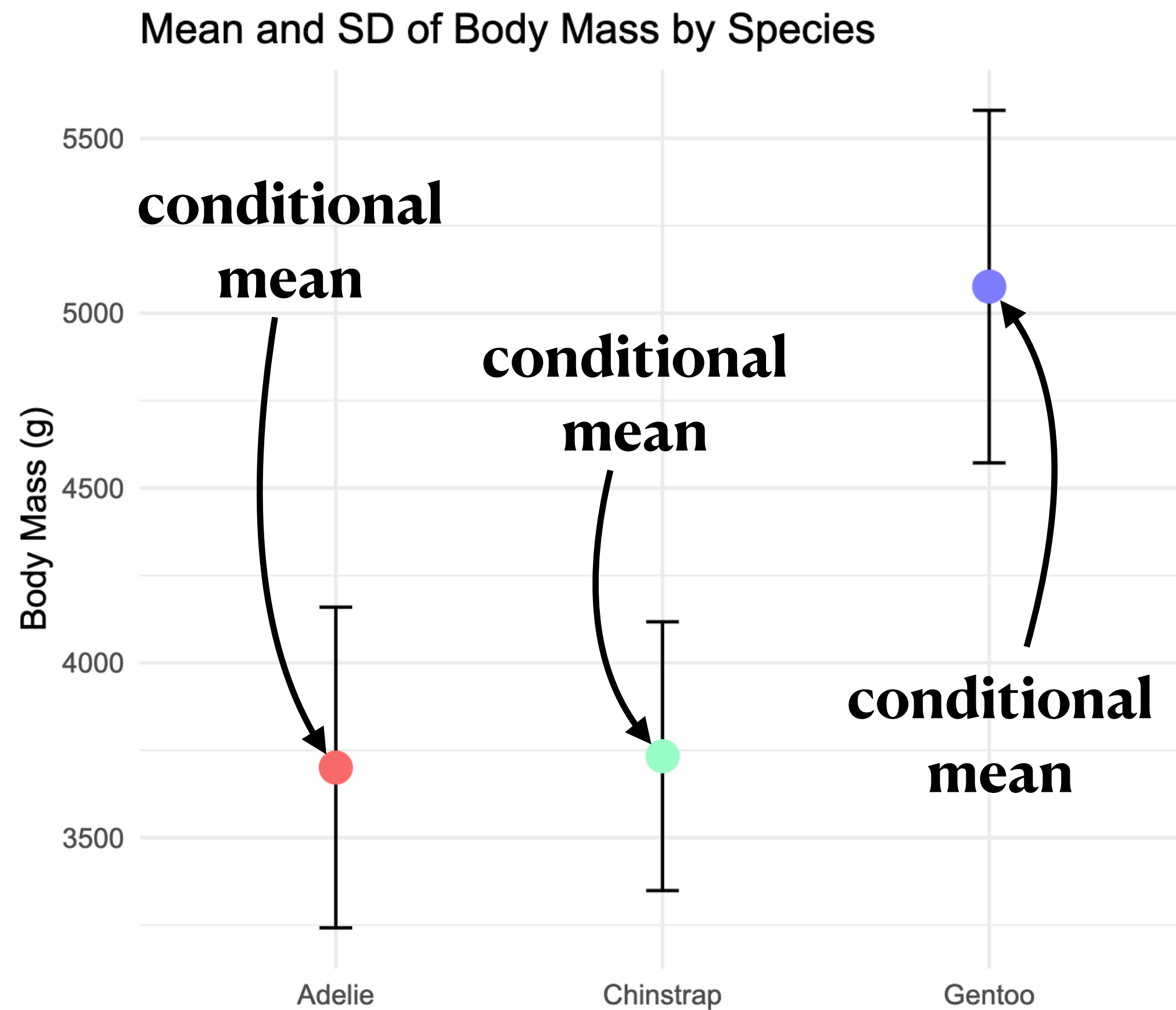
💣 This is one of the things I do differently from the textbook! 💣



Slope Interpretation

Comparing two groups of penguins of different species, the **average** body mass of the Gentoo group is 1342.9 g greater than the Chinstrap group.

Comparing two groups of penguins that differ by 1 mm in flipper length, the **average** body mass of the group with longer flippers is 49.7 g greater.



Least-Squares Regression

Interpretation

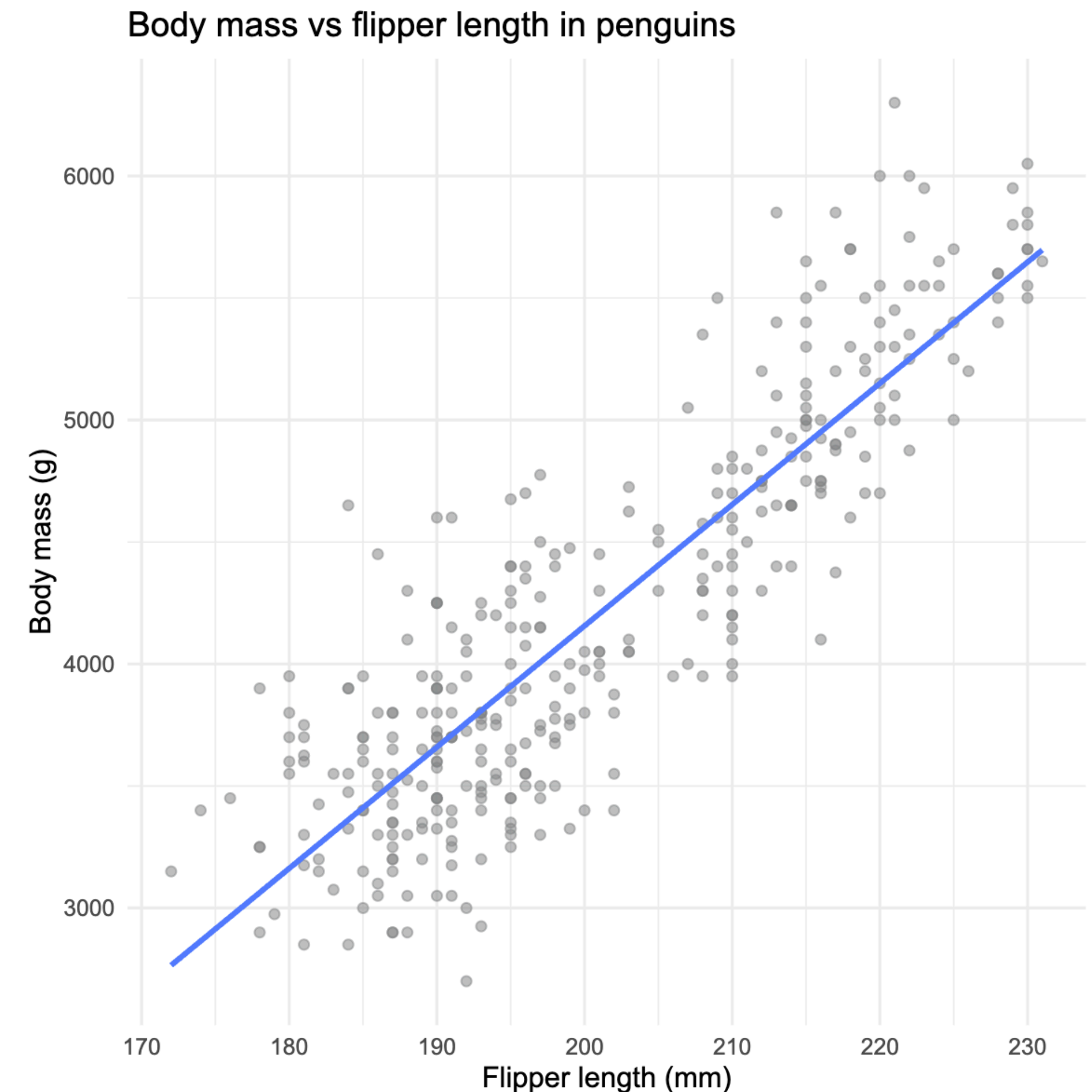
$$\hat{y} = b_0 + b_1x$$

(avg. body mass) = $-5781 + 49.7$ (flipper length)

Slope interpretation: **Comparing two groups** of penguins that differ by 1 mm in flipper length, the **average** body mass of the group with longer flippers is 49.7 g greater.

Anatomy of a good slope interpretation:

- Explicitly frames the slope as a comparison between groups differing in x by 1
- Clearly specifies that we are speaking in terms of conditional *means* or *averages*
- Interprets the number *in the scientific context*



Least-Squares Regression

Prediction

- Suppose you work in a clinic and want to predict your next patient's systolic blood pressure
 - If you know nothing about the patient, a reasonable prediction is the general average systolic blood pressure, say about 120 mmHg
- Now suppose you see the patient walk in and estimate that they are about 80 years old
 - Now you have more information: since older adults tend to have higher blood pressure on average, it may be more reasonable to predict this patient's blood pressure using the **average systolic blood pressure among older adults**, say about 130 mmHg.
- We can use the same idea for prediction with the conditional mean in linear regression

Least-Squares Regression

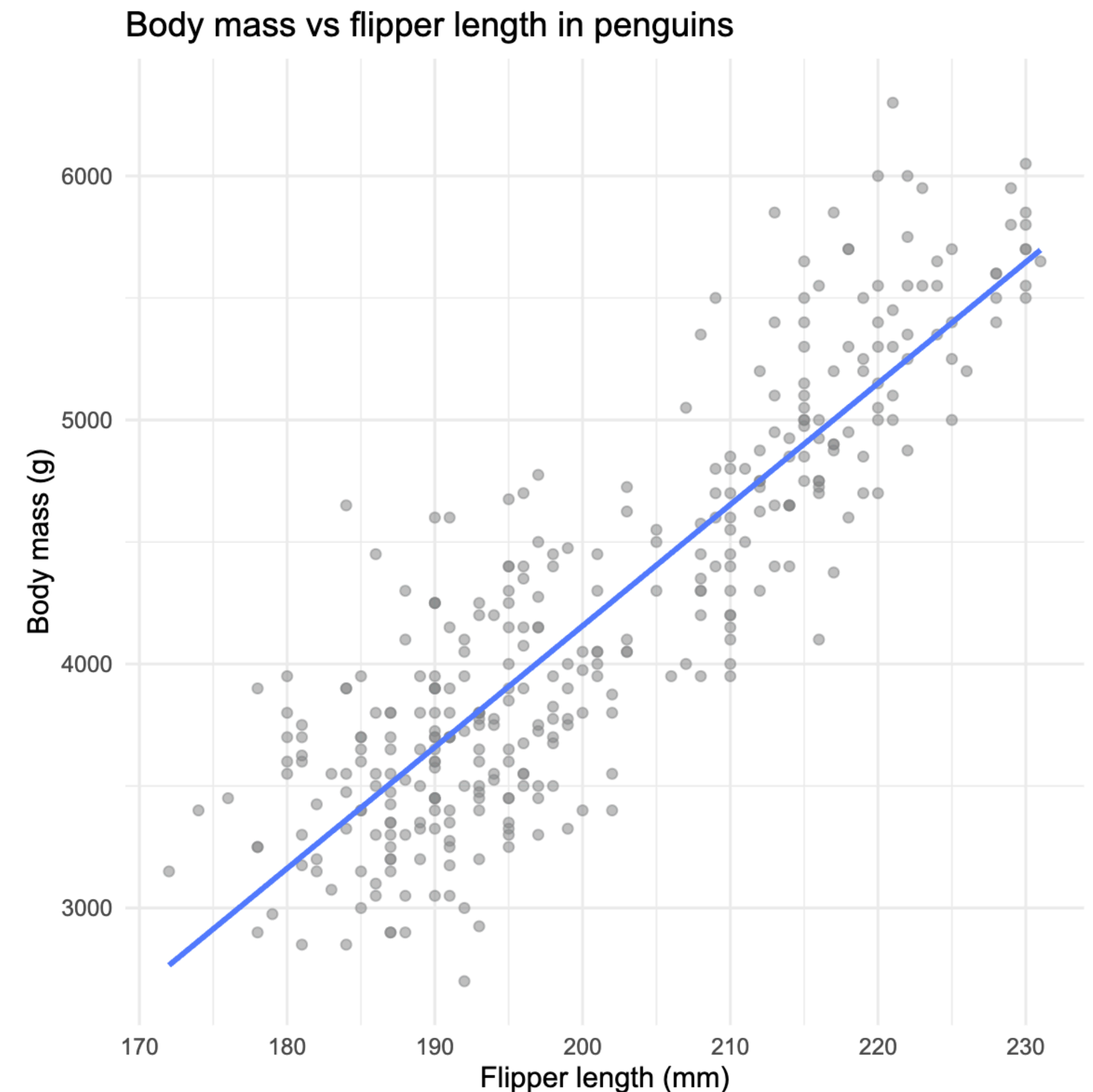
Prediction

$$\hat{y} = b_0 + b_1x$$

(avg. body mass) = $-5781 + 49.7$ (flipper length)

What would you predict for the body mass of a penguin whose flipper length is 200 mm?

$$\begin{aligned}\hat{y} &= -5781 + 49.7(200) \\ &= -5781 + 9940 \\ &= 4159 \text{ g}\end{aligned}$$



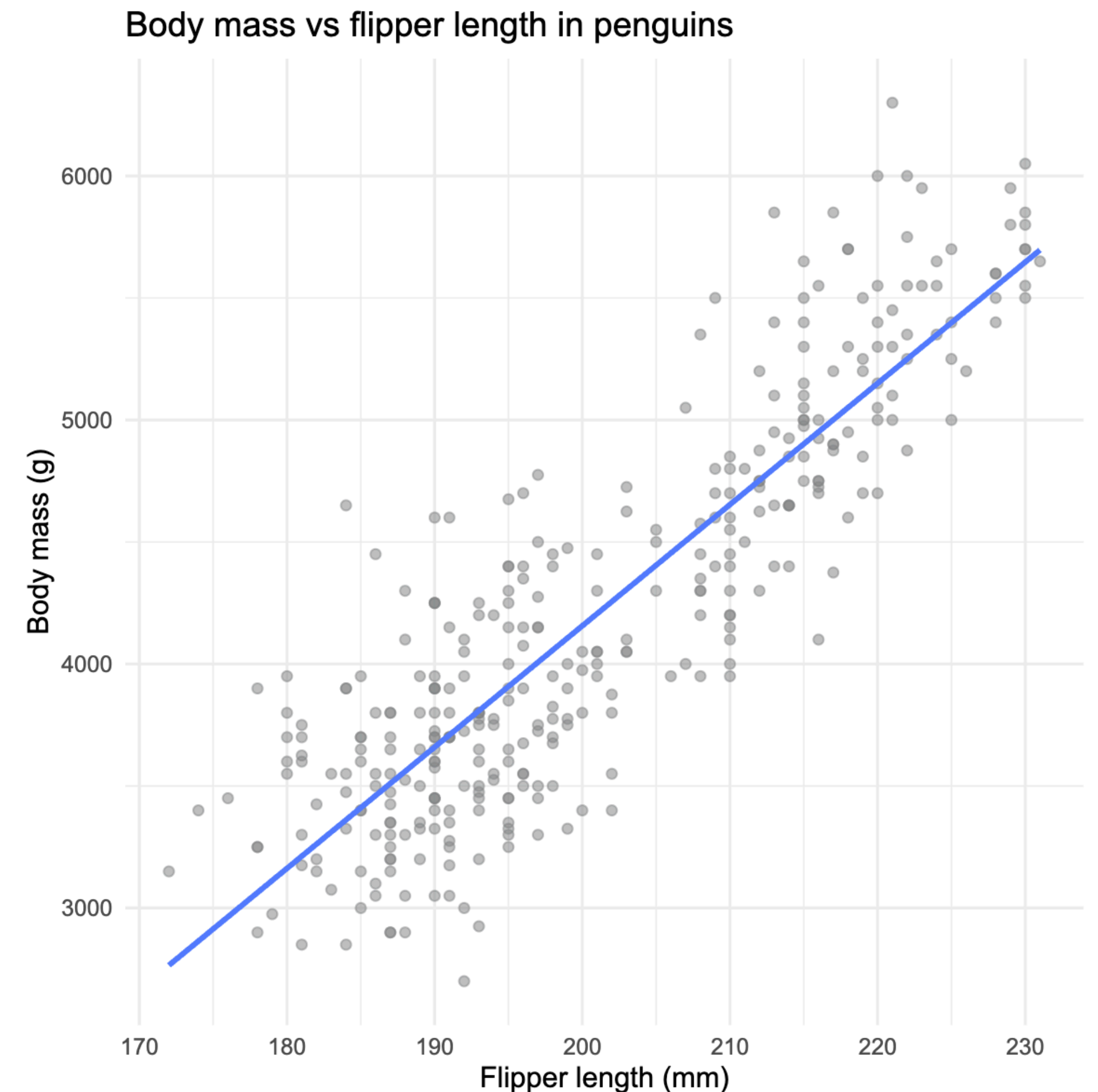
Least-Squares Regression

Prediction

$$\hat{y} = b_0 + b_1x$$

(avg. body mass) = $-5781 + 49.7$ (flipper length)

What would you predict for the body mass of a penguin whose flipper length is 400 mm?



Least-Squares Regression

Prediction

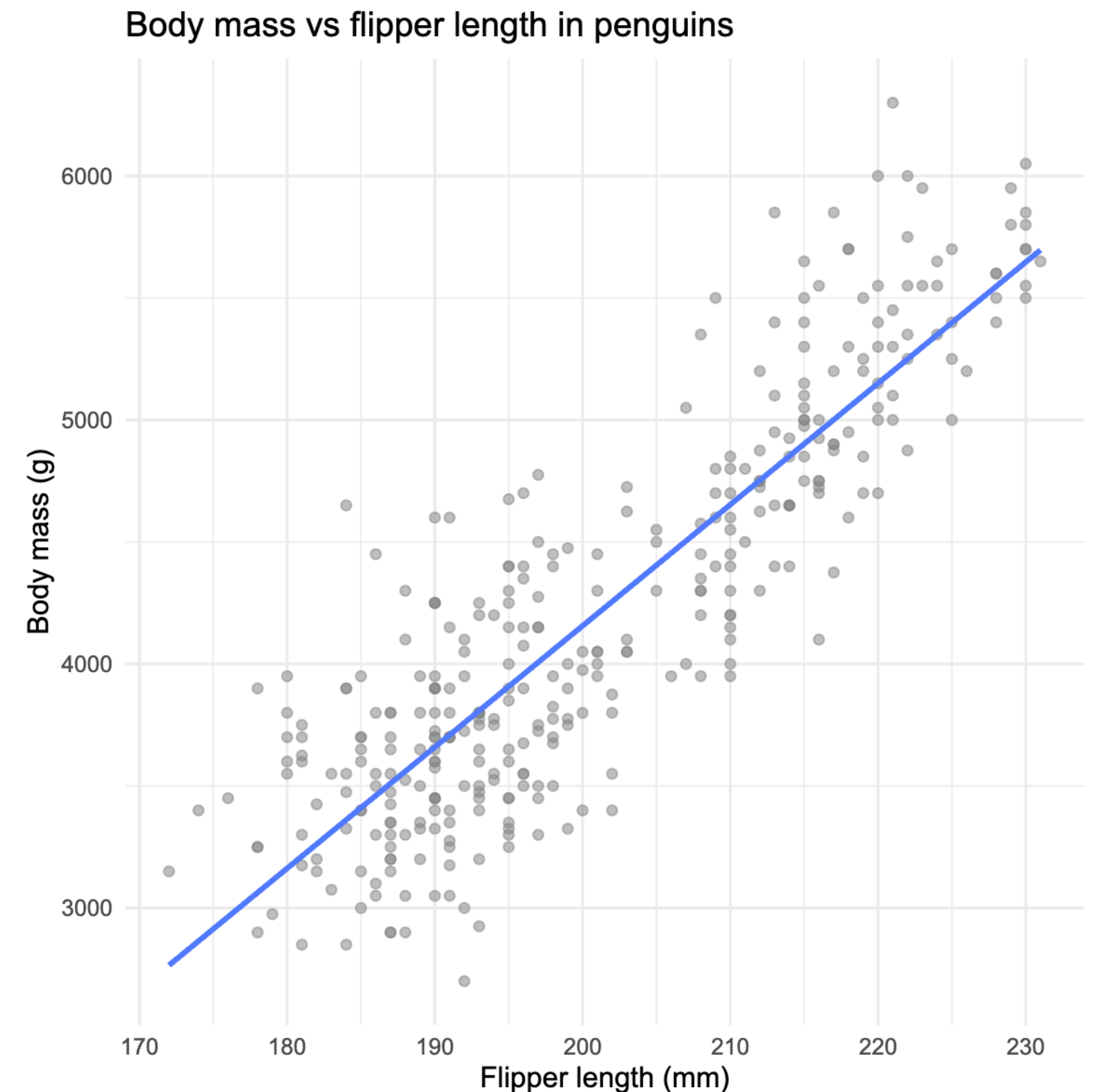
$$\hat{y} = b_0 + b_1x$$

(avg. body mass) = $-5781 + 49.7$ (flipper length)

What would you predict for the body mass of a penguin whose flipper length is 400 mm?

What would you predict for the systolic blood pressure of a patient who is 140 years old?

Extrapolation!



Least-Squares Regression

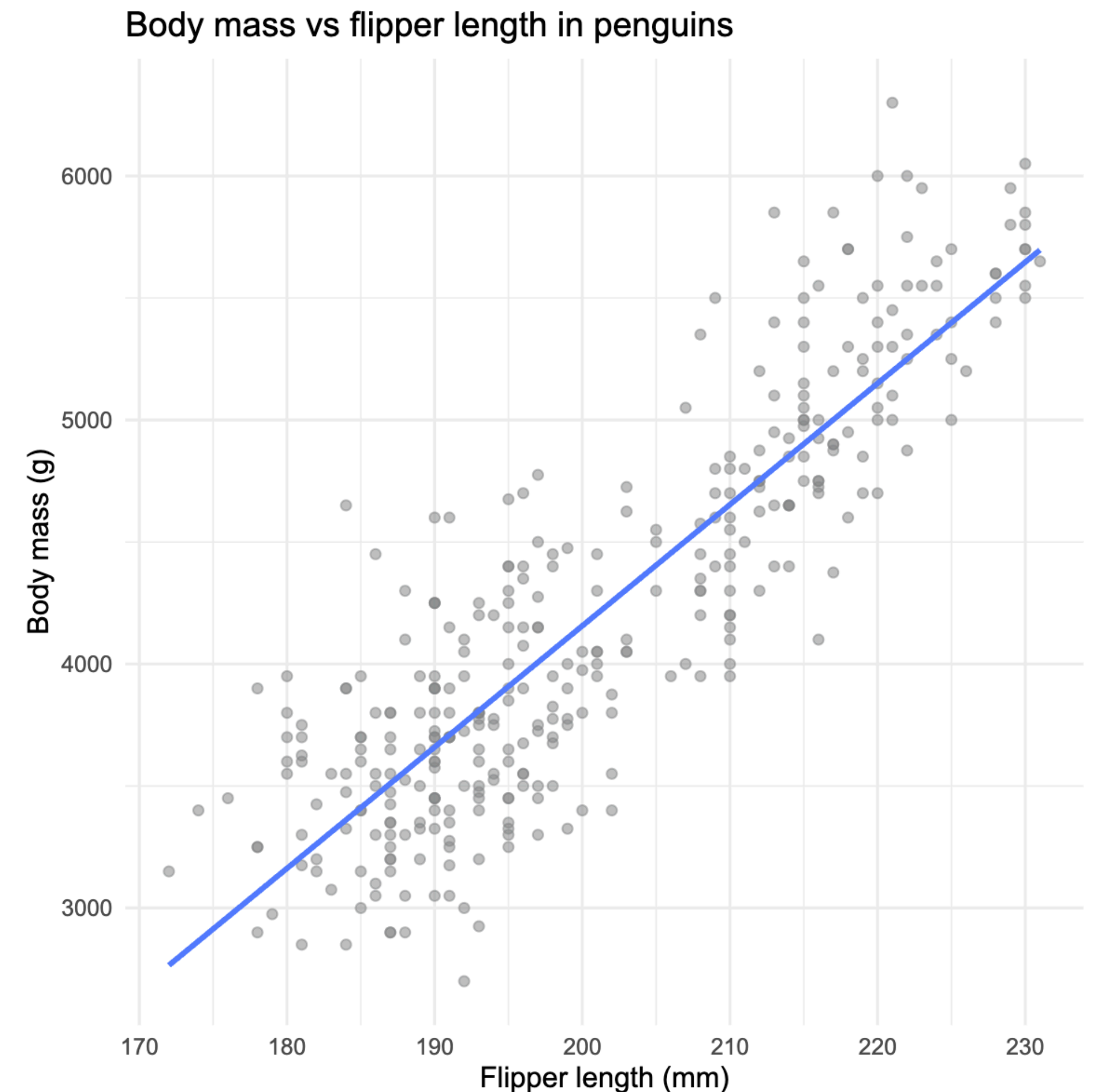
Prediction

$$\hat{y} = b_0 + b_1x$$

(avg. body mass) = $-5781 + 49.7$ (flipper length)

What would you predict for the body mass of a penguin whose flipper length is 180 mm?

PollEv.com/stats8



Week 3: Association

